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CLINICAL STUDY PROTOCOL TITLE PAGE

A Multicenter, Open-Label, Phase 2 Study to Investigate the Efficacy and Safety of Sonrotoclax Combined With Zanubrutinib Compared With Zanubrutinib Monotherapy in Adult Patients With Previously Untreated Chronic Lymphocytic Leukemia

Brief Title:

A Phase 2 Study to Investigate Sonrotoclax Combined With Zanubrutinib Versus Zanubrutinib Alone in Patients With Previously Untreated Chronic Lymphocytic Leukemia

Protocol Number:	BGB-11417-204/EU-1
Amendment Number:	Original Protocol
Investigational Medicinal Product(s):	Sonrotoclax, Zanubrutinib
Regulatory Agency Identification Number(s):	United States IND 151311 EudraCT Number 2024-513970-23-00
Sponsor:	BeiGene, Ltd. c/o BeiGene USA, Inc. 1840 Gateway Drive, 3rd Floor San Mateo, California 94404 USA
Approval Date:	*See electronic signature*
Contact Information:	BeiGene Medical Officer Phone: 1.877.828.5568 Email: clinicaltrials@beigene.com

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INVESTIGATOR SIGNATURE PAGE

I have read the protocol, appendices, and accessory materials related to study BGB-11417-204 and agree to the following:

- To conduct this study as described by the protocol and any accessory materials.
- To protect the rights, safety, and welfare of the patients under my care.
- To provide oversight to all personnel to whom study activities have been delegated. This includes personnel at my site as well as personnel working in any facility where study activities are my responsibility.
- To control all investigational products provided by the sponsor and maintain records of the disposition of those products.
- To conduct the study in accordance with all applicable laws and regulations, the requirements of the ethics committee of record for my clinical site, and current GCP as outlined by ICH E6(R2).
- To obtain approval for the protocol and all written materials provided to patients before initiating the study at my site.
- To obtain informed consent – and updated consent in the event of new information or amendments – from all patients enrolled at my study site before initiating any study-specific procedures or administering investigational products to those patients.
- To maintain records of each patient's participation and all data required by the protocol in an accurate and timely manner.

Acceptance of this protocol constitutes my agreement that no confidential information contained herein will be published or disclosed without prior written approval from BeiGene, Ltd. or one of its affiliates, unless and only to the extent required by applicable laws and regulations.

Name: [Last Name, First Name]	**Title: [**Title at Institution]	**Institution: [**Address]
Signature:		**Date: [**DD Month YYYY]

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DOCUMENT HISTORY

Amendment Version Number	Approval Date	Type of Protocol Amendment	Types of Changes
Original protocol	31 May 2024	Not applicable	Not applicable

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PROTOCOL AMENDMENT SUMMARY OF CHANGES

This protocol amendment replaces the original protocol. This amendment is considered non-substantial based on the criteria set forth in Regulation (EU) No 536/2014 of the European Parliament and the Council of the EU because it neither significantly impacts the safety or physical/mental integrity of participants nor the scientific value of the study.

The primary purpose of this amendment is to address EMA's requests. The changes made in this amendment are described in the table below. Editorial and formatting changes are not included in this summary.

Protocol Amendment Summary of Changes Table

Section Number and Title	Summary of Change	Brief Rationale for Change
Section 10.1. Regulatory and Ethical Considerations	Added statement that the study will comply with Regulation [EU] No 536/2014	To comply with EU Health Authority's (EMA) regulations
Section 10.1. Regulatory and Ethical Considerations	Added description of serious breach procedure	To comply with EU Health Authority's (EMA) regulations
Section 10.1. Regulatory and Ethical Considerations	Added description of safety reporting via Eudravigilance database in accordance with Article 40 CTR	To comply with EU Health Authority's (EMA) regulations
10.11. Publications and Data Presentation Policy	Added description of study result submission to database per CTR Annex IV	To comply with EU Health Authority's (EMA) regulations

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LIST OF ABBREVIATIONS AND DEFINITIONS OF TERMS

Abbreviation or Term	Definition/Explanation
ALC	absolute lymphocyte count
ALT	alanine aminotransferase
AST	aspartate aminotransferase
AUC	area under the concentration-time curve
AxMP	auxiliary medicinal product
Bcl-2	B-cell lymphoma-2
BCR	B Cell Receptor
BTK	Bruton tyrosine kinase
CBC	complete blood count
CDE	Center for Drug Evaluation
CIRS	Cumulative Illness Rating Scale
CLL	chronic lymphocytic leukemia
C_{max}	maximum observed plasma concentration
CR	complete response
CRi	complete response with incomplete bone marrow recovery
CRR	complete response rate
CSR	Clinical study report
CT	computed tomography
CTR	Clinical Trials Regulation
CYP	cytochrome P450
DOR	duration of response
eCRF	electronic case report form
EDC	electronic data capture
EMA	European Medicines Agency
EORTC	European Organisation for Research and Treatment of Cancer
E-R	exposure-response
EU	European Union
EudraCT	European Union Drug Regulating Authorities Clinical Trials
FDA	Food and Drug Administration
FIH	first-in-human

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Abbreviation or Term	Definition/Explanation
GCP	Good Clinical Practice
GHS	global health status
GI	gastrointestinal
HBcAb	hepatitis B core antibody
HBsAg	hepatitis B surface antigen
HBV	hepatitis B virus
HCV	hepatitis C virus
HCVAb	hepatitis C antibody
HIVAb	HIV antibody
HR	hazard ratio
HRQoL	health-related quality of life
IB	Investigator's brochure
ICF	informed consent form
ICH	International Council for Harmonisation
IEC	Independent Ethics Committee
IGHV	immunoglobulin heavy-chain variable
IND	Investigational New Drug
IRB	Institutional Review Board
IRC	Independent Review Committee
IRT	Interactive Response Technology
ITT	Intent-to-Treat
I+V	ibrutinib + venetoclax
iwCLL	International Workshop on Chronic Lymphocytic Leukemia
LDi	longest diameter
LDT	lymphocyte doubling time
MCL	mantle cell lymphoma
MRI	magnetic resonance imaging
NCI-CTCAE	National Cancer Institute-Common Terminology Criteria for Adverse Events
NGS	next-generation sequencing
ORR	overall response rate
OS	overall survival
PB	peripheral blood

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Abbreviation or Term	Definition/Explanation
PCR	polymerase chain reaction
PD	Progressive disease
PFS	progression-free survival
PK	pharmacokinetic(s)
PR	partial response
PRO	patient-reported outcome
PTFU	post-treatment follow-up
PTFU 1	first Post-treatment Follow-Up Visit
QLQ-C30	Quality of Life Questionnaire Core 30
QLQ-CLL17	Quality of Life Questionnaire – Chronic Lymphocytic Leukemia Module 17 Items
QoL	quality of life
QTcF	QT interval corrected by Fridericia's method
R/R	relapsed/refractory
SAP	statistical analysis plan
SLL	small lymphocytic lymphoma
SMC	safety monitoring committee
SZ	sonorotoxlax + zanubrutinib
TLS	tumor lysis syndrome
TN	treatment-naïve
TTR	time to response
ULN	upper limit of normal
uMRD	undetectable minimal residual disease
uMRD4	undetectable minimal residual disease at $< 10^{-4}$ sensitivity
US(A)	United States (of America)
VO	venetoclax + obinutuzumab
Z	zanubrutinib monotherapy

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1. PROTOCOL SUMMARY

1.1. Synopsis

Protocol Synopsis	
Study Title	A Multicenter, Open-Label, Phase 2 Study to Investigate the Efficacy and Safety of Sonrotoclax Combined With Zanubrutinib Compared With Zanubrutinib Monotherapy in Adult Patients With Previously Untreated Chronic Lymphocytic Leukemia
Brief Study Title	A Phase 2 Study to Investigate Sonrotoclax combined with Zanubrutinib Versus Zanubrutinib alone in Patients With Previously Untreated Chronic Lymphocytic Leukemia
Brief Summary	The purpose of this study is to support the registration plan of sonrotoclax plus zanubrutinib treatment in patients with previously untreated CLL. This study is designed to assess the contribution of sonrotoclax to the efficacy outcome of the combination of zanubrutinib and sonrotoclax. Study details include: • The study duration will be up to approximately 60 to 68 months. • The treatment duration will be up to approximately 15 months for patients in Arm A and continuous treatment of zanubrutinib until PD or intolerance for patients in Arm B.
Sponsor	BeiGene, Ltd.
Amendment Version	Not applicable
Regulatory Agency Identifier Number(s)	United States IND 151311 EudraCT Number 2024-513970-23-00
Rationale	Chronic lymphocytic leukemia (CLL) is a malignant disorder of B lymphocytes. CLL is considered a treatable but essentially incurable disease. Although BTK inhibitor monotherapy is effective at disease control, it does not induce deep remissions in most patients, thus committing patients to continuous use complicated by toxic effects and the emergence of resistance. Bcl-2 inhibitor can induce rapid onset apoptosis of CLL cells. The combination of Bcl-2 and BTK inhibition, venetoclax + ibrutinib, has been explored as time-limited treatment and found to be highly effective with deep remissions achieved in clinical trials of CLL patients. However, the toxicity profile for venetoclax + ibrutinib suggests the need for a more tolerable BTK/Bcl-2 inhibitor combination. Zanubrutinib is a more selective covalent BTK inhibitor, having demonstrated a more tolerable safety profile, and better efficacy compared to ibrutinib in CLL. BGB-11417 (sonrotoclax) is a highly potent and selective Bcl-2 protein inhibitor. In the previous clinical study, BGB-11417-101 the combination of sonrotoclax + zanubrutinib revealed robust efficacy and safety. This data compares favorably to the published data with ibrutinib + venetoclax and led to the development of a Phase 3 study in patients with previously untreated CLL for registrational purposes (BGB-11417-301). Zanubrutinib monotherapy is approved for the treatment of CLL. In the phase 3 SEQUOIA study, zanubrutinib resulted in higher ORR and statistically significant improvement in PFS compared to bendamustine + rituximab (BR) in patients with untreated CLL without del(17p)/*TP53* mutation. Zanubrutinib monotherapy is recommended for frontline CLL treatment by clinical guidelines, which supports its suitability as the control arm therapy.

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	To support the registration plan of sonrotoclax plus zanubrutinib treatment in patients with previously untreated CLL, this phase 2 study is designed to assess the contribution of sonrotoclax to the efficacy outcome of the combination of zanubrutinib and sonrotoclax. Further safety information will be provided to confirm the safety of such combination.
Objectives and Endpoints	
Objectives	**Endpoints**
Primary: To compare efficacy between Arm A (sonrotoclax plus zanubrutinib) versus Arm B (zanubrutinib) measured by the proportion of patients that achieve a complete response (CR) or complete response with incomplete bone marrow recovery (CRi)	**Primary:** Best CR/CRi rate up to the first Post-Treatment Follow-up (PTFU1) Visit for Arm A and up to the C16D1 Visit for Arm B per Independent Review Committee (IRC) response assessment using the 2018 International Workshop on Chronic Lymphocytic Leukemia (Hallek et al 2018) guidelines with modification for treatment-related lymphocytosis (Cheson et al 2012) for patients with CLL
Key Secondary: To compare efficacy between Arm A (sonrotoclax plus zanubrutinib) versus Arm B (zanubrutinib) by rate of undetectable minimal residual disease at $< 10^{-4}$ sensitivity (uMRD4)	**Key Secondary:** uMRD4 rate measured in both peripheral blood and bone marrow aspirate at PTFU 1 Visit for Arm A and at C16D1 Visit for Arm B based on next-generation sequencing (NGS) (clonoSEQ®)
Secondary: To compare the efficacy between Arm A versus Arm B through CR/CRi per investigator response assessment; overall response rate (ORR), duration of response (DOR), and time to response (TTR) per IRC and investigator assessment; progression-free survival (PFS) per investigator assessment, and overall survival (OS).	**Secondary:** • CR/CRi rate per investigator response assessment • ORR per IRC and investigator response assessment • Landmark DOR event-free rate at 12 months per IRC response assessment • DOR per investigator response assessment • TTR per IRC and investigator response assessment • Landmark PFS rate at 24 months and overall PFS per investigator response assessment • OS
Secondary: To compare the safety and tolerability of Arm A (sonrotoclax plus zanubrutinib) versus Arm B (zanubrutinib)	**Secondary:** Safety will be assessed by monitoring and recording of all treatment emergent adverse events (AEs) graded by National Cancer Institute-Common Terminology Criteria for Adverse Events (NCI-CTCAE) v5.0 and the iwCLL Grading Scale for Hematologic Toxicities in CLL Studies as appropriate. Tumor lysis syndrome (TLS) will be monitored per Howard criteria.
Study Design:	This study is a multicenter, global Phase 2, open-label, randomized trial to evaluate the efficacy and safety of a fixed duration of sonrotoclax in combination with zanubrutinib (SZ) compared to continuous zanubrutinib monotherapy (Z) (as an active comparator) in adult patients with previously untreated Chronic Lymphocytic Leukemia (CLL) without del17p or known *TP53* mutation. Eligible patients will be randomized 2:1 to receive either 3 cycles of lead-in zanubrutinib monotherapy followed by sonrotoclax in

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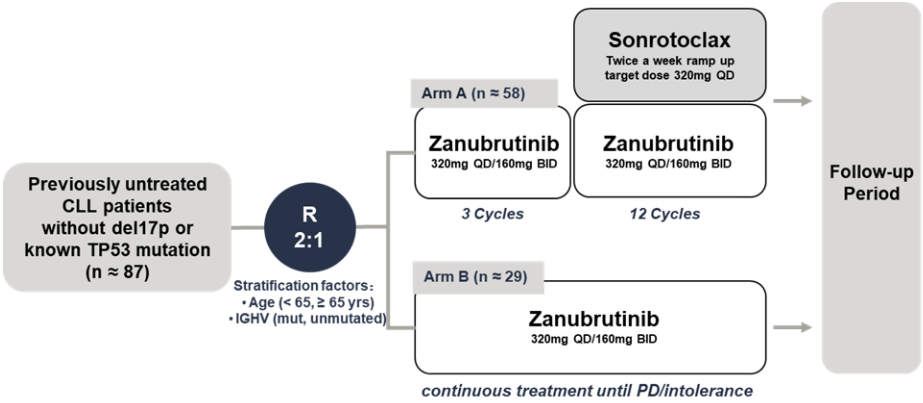
	combination with zanubrutinib for 12 cycles (Arm A), or to receive zanubrutinib monotherapy continuously until disease progression or intolerance (Arm B).  The sample size of the study will be approximately 87 patients across approximately 40 sites. Stratification factors include age (age < 65 years versus age ≥ 65 years) and *IGHV* status (mutated versus unmutated). Each patient will participate in a screening period of up to 35 days. The first dose of study drug is to be administered ≤ 5 calendar days after date of randomization. Patients will be randomized to receive open-label study treatment with 1 of the following: • Arm A: SZ • Arm B: Z A Safety Monitoring Committee (SMC) and an Independent Review Committee (IRC) have been assembled for this study. The SMC will review all available safety, efficacy, PK, and exploratory data and make recommendations. The IRC is to provide an independent and blinded review of imaging and clinical data necessary to assess tumor response in this study.
Description of Patients:	Patients aged ≥ 18 years will have a confirmed diagnosis of CLL, based on Hallek et al 2018, requiring treatment as defined by ≥ 1 of the following: progressive bone marrow failure; massive, progressive, or symptomatic splenomegaly; massive, progressive, or symptomatic lymphadenopathy; progressive lymphocytosis with rapid doubling time; symptomatic or functional extranodal involvement; and/or constitutional symptoms. Eligible patients must have ≥ 1 measurable lymph node based on computed tomography/magnetic resonance imaging and have had no prior systemic treatment for CLL; no del17p or known TP53 mutation; no history of or known prolymphocytic leukemia or history of, or currently suspected, Richter's transformation at time of consideration for study; no ongoing clinically significant cardiovascular disease; and no active infection including hepatitis B or C virus or HIV.
Number of Patients	Approximately 87 patients will be randomly assigned with a 2:1 ratio for an estimated 58 patients randomized to Arm A and 29 patients to Arm B.
Study Drug/Treatments	**Test Product, Dose, and Mode of Administration (Arm A):** • Sonrotoclax, orally, with food once daily following a twice a week ramp-up schedule until the target dose of 320 mg is reached. • Zanubrutinib, orally, with or without food at a 320 mg total daily dose (either 320 mg once daily or 160 mg twice daily). Zanubrutinib is taken orally at a 320mg total daily dose (320mg QD or 160mg BID) starting on Cycle 1, Day 1, alone for the first 3 cycles, then along with sonrotoclax for 12 cycles. Zanubrutinib is taken with or without food. On Cycle 4, Day 1, patients will

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	initiate sonrotoclax administration with food once daily using the following twice a week ramp-up schedule until the target dose of 320 mg is reached (gradually increasing incremental doses of 1 mg, 2 mg, 5 mg, 10 mg, 20 mg, 40 mg, 80 mg, and 160 mg, increasing twice a week over 1 month to reach the target dose 320 mg). The ramp-up doses and visits may be adjusted per the available data in conjunction with SMC review. Thereafter, patients will continue sonrotoclax in combination with zanubrutinib through Cycle 15, Day 28. **Active Comparator, Dose, and Mode of Administration (Arm B):** Patients in Arm B will receive a continuous zanubrutinib monotherapy. Zanubrutinib will be given orally at a 320mg total daily dose (320mg QD or 160mg BID) with or without food, until disease progression or unacceptable toxicity, whichever occurs first.
Ethical Considerations	Accounting for the measures taken to minimize the risk to patients participating in this study, the potential risks identified in association with sonrotoclax and zanubrutinib are justified by the anticipated benefits that may be afforded to patients with previously untreated CLL. The data available for the combination of SZ for the treatment of patients with previously untreated CLL shows this treatment is safe and tolerable and has compelling efficacy. This supports further investigation of this combination in this randomized study.

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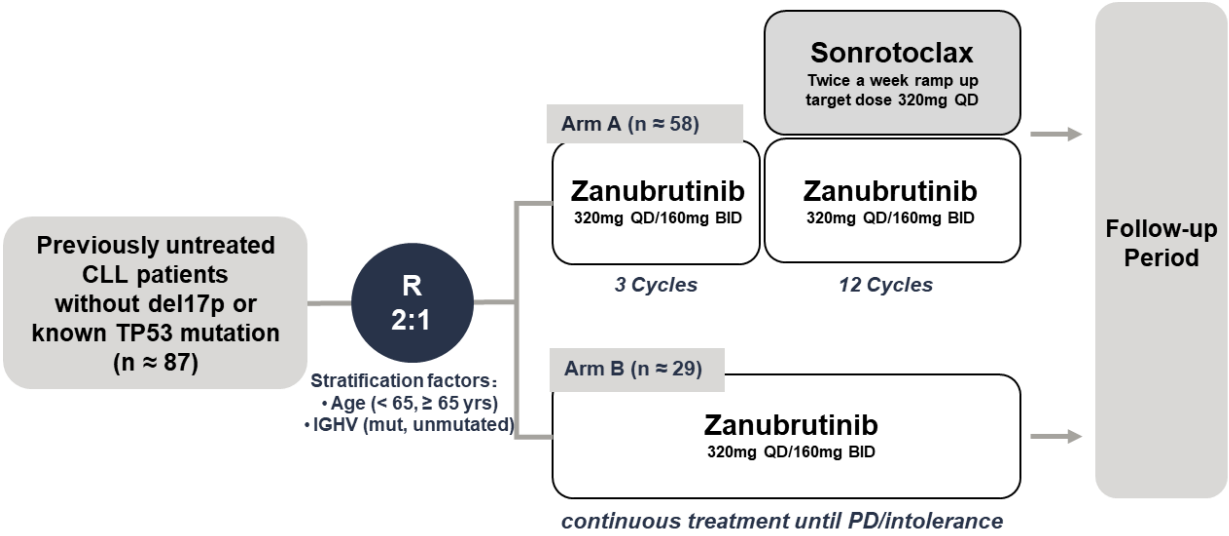
1.2. Study Schema

Figure 1: Study Schema



Abbreviations: BID, twice per day; CLL, chronic lymphocytic leukemia; *IGHV*, immunoglobulin heavy-chain variable region gene; mut, mutated; PD, progressive disease; QD, once per day; R, randomization.

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1.3. Schedule(s) of Activities

1.3.1. Schedule of Activities

Table 1: Schedule of Activities (Screening and Treatment Period) – Arm A

Study Procedures	**Protocol Section**	**Screening**	**Enrollment / Randomization [1]**	**Treatment (1 cycle = 28 days) Arm A = 15 treatment cycles [2]**								
Treatment Cycles		**–**	**–**	**C1**	**C2**	**C3**	**C4**	**C5**	**C6**	**C7**	**C10**	**C13**
Visit Day		**-35 to date of enrollment**	**-5 to 0 before C1D1**	**D1**	**D1**	**D1**	**D1 (See Table 4 for ramp-up visit schedule)**	**D1**	**D1**	**D1**	**D1**	**D1**
Visit Window (day)		**–**	**–**		**± 2**	**± 2**	**± 2**	**± 2**	**± 2**	**± 2**	**± 7**	**± 7**
Administrative												
Informed consent and screening number [3]	Section 10.3	X										
Medical and cancer history and CIRS	Section 8.1.2	X										
Demographic data	Section 8.1.1	X										
IRT enrollment and randomization [4]	Section 6.3		X									
Administration of Study Drugs												
Zanubrutinib dispensing and accountability [5]	Section 6.2 Section 6.5			X	X	X	X	X	X	X	X	X
Sonrotoclax dispensing and accountability [6]	Section 6.2 Section 6.5						X	X	X	X	X	X
Safety Assessments												
Vital signs (temperature, BP, and heart rate/pulse rate) [7]	Section 8.3.1	X		X	X	X	See Table 3		X	X	X	X
Physical examination [8]	Section 8.3.1	X		X	X	X	X	X	X	X	X	X
Cardiac function [9]	Section 8.3.2	X		X (as clinically indicated)								

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Study Procedures	**Protocol Section**	**Screening**	**Enrollment / Randomization [1]**	**Treatment (1 cycle = 28 days) Arm A = 15 treatment cycles [2]**								
Treatment Cycles		**–**	**–**	**C1**	**C2**	**C3**	**C4**	**C5**	**C6**	**C7**	**C10**	**C13**
Visit Day		**-35 to date of enrollment**	**-5 to 0 before C1D1**	**D1**	**D1**	**D1**	**D1 (See Table 4 for ramp-up visit schedule)**	**D1**	**D1**	**D1**	**D1**	**D1**
Visit Window (day)		**–**	**–**		**± 2**	**± 2**	**± 2**	**± 2**	**± 2**	**± 2**	**± 7**	**± 7**
ECOG Performance Status	Section 8.3.7.3	X		X	X	X	X	X	X	X	X	X
12-Lead ECG (local read) [10] /QTcF	Section 8.3.3	X		X			X	X		X	X	X
TLS risk category determination [11]	Section 8.4.8			X			X					
Concomitant medications review	Section 8.3.7.1	X		X	X	X	X	X	X	X	X	X
AE review [12]	Section 8.3.7.2 Section 8.4.1	X	X	X	X	X	X	X	X	X	X	X
Efficacy Assessments												
Physical examination of liver, spleen, and lymph nodes [13]	Section 8.2.2	X		X			X			X	X	X
Disease-related constitutional symptoms	Section 8.2.1	X		X			X			X	X	X
CT with intravenous contrast [14]/MRI	Section 8.2.3	X					X				X	
Bone marrow biopsy [15]	Section 8.2.4			X								
Disease status/ response assessment [16]	Section 8.2	X		X			X			X	X	X
PRO questionnaires [17]	Section 8.5			X			X			X	X	X
Laboratory Assessments												
Hematology [18]	Section 8.3.4	X		X	X	X	See Table 3		X	X	X	X
Serum chemistry [19, 20]	Section 8.3.4	X		X	X	X	X	X	X	X	X	X
TLS laboratory risk assessment and monitoring [21]	Section 8.3.4						See Table 3					

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Study Procedures	**Protocol Section**	**Screening**	**Enrollment / Randomization [1]**	**Treatment (1 cycle = 28 days) Arm A = 15 treatment cycles [2]**								
Treatment Cycles		**–**	**–**	**C1**	**C2**	**C3**	**C4**	**C5**	**C6**	**C7**	**C10**	**C13**
Visit Day		**-35 to date of enrollment**	**-5 to 0 before C1D1**	**D1**	**D1**	**D1**	**D1 (See Table 4 for ramp-up visit schedule)**	**D1**	**D1**	**D1**	**D1**	**D1**
Visit Window (day)		**–**	**–**		**± 2**	**± 2**	**± 2**	**± 2**	**± 2**	**± 2**	**± 7**	**± 7**
Serum immunoglobulins [22]	Section 8.3.4	X					X			X	X	X
Coagulation tests [23]	Section 8.1.7	X										
Coombs test	Section 8.1.8	X										
IGHV mutation analysis [24]	Section 8.7.1	X										
del17p and cytogenetics [25]	Section 8.7.1	X										
TP53 mutation [25]	Section 8.7.1	X										
Complex karyotype [25]	Section 8.7	X										
PB sample collection for Flow Cytometry MRD [26]	Section 8.2.6	X					X			X	X	X
BMA sample collection for Flow Cytometry MRD [26]	Section 8.2.6			At CR/CRi								
PB sample collection for NGS MRD [26]	Section 8.2.6	X								X		
Diagnostic flow cytometry for CLL markers [27]	Section 8.1.5	X										
Resistance mechanism biomarker sample [28]	Section 4.2.5.4			At C1D1 pre-dose, PTFU 1 Visit, and unequivocal disease progression								
Hepatitis B, C, and HIV testing [29]	Section 8.1.3 Section 8.1.6	X										
HBV DNA and HCV RNA monitoring by PCR (if applicable) [30]	Section 8.1.3			Every cycle								
Urinalysis [31]	Section 8.1.10	X										

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Study Procedures	Protocol Section	Screening	Enrollment / Randomization [1]	Treatment (1 cycle = 28 days) Arm A = 15 treatment cycles [2]								
Treatment Cycles		–	–	**C1**	**C2**	**C3**	**C4**	**C5**	**C6**	**C7**	**C10**	**C13**
Visit Day		**-35 to date of enrollment**	**-5 to 0 before C1D1**	**D1**	**D1**	**D1**	**D1 (See Table 4 for ramp-up visit schedule)**	**D1**	**D1**	**D1**	**D1**	**D1**
Visit Window (day)		–	–		**± 2**	**± 2**	**± 2**	**± 2**	**± 2**	**± 2**	**± 7**	**± 7**
Pregnancy test (if applicable) [32]	Section 8.3.5	X		Every 4 weeks								

1.3.1.1. Footnotes for Table 1: Schedule of Activities (Screening and Treatment Period) – Arm A

Abbreviations: AE, adverse event; AESI, adverse event of special interest; ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate aminotransferase; BMA, bone marrow aspirate; BP, blood pressure; CIRS, Cumulative Illness Rating Scale; CLL, chronic lymphocytic leukemia; CR, complete response; CRi, complete response with incomplete bone marrow recovery; CT, computed tomography; eCRF, electronic case report form; ECG, electrocardiogram; EORTC, European Organization For Research and Treatment of Cancer; GHPS, gated heart pool scan; HBcAb, hepatitis B core antibody; HBsAb, hepatitis B surface antibody; HBsAg, hepatitis B surface antigen; HBV, hepatitis B virus; HCV, hepatitis C virus; HCVAb, hepatitis C virus antibody; HIVAb, human immunodeficiency virus antibody; Ig, immunoglobulin; *IGHV*, immunoglobulin heavy-chain variable region gene; IRT, Interactive Response Technology; MRI, magnetic resonance imaging; MRD, minimal residual disease; MUGA, multigated acquisition scan; NGS, next-generation sequencing; PB, peripheral blood; PCR, polymerase chain reaction; PD, disease progression; PRO, patient-reported outcome; PTFU, Post-treatment Follow-up; PTFU 1, first Post-treatment Follow-Up Visit; QLQ-C30, Quality of Life Questionnaire – Core 30; QLQ-CLL17, Quality of Life Questionnaire – Chronic Lymphocytic Leukemia Module 17 Items; QTcF, QT interval corrected by Fridericia's method; SAE, serious adverse event; SZ, sonrotoclax + zanubrutinib; TLS, tumor lysis syndrome; Z, zanubrutinib.

[1] The time to C1D1 should be within 5 calendar days after randomization/treatment assignment.

[2] All scheduled activities should occur predose at Day 1 of each cycle unless specified.

[3] This must occur before any study specific procedures and may be obtained before the 35-day screening window. Consent must be obtained on the current version of the form approved by the independent ethics committee.

[4] Patients will be randomized into 1 of 2 arms. Central randomization (2:1) will be used to assign patients to 1 of the following study drug treatments: Arm A (SZ) or Arm B (Z).

[5] Zanubrutinib will be permanently discontinued after Cycle 15 for the Arm A of SZ. A drug administration diary for zanubrutinib will be provided to each patient to record the study drug dose taken each day. Any missed doses should be recorded with explanation in the diary. The diary should be brought to the clinic for each visit and will be reviewed by the study coordinator on a regular basis. The complete final patient dosing diary will be provided to the study staff upon the completion of treatment at the PTFU 1 Visit. The patient diary is not applicable to the PTFU period.

[6] The first dose of sonrotoclax must be administered on C4D1 (no window), however the assessments for C4D1 may be performed within the noted window unless otherwise specified. Sonrotoclax will be permanently discontinued after Cycle 15. Any missed doses should be recorded with explanation in the diary. The diary should be brought to the clinic for each visit and will be reviewed by the study coordinator on a regular basis. The complete final patient dosing diary will be provided to the study staff at upon the completion of treatment at the PTFU 1 Visit. The patient diary is not applicable to the PTFU period. If the ramp-up period is prolonged, C6D1 should be delayed until the first target dose of sonrotoclax is administered.

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[7] In addition to the vital signs required for each cycle visit, vital signs will be performed at laboratory collection timepoints for the initial sonrotoclax dose and each dose increase.

[8] Assess systems per standard of care at the study site and as clinically indicated by symptoms. Includes weight (height at Screening Visit only). The assessment of vital signs and a focused physical examination on Cycle 1 Day 1 may be skipped if performed within the last 7 days.

[9] An echocardiogram, MUGA, or GHPS will be performed within 90 days before randomization (ie, a standard-of-care procedure performed before consent was received for this study may be used if within window), and as medically necessary.

[10] A 12-lead ECG will be performed in triplicate (≥ 1 minute apart) at screening. The QTcF value may be calculated as the numerical average of up to 3 separate readings for eligibility. During Treatment period visits a single 12-lead ECG read will be performed at the scheduled visits and prior to any study drug(s) administration on that day.

[11] All patients must have an assessment of TLS risk category completed at C1D1 and C4D1 before the first dose of sonrotoclax. Imaging and hematology laboratory results are required for TLS risk assessment. Risk category must be determined with sufficient time to allow initiation of required TLS medication.

[12] After informed consent has been signed but prior to the administration of the study drug or screen failure, only SAEs should be collected. After initiation of study drug, all AEs, SAEs and AESI, regardless of relationship to study drug(s), will be collected and reported until 30 days after the last dose of study drug, or before the start of the next line of CLL treatment, whichever occurs earlier. After this period, the investigator should report any SAEs that are believed to be related to prior study drug treatment. For all patients, any new second primary malignancy, regardless of severity and relationship to study drug(s), should be recorded until the end of the study.

[13] This procedure can occur at the same time as the physical examination under “Safety Assessment” so long as the visit date adheres to the required visit window for response evaluation.

[14] All patients must have baseline imaging within 35 days before randomization. A CT scan with intravenous contrast (unless severe contrast allergy or likely contrast nephropathy) or MRI of the neck, chest, abdomen, pelvis, and any other disease sites should be performed. Imaging of the neck, chest, abdomen, and pelvis is required for all patients at screening, C4D1, C10D1, and at the PTFU 1 Visit. Imaging studies will also be performed for confirmation of suspected occurrence of CR or CRi and promptly at the time of suspected progressive disease based on physical examination findings. Imaging performed outside of the optimal procedure window specified in the Schedule of Activities will not necessarily be assessed as a protocol deviation; the sponsor will make the final determination. Copies of all scans will be sent to the Independent Review Committee for potential response assessment up to the PTFU 1 visit.

[15] During the treatment period, bone marrow biopsy and aspirate are required if clinical and laboratory results demonstrate a potential CR/CRi, regardless of minimal residual disease status. The bone marrow biopsy will be centrally tested and reviewed.

[16] Disease status is the baseline assessment of CLL before the first dose of study drug. After the start of study drug: For a timepoint without imaging provide a clinical response evaluation; for a timepoint including imaging provide an overall response evaluation.

[17] PROs (EORTC QLQ-C30 and QLQ-CLL17) will be collected at C1D1and every 3 cycles thereafter (C4D1, C7D1, C10D1 and C13D1) prior to dosing and any health-related procedures or discussion.

[18] Complete blood count will be evaluated by local laboratory, including hemoglobin, hematocrit, platelet count, red blood cell count, and white blood cell count with differential, which includes neutrophils, lymphocytes, monocytes, eosinophils, and basophils. Screening blood tests performed within 72 hours of the first study drug administration do not need to be repeated in Cycle 1. It is not necessary to duplicate the hematology laboratory tests under “Safety Assessment” and “Efficacy Assessment” so long as the date of laboratory test results adheres to the required visit windows for both safety and efficacy assessments.

[19] Serum chemistry will be evaluated by local laboratory. Serum chemistries include sodium, potassium, chloride, bicarbonate or CO_2 (or if neither are available, CO_2 combining power), glucose, blood urea nitrogen (or urea), creatinine, calcium, phosphate/phosphorus, magnesium, total bilirubin, total protein, albumin, ALT, AST, lactate dehydrogenase, ALP, and uric acid.

[20] β-2 microglobulin and direct antiglobulin test will only be done at screening and will be performed locally.

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[21] For sonrotoclax ramp-up TLS monitoring: Refer to Table 3 timepoints for select serum chemistry (including uric acid, creatinine, phosphorus, potassium and calcium) and complete blood count including hemoglobin, hematocrit, platelet count, red blood cell count, and white blood cell count with differential, which includes neutrophils, lymphocytes, monocytes, eosinophils, and basophils. These samples are to be sent immediately to the laboratory, and the results must be reviewed promptly by the investigator. Additional laboratory assessments may be performed per investigator discretion. The time of sonrotoclax administration at initiation and at each dose increase will be recorded.

[22] Serum immunoglobulin evaluations by local laboratory include IgG, IgM, and IgA. For Cycle 4, serum immunoglobulin is only required on cycle 4 day 1.

[23] The coagulation profile includes prothrombin time, which will also be reported as international normalized ratio, and activated partial thromboplastin time. The coagulation profile will be performed at screening only and thereafter as medically indicated by local laboratory.

[24] Prospective baseline biomarker test results for *IGHV* mutations from central laboratory are required for randomization. All patients will provide PB samples at the time of screening to study the mutation status of *IGHV* by next-generation sequencing using a specialized central laboratory.

[25] A PB sample will be collected during screening to perform central laboratory assessment to study metaphase complex karyotype, CLL cytogenetics (eg, del17p, del11q, del13q and trisomy 12) and the *TP53* mutation status.

[26] All patients will have PB samples collected at screening, C4D1, C7D1, C10D1, C13D1, PTFU 1 and every 16 weeks from PTFU 1 Visit until PD and BMA sample collected at CR/CRi for flow cytometry MRD assessment. All patients will have PB collected at screening, C7D1, and PTFU 1 Visit; BMA sample collected at PTFU 1 Visit for NGS MRD assessment (Table 23).

[27] Peripheral blood samples will be collected during screening to perform central laboratory assessment to study CLL diagnostic by flow.

[28] To study the resistance mechanism of SZ, all patients will be asked to provide a PB sample at C1D1 predose, PTFU 1 Visit, at the time when unequivocal progressive disease occurs. These PB samples will be used for exploratory biomarker assessment such as mutation detection and/or molecular profiling to understand potential resistance mechanisms in patients receiving SZ.

[29] Hepatitis B and C serologies (includes HBsAg, HBsAb, HBcAb and HCVAb) will be tested at screening. An HIV serology (HIVAb) test will be conducted at screening unless previous HIV test results from ≤ 4 weeks before screening are available. These tests are performed by local laboratory.

[30] Patients who are HBsAg negative, HBcAb positive, and HBV DNA negative and not on prophylactic antiviral treatment must undergo HBV DNA monitoring by PCR at least once on every cycle. These patients should be considered for prophylactic antiviral treatment in consultation with a local HBV expert. If a patient is being treated prophylactically with antivirals, HBV DNA monitoring by PCR must be done at least every third cycle. Patients positive for HCV antibody but negative for HCV RNA must have HCV RNA testing every cycle during the study treatment period. After the end of treatment, monitoring of HBV DNA/HCV RNA by PCR should continue every 12 weeks for 12 months starting from the last dose of study treatment.

[31] Urinalysis will include pH, glucose, proteins, ketones, and specific gravity.

[32] All women of childbearing potential (including those who have had a tubal ligation) will have a serum pregnancy test ≤ 7 days before the first dose of study drug as part of the screening process. Laboratory-based, highly sensitive urine or serum pregnancy tests will also be performed with the results reviewed by the investigators on Cycle 1 Day 1 prior to dosing. Either serum or urine pregnancy tests will be performed every 4 weeks during the treatment period at specified visits. Patients are allowed to take the pregnancy tests locally in the cycles with no scheduled visits, but the investigators should be informed of the test results and should record them in the eCRF timely. Either serum or urine pregnancy tests will continue to be performed every 4 weeks until ≥ 90 days after the last dose of study drug. This will require additional visits after the Safety Follow-up Visit. If a urine pregnancy test is positive, it must be confirmed by a serum pregnancy test. Any patient with a positive serum pregnancy test result will immediately discontinue study treatment.

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Table 2: Schedule of Activities (Post-treatment and Survival Follow-up) – Arm A

Study Period or Visit	**Protocol Section**	**Safety Follow-up Visit [1] (30 days after the last dose of study drug[s])**	**Post-treatment Follow-up Period [2]**		**Survival Follow-up Period [3]**
Visit			**PTFU 1 (The day after the last dose of study drug[s])**	**Every 16 weeks beginning from PTFU 2 (PTFU 2, 3, etc)**	**Every 16 weeks**
Day		**–**	**1**	**1**	**1**
Visit Window (Days)		**± 7**	**+ 14**	**± 14**	**± 14**
Safety assessments					
Vital signs (temperature, BP, and heart rate/pulse rate)	Section 8.3.1	X	X		
Physical examination [4]	Section 8.3.1	X	X		
ECOG Performance Status	Section 8.3.7.3	X	X		
12-Lead ECG (local read) [5] /QTcF	Section 8.3.3	X	X		
AE review [6]	Section 8.3.7.2	X	X	X	X
Administration of study drug(s)					
Zanubrutinib accountability [7]	Section 6.5		X		
Sonrotoclax accountability [8]	Section 6.5		X		
Efficacy Assessments					
Disease-related constitutional symptoms	Section 8.2.1		X	X	
Physical examination of liver, spleen, and lymph nodes [9]	Section 8.2.2		X	X	
CT with intravenous contrast / MRI [10]	Section 8.2.3		X		

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Study Period or Visit	**Protocol Section**	**Safety Follow-up Visit [1]**	**Post-treatment Follow-up Period [2]**		**Survival Follow-up Period [3]**
Visit		**(30 days after the last dose of study drug[s])**	**PTFU 1 (The day after the last dose of study drug[s])**	**Every 16 weeks beginning from PTFU 2 (PTFU 2, 3, etc)**	**Every 16 weeks**
Day		**–**	**1**	**1**	**1**
Visit Window (Days)		**± 7**	**+ 14**	**± 14**	**± 14**
Bone marrow biopsy [11]	Section 8.2.4		X	X [11]	
Response assessment [12]	Section 8.2		X	X	
Survival status data collection [13]	Section 8.2.8				X
PRO questionnaires [14]	Section 8.5	X	X	X	
Laboratory Assessments					
Hematology [15]	Section 8.3.4	X	X	X	
Serum chemistry [16]	Section 8.3.4	X	X		
Serum immunoglobulin [17]	Section 8.3.4	X	X		
PB Sample collection for Flow Cytometry MRD [18]	Section 8.2.6		X	X	
BMA Sample collection for Flow Cytometry MRD [18]	Section 8.2.6		At CR/CRi		
PB Sample collection for NGS MRD [18]	Section 8.2.6		X		
BMA Sample collection for NGS MRD [18]	Section 8.2.6		X		
Resistance mechanism biomarker sample [19]	Section 4.2.5.4	PTFU 1 Visit and at unequivocal disease progression			

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Study Period or Visit	Protocol Section	Safety Follow-up Visit [1] (30 days after the last dose of study drug[s])	Post-treatment Follow-up Period [2]		Survival Follow-up Period [3]
Visit			**PTFU 1 (The day after the last dose of study drug[s])**	**Every 16 weeks beginning from PTFU 2 (PTFU 2, 3, etc)**	**Every 16 weeks**
Day		**–**	**1**	**1**	**1**
Visit Window (Days)		**± 7**	**+ 14**	**± 14**	**± 14**
HBV DNA and HCV RNA monitoring by PCR (if applicable) [20]	Section 8.1.3	Every 12 weeks			
Pregnancy test (if applicable) [21]	Section 8.3.5	Every 4 weeks			

1.3.1.2. Footnotes for Table 2: Schedule of Activities (Post-treatment and Survival Follow-up) – Arm A

Abbreviations: AE, adverse event; AESI, adverse event of special interest; ALC, absolute lymphocytic count; BMA, bone marrow aspirate; BP, blood pressure; CLL, chronic lymphocytic leukemia; CR, complete response; CRi, complete response with incomplete bone marrow recovery; CT, computed tomography; ECG, electrocardiogram; EORTC, European Organisation for Research and Treatment of Cancer; eCRF, electronic case report form; Hb, hemoglobin; HBcAb, hepatitis B core antibody; HBsAg, hepatitis B surface antigen; HBV, hepatitis B virus; HCVAb, hepatitis C virus antibody; HIVAb, HIV antibody; Ig, immunoglobin; MRD, minimal residual disease; MRI, magnetic resonance imaging; NGS, next-generation sequencing; PB, peripheral blood; PCR, polymerase chain reaction; PD, disease progression; PRO, patient-reported outcome; PTFU, post-treatment follow-up; PTFU 1, first Posttreatment Follow-up Visit; QLQ-C30, Quality of Life Questionnaire – Core 30; QLQ-CLL17, Quality of Life Questionnaire – Chronic Lymphocytic Leukemia Module 17 Items; QTcF, QT interval corrected by Fridericia's method; SAE, serious adverse event; SZ, sonrotoclax + zanubrutinib.

[1] Approximately 30 days after permanent treatment discontinuation or before initiation of a new anticancer therapy, whichever occurs first.

[2] Patients with no assessed unequivocal disease progression at permanent treatment discontinuation will enter PTFU period the day after the last dose of study drug(s). PTFU 1 Visit will be performed the day after the last dose of study drug. Subsequent PTFU visits (eg, PTFU 2, PTFU 3) will be performed every 16 weeks. Patients remain on PTFU period until unequivocal disease progression has been assessed.

[3] Patients with assessed unequivocal disease progression will enter the Survival Follow-up period until death, withdrawal of patient consent, loss to follow-up, or study closure whichever occurs first. Survival Follow-up visits will be performed every 16 weeks.

[4] Assess systems per standard of care at the study site and as clinically indicated by symptoms. Includes weight.

[5] A 12-lead ECG will be performed in triplicate (≥ 1 minute apart) at Safety Follow-up Visit.

[6] All AEs, SAEs, and AESIs, regardless of relationship to study drug, will be collected and reported until 30 days after the last dose of study drug, the first day of CLL treatment, or until disease progression, whichever occurs earlier. After this period, the investigator should report any SAEs that are believed to be related to prior study drug treatment. For all patients, any new second primary malignancy, regardless of severity and relationship to study drug, should be recorded until the end of the study.

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[7] Zanubrutinib will be permanently discontinued after Cycle 15. The complete final patient dosing diary will be provided to the study staff at the completion of treatment on PTFU 1 Visit. The patient diary is not applicable during the PTFU period.

[8] Sonrotoclax will be permanently discontinued after Cycle 15. The complete final patient dosing diary will be provided to the study staff at the completion of treatment on PTFU 1 Visit. The patient diary is not applicable during the PTFU period.

[9] This procedure can occur at the same time as the physical examination under "Safety Assessment" so long as the visit date adheres to the required visit window for response evaluation.

[10] All patients must have baseline imaging within 35 days before randomization. A CT scan with intravenous contrast/MRI of the neck, chest, abdomen, pelvis, and any other disease sites should be performed. Imaging of the neck, chest, abdomen, and pelvis is required for all patients at Screening, C4D1, C10D1, and at the PTFU 1 Visit. Imaging studies will also be performed for confirmation of suspected occurrence of CR or CRi and also promptly at the time of suspected progressive disease based on physical examination findings. Imaging performed outside of the optimal procedure window specified in the Schedule of Activities will not necessarily be assessed as a protocol deviation; the sponsor will make the final determination. Copies of all scans will be sent to the Independent Review Committee for potential response assessment up to the PTFU 1 visit.

[11] A bone marrow biopsy is required at the PTFU 1 Visit for patients who have achieved ALC < 4.0 x 10^9/L unless the patient already has obtained bone marrow biopsy that confirmed CR/CRi. After the PTFU 1 Visit, a bone marrow core biopsy is required under the following conditions during the PTFU period starting after PTFU 1 Visit: if clinical and laboratory results demonstrate a potential CR or CRi, perform a bone marrow biopsy to confirm a CR or CRi if not already confirmed; in cases of progression of cytopenia unrelated to autoimmune cytopenia or study treatment, perform a bone marrow biopsy to confirm progressive disease. The bone marrow biopsy will be centrally tested and reviewed.

Note: Cytopenic disease progression is documented by a decrease of Hb levels > 2 g/dL or to < 10 g/dL or by a decrease of platelet counts ≥ 50% or to < 100,000/µL which occurs ≥ 3 months after the treatment, if the marrow biopsy demonstrates an infiltrate of clonal CLL cells. During therapy, cytopenia cannot be used to define disease progression.

[12] For a timepoint without imaging provide a clinical response evaluation and for a timepoint including imaging provide an overall response evaluation.

[13] Patients in survival follow-up will be followed for survival. Contact may be a visit in person or via phone (with the patient's guardian, if applicable). Additionally, during the survival follow-up period, subsequent new anticancer treatment information, including the reason for starting CLL anticancer treatment, name of each medication along with start date and end date, and date of progression after receiving the subsequent new anticancer treatment will be collected, as well as data regarding second primary malignancy and death.

[14]. PROs (EORTC QLQ-C30 and QLQ-CLL17) will be collected at safety follow up, PTFU 1and every 16 weeks thereafter. Patients must complete PROs prior to any health-related procedures or discussion.

[15] Complete blood count and differential will be evaluated by a local laboratory. Complete blood count includes hemoglobin, hematocrit, platelet count, red blood cell count, and white blood cell count with differential, which includes neutrophils, lymphocytes, monocytes, eosinophils, and basophils. It is not necessary to duplicate the hematology laboratory tests under "Safety Assessment" and "Efficacy Assessment" so long as the date of laboratory test results adheres to the required visit windows for both safety and efficacy assessments.

[16] Serum chemistry will be evaluated by a local laboratory and includes sodium, potassium, chloride, bicarbonate or CO_2 (or if neither are available, CO_2 combining power), uric acid, etc. Note: Serum chemistry testing is not required in the PTFU period except at the Safety Follow-up Visit.

[17] Serum immunoglobulin evaluations include IgG, IgM, and IgA.

[18] All patients will have PB samples collected at screening, C4D1, C7D1, C10D1, C13D1, PTFU1 visit and every 16 weeks from PTFU 1 Visit until PD and BMA samples collected at CR/CRi for Flow Cytometry MRD assessment. All patients will have PB collected at screening, C7D1, PTFU 1 Visit and BMA samples collected at PTFU 1 Visit for NGS MRD assessment (Table 23).

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[19] To study the resistance mechanisms of SZ, all patients in treatment arm will be asked to provide a PB sample at C1D1 predose, PTFU 1 Visit, and at the time when unequivocal progressive disease occurs. These peripheral blood samples will be used for exploratory biomarker assessment, such as mutation detection and/or molecular profiling, to understand potential resistance mechanisms in patients receiving SZ.

[20] Patients who are HBsAg negative, HBcAb positive, and HBV DNA negative and not on prophylactic antiviral treatment must undergo HBV DNA monitoring by PCR at least once on every cycle. These patients should be considered for prophylactic antiviral treatment in consultation with a local HBV expert. If a patient is being treated prophylactically with antivirals, HBV DNA monitoring by PCR must be done at least every third cycle. Patients positive for HCV antibody but negative for HCV RNA must have HCV RNA testing every cycle during the study treatment period. After the end of treatment, monitoring of HBV DNA/HCV RNA PCR should continue every 12 weeks for 12 months starting from the last dose of study treatment.

[21] Either serum or urine pregnancy tests will be performed every 4 weeks during the treatment period at specified visits. Patients are allowed to take the pregnancy tests locally in the cycles with no scheduled visits, but the investigators should be informed of the test results and should record them in the eCRF timely. Either serum or urine pregnancy tests will continue to be performed every 4 weeks until ≥ 90 days after the last dose of study drug. This will require additional visits after the Safety Follow-up Visit. If a urine pregnancy test is positive, it must be confirmed by a serum pregnancy test.

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Table 3: Twice a Week Ramp-up Schedule for Arm A - TLS Monitoring Schedules

Day	**All Ramp-up Visits and Cycle 5 Day 1 (see Table 4 for schedule)**		
Hours	**Predose**	**0**	**4-6hours postdose**
Window [1]	**Within 24 hours**	**N/A**	**± 1 hour**
Vital signs	X		
TLS laboratory assessments (send immediately) [2, 3]	X		X [3]
Hematology [4]	X		
Study drug dispensation/administration [5]		X	

1.3.1.3. Footnotes for Table 3: Twice a Week Ramp-up Schedule for Arm A- TLS Monitoring Schedules

Abbreviations: ECG, electrocardiogram; N/A, not applicable; PK, pharmacokinetic; TLS, tumor lysis syndrome.
Note: Actual drug dosing and PK sampling times must be documented by the sites and will be captured in the database. It is important that PK sampling occurs as close as possible to the scheduled time. To achieve this, some of the other assessments scheduled at the same time need to be initiated before or after the timepoint to allow for completion of these measurements in enough time for the PK sampling to be taken at the designated timepoint. When coinciding with blood draws and other procedures (eg, vital sign measurements) at the same timepoint, ECG assessment should be performed before these procedures.
[1] Sampling window: All efforts will be made to obtain the blood samples at the exact nominal time relative to dosing. Hour 0 indicates the time of sonrotoclax dosing.
[2] Mandatory TLS laboratory assessments are potassium, phosphorus, uric acid, calcium, and serum creatinine. These assessments must be run immediately at the timepoints indicated. More TLS laboratory assessments could be added on a scheduled visit day or on an unscheduled visit, per the investigator's decision if clinically indicated. Predose laboratory values are not required to be reviewed before the first dose (Ramp-up Week 1 Day 1) if laboratory values from the previous laboratory assessments were within normal limits.
[3] Patients should not leave clinic until 4 to 6 hours postdose TLS laboratory test results have been reviewed.
[4] Hematology assessments are as described in Table 24. Hematology samples will be used for TLS monitoring and pharmacodynamic analysis.
[5] Study drug should be dispensed to patients to ensure that enough drug is available to get to their next study visit. Enough medication should be dispensed to allow for visits to occur at the end of the visit window.

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Table 4: Twice a Week Ramp-up Schedule Visit Guidance for Arm A

Visits should occur as indicated in the Schedule of Activities for Arm A (Table 1).

Twice a Week Doubling: Dosing, and Visit Schedule for sonrotoclax

Week	**Cycle 4-RW1**							**Cycle 4-RW2**						
Days	**D1**	**D2**	**D3**	**D4**	**D5**	**D6**	**D7**	**D8**	**D9**	**D10**	**D11**	**D12**	**D13**	**D14**
Daily Dose (mg)	**1**	**1**	**1**	**2**	**2**	**2**	**2**	**5**	**5**	**5**	**10**	**10**	**10**	**10**
Study Visit	X	-	-	(X)	-	-	-	X	-	-	(X)	-	-	-
Week	**Cycle 4-RW3**							**Cycle 4-RW4**						
Days	**D15**	**D16**	**D17**	**D18**	**D19**	**D20**	**D21**	**D22**	**D23**	**D24**	**D25**	**D26**	**D27**	**D28**
Daily Dose (mg)	**20**	**20**	**20**	**40**	**40**	**40**	**40**	**80**	**80**	**80**	**160**	**160**	**160**	**160**
Study Visit	X	-	-	(X)	-	-	-	X	-	-	(X)	-	-	-

1.3.1.4. Footnotes for Table 4: Twice a Week Ramp-up Schedule Visit Guidance for Arm A

Abbreviations: D, Day; RW, ramp-up week; W, week; (X), optional visit.

Patients will reach the sonrotoclax target dose of 320 mg once per day on Cycle 5 Day 1 per scheduled dosing and will then continue receiving this dose.

Study Visit Schedule

During the ramp-up schedules, patients are not required to be hospitalized but may be hospitalized per investigator's discretion. Elective hospitalization is not considered to be a serious adverse event. If there is a limited reduction of circulating tumor cells or nodal disease burden at the lower dose levels, patients may be considered for hospitalization per the investigator's decision. Also additional tumor lysis syndrome laboratory assessments may be performed per the investigator's discretion.

Visits of Day 4 of each ramp-up week are not required but should be considered optional visits based on the investigator's discretion.

The ramp-up doses and visits may be adjusted per the available data in conjunction with Safety Monitoring Committee review and recommendation.

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Table 5: Schedule of Activities (Screening and Treatment Period) – Arm B

Study Procedures	Protocol Section	Screening	Enrollment / Randomization[1]	Treatment (1 cycle = 28 days)[2]					
Treatment Cycles/ Visit		–	–	**C1**	**C4**	**C7**	**C10**	**C13**	**C16 then every 4 cycles until disease progression**
Visit Days		**-35 to date of enrollment**	**-5 to 0 before C1D1**	**D1**	**D1**	**D1**	**D1**	**D1**	**D1**
Visit Window (days)		–	–		**± 7**	**± 7**	**± 7**	**± 7**	**±14 (+14 for C16D1)**
Administrative									
Informed consent and screen number [3]	Section 10.3	X							
Medical and cancer history and CIRS	Section 8.1.2	X							
Demographic data	Section 8.1.1	X							
IRT enrollment and randomization [4]	Section 6.3		X						
Administration of study drug(s)									
Zanubrutinib dispensing and accountability [5]	Section 6.2 Section 6.5			X	X	X	X	X	X
Safety Assessments									
Vital signs (temperature, BP, and heart rate/pulse rate)	Section 8.3.1	X		X	X	X	X	X	X
Physical examination [6]	Section 8.3.1	X		X	X	X	X	X	X
Cardiac function [7]	Section 8.3.2	X		X (as clinically indicated)					
ECOG performance status	Section 8.3.7.3	X		X	X	X	X	X	X
12-Lead ECG (local read) [8] /QTcF	Section 8.3.3	X		X	X	X	X	X	X (C16D1 only)
Concomitant medications review	Section 8.3.7.1	X		X	X	X	X	X	X
AE review [9]	Section 8.3.7.2	X	X	X	X	X	X	X	X
Efficacy Assessments									

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Study Procedures	**Protocol Section**	**Screening**	**Enrollment / Randomization[1]**	**Treatment (1 cycle = 28 days)[2]**					
Treatment Cycles/ Visit		**–**	**–**	**C1**	**C4**	**C7**	**C10**	**C13**	**C16 then every 4 cycles until disease progression**
Visit Days		**-35 to date of enrollment**	**-5 to 0 before C1D1**	**D1**	**D1**	**D1**	**D1**	**D1**	**D1**
Visit Window (days)		**–**	**–**		**± 7**	**± 7**	**± 7**	**± 7**	**±14 (+14 for C16D1)**
Physical examination of liver, spleen, and lymph nodes [10]	Section 8.2.2	X		X	X	X	X	X	X
Disease-related constitutional symptoms	Section 8.2.1	X		X	X	X	X	X	X
CT with intravenous contrast [11]/MRI	Section 8.2.3	X			X		X		X (C16D1 only)
Bone marrow biopsy [12]	Section 8.2.4			X					
Disease status/ response assessment [13]	Section 8.2	X		X	X	X	X	X	X
PRO questionnaires [14]	Section 8.5			X	X	X	X	X	X
Laboratory Assessments									
Hematology [15]	Section 8.3.4	X		X	X	X	X	X	X
Serum chemistry [16, 17]	Section 8.3.4	X		X	X	X	X	X	X
Serum immunoglobulins [18]	Section 8.3.4	X			X	X	X	X	X (C16D1 only)
Coagulation [19]	Section 8.1.7	X							
Coombs test	Section 8.1.8	X							
IGHV mutation analysis [20]	Section 8.7.1	X							
del17p and cytogenetics [21]	Section 8.7.1	X							
TP53 mutation [21]	Section 8.7.1	X							
Complex karyotype [21]	Section 8.7	X							
PB Sample collection for NGS MRD [22]	Section 8.2.6	X							X (C16D1 only)

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Study Procedures	Protocol Section	Screening	Enrollment / Randomization[1]	Treatment (1 cycle = 28 days)[2]					
Treatment Cycles/ Visit		–	–	**C1**	**C4**	**C7**	**C10**	**C13**	**C16 then every 4 cycles until disease progression**
Visit Days		**-35 to date of enrollment**	**-5 to 0 before C1D1**	**D1**	**D1**	**D1**	**D1**	**D1**	**D1**
Visit Window (days)		–	–		**± 7**	**± 7**	**± 7**	**± 7**	**±14 (+14 for C16D1)**
BMA Sample collection for NGS MRD [22]	Section 8.2.6								X (C16D1 only)
Diagnostic flow cytometry for CLL markers [23]	Section 8.1.5	X							
Resistance mechanism biomarker sample [24]	Section 4.2.5.4			At C1D1 predose and unequivocal disease progression					
Hepatitis B, C and HIV testing [25]	Section 8.1.3 Section 4.2.5.4	X							
HBV DNA and HCV RNA monitoring by PCR (if applicable) [26]	Section 8.1.3			Every cycle					
Urinalysis [27]	Section 8.1.10	X							
Pregnancy test (if applicable) [28]	Section 8.3.5	X		Every 4 weeks					

1.3.1.5. Footnotes for Table 5: Schedule of Activities (Screening and Treatment Period) – Arm B

Abbreviations: AE, adverse event; AESI, adverse event of special interest; BM, bone marrow; BMA, bone marrow aspirate; BP, blood pressure; CIRS, Cumulative Illness Rating Scale; CLL, chronic lymphocytic leukemia; CR, complete response; CRi, complete response with incomplete bone marrow recovery; CT, computed tomography; ECG, electrocardiogram; GHPS, gated heart pool scan; HBcAb, hepatitis B core antibody; HBV, hepatitis B virus; HBsAb, hepatitis B surface antibody; HBsAg, hepatitis B surface antigen; HCV, hepatitis C virus; HCVAb, hepatitis C antibody; Ig, immunoglobulin; IRT, Interactive Response Technology; IV, intravenous; LVEF, left ventricular ejection fraction; MRI, magnetic resonance imaging; MRD, minimal residual disease; MUGA, multigated acquisition scan; NGS, next-generation sequencing; PB, peripheral blood; PCR, polymerase chain reaction; PTFU, posttreatment follow-up period; SAE, serious adverse event; SZ, sonrotoclax + zanubrutinib; Z, zanubrutinib.

[1] The time to C1D1 should be within 5 calendar days after randomization/treatment assignment.

[2] All scheduled assessments should occur predose at Day 1 of each cycle unless specified.

[3] This must occur before any study-specific procedures and may be obtained before the 35-day screening window. Consent must be obtained on the current version of the form approved by the ethics committee.

[4] Patients will be randomized into 1 of 2 arms. Central randomization (2:1) will be used to assign patients to 1 of the following study drug treatments: Arm A (SZ) or Arm B (Z).

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[5] The zanubrutinib drug administration diary will be provided to each patient to record the study drug dose taken each day. Any missed doses with explanation should be recorded in the diary. The diary should be brought to the clinic for each visit and will be reviewed by the study coordinator on a regular basis. The complete final patient dosing diary will be provided to the study staff after the permanent discontinuaiton of treatment.

[6]. Assess systems per standard of care at the study site and as clinically indicated by symptoms. Includes weight (height at Screening only). The assessment of vital signs and a focused physical examination on Cycle 1 Day 1 may be skipped if performed within the last 7 days.

[7] An echocardiogram, MUGA, or GHPS will be performed within 90 days before randomization (ie, a standard of care procedure performed before consent for this study may be used if within window), and as medically necessary.

[8] A 12-lead ECG will be performed in triplicate (≥ 1 minute apart) at screening. The QTcF value may be calculated as the numerical average of up to 3 separate readings for eligibility. During Treatment period visits a single 12-lead ECG read will be performed every 12 weeks until C16D1.

[9] After informed consent has been signed but prior to the administration of the study drug or screen failure, only SAEs should be collected. After initiation of study drug, all AEs and SAEs, and AESI regardless of relationship to study drug, will be collected and reported until 30 days after the last dose of study drug, or before the first day of next line CLL treatment, whichever occurs earlier. After this period, the investigator should report any SAEs that are believed to be related to prior study drug treatment. For all patients, any new second primary malignancy, regardless of severity and relationship to study drug, should be recorded until the end of the study.

[10] This procedure can occur at the same time as the physical examination under “Safety Assessment” so long as the visit date adheres to the required visit window for response evaluation.

[11] All patients must have baseline imaging within 35 days before randomization. A CT scan with intravenous contrast/MRI of the neck, chest, abdomen, pelvis, and any other disease sites should be performed. Imaging of the neck, chest, abdomen, and pelvis is required for all patients at Screening, C4D1, C10D1, and C16D1. Imaging studies will also be performed for confirmation of suspected occurrence of CR or CRi and also promptly at the time of suspected progressive disease based on physical examination findings. Imaging performed outside of the optimal procedure window specified in the Schedule of Assessments will not necessarily be assessed as a protocol deviation; the sponsor will make the final determination. Copies of all scans will be sent to the independent review committee for potential response assessment up to C16D1 visit.

[12] During Treatment Period and Post treatment follow-up Period bone marrow biopsy and aspirate are required if clinical and laboratory results demonstrate a potential CR/CRi, regardless of minimal residual disease status. A bone marrow biopsy is required at C16D1 for patients who have achieved ALC < 4.0 x 10^9/L unless the patient already has obtained bone marrow biopsy that confirmed CR/CRi. The bone marrow biopsy will be centrally tested and reviewed.

[13] Disease status is the baseline assessment of CLL before the first dose of study drug. After the start of study drug: for a timepoint without imaging provide a clinical response evaluation and for a timepoint including imaging provide an overall response evaluation.

[14] PROs (EORTC QLQ-C30 and QLQ-CLL17) will be collected C1D1 and every 3 cycles up to C16D1, and every 16 weeks (4 cycles) thereafter Complete prior to any health-related procedures or discussion.

[15] Complete blood count will be done at local lab, including hemoglobin, hematocrit, platelet count, red blood cell count, and white blood cell count with differential, which includes neutrophils, lymphocytes, monocytes, eosinophils, and basophils. Screening blood tests performed within 72 hours of the first study drug administration do not need to be repeated in Cycle 1. It is not necessary to duplicate the hematology laboratory tests under “Safety Assessment” and “Efficacy Assessment” so long as the date of laboratory test results adheres to the required visit windows for both safety and efficacy assessments.

[16] Serum chemistry will be evaluated by local laboratory. Serum chemistries include sodium, potassium, chloride, bicarbonate or CO2 (or if neither are available, CO2 combining power), glucose, blood urea nitrogen (or urea), creatinine, calcium, phosphate/phosphorus, magnesium, total bilirubin, total protein, albumin, ALT, AST, lactate dehydrogenase, ALP, and uric acid.

[17] β-2 microglobulin and direct antiglobulin tests. will only be done at screening and will be performed locally.

[18] Serum immunoglobulin evaluations by local laboratory include IgG, IgM, and IgA.

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[19] The coagulation profile includes prothrombin time, which will also be reported as international normalized ratio, and activated partial thromboplastin time. The coagulation profile will be performed at screening only and thereafter as medically indicated by local labrotory.

[20] Prospective baseline biomarker test results for IGHV mutations from central laboratory are required for randomization. All patients will provide PB samples at the time of screening to assess the mutation status of IGHV by next-generation sequencing using a specialized central laboratory.

[21] A peripheral blood sample will be collected during screening to perform central laboratory assessment to study metaphase complex karyotype, CLL cytogenetics (eg, del17p, del11q, del13q and Trisomy 12) and TP53 mutation.

[22] Patients will have PB collected at screening and C16D1; BMA sample collected at C16D1 for NGS MRD assessment (Table 23).

[23] Peripheral blood samples will be collected during screening to perform central laboratory assessment to study CLL diagnostic by flow.

[24] To study the resistance mechanisms of Z, all patients will be asked to provide a peripheral blood sample at C1D1 predose, at the time when unequivocal progressive disease occurs. These peripheral blood samples will be used for exploratory biomarker assessment such as molecular profiling to understand potential resistance mechanisms in patients receiving Z.

[25] Hepatitis B and C serologies (includes HBsAg, HBsAb, HBcAb and HCVAb). An HIV serology (HIVAb) test will be conducted at screening unless previous HIV test results from ≤ 4 weeks before screening are available. These tests are performed by local laboratory.

[26] Patients who are HBsAg negative, HBcAb positive, and HBV DNA negative and not on prophylactic antiviral treatment must undergo HBV DNA monitoring by PCR at least once on every cycle during treatment period. These patients should be considered for prophylactic antiviral treatment in consultation with a local HBV expert. If a patient is being treated prophylactically with antivirals, HBV DNA monitoring by PCR must be done at least every third cycle. Patients positive for HCV antibody but negative for HCV RNA must have HCV RNA testing every cycle during the study treatment period. After the end of treatment, monitoring of HBV DNA/HCV RNA PCR should continue every 12 weeks for 12 months starting from the last dose of study treatment.

[27] Urinalysis will include pH, glucose, proteins, ketones, and specific gravity, and a 24-hour urine collection for creatinine clearance test may be performed per investigator's judgement to determine creatinine clearance at screening.

[28] All women of childbearing potential (including those who have had a tubal ligation) will have a serum pregnancy test ≤ 7 days before the first dose of study drug as part of the screening process. Laboratory-based, highly sensitive urine or serum pregnancy tests will also be performed with the results reviewed by the investigators on Cycle 1 Day 1 prior to dosing. Either serum or urine pregnancy tests will be performed every 4 weeks during the treatment period at specified visits. Patients are allowed to take the pregnancy tests locally in the cycles with no scheduled visits, but the investigators should be informed of the test results and should record them in the eCRF timely. Either serum or urine pregnancy tests will continue to be performed every 4 weeks until ≥ 90 days after the last dose of study drug. If a urine pregnancy test is positive, it must be confirmed by a serum pregnancy test.

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Table 6: Schedule of Activities (Post-treatment and Survival Follow-up) – Arm B

Study Procedures	Protocol Section	Safety Follow-up Visit [1]	Post-treatment Follow-up Period [2]	Survival Follow up Period [3]
Treatment Cycles/ Visit		**30 days from the last dose of study drug**	**Every 16 weeks**	**Every 16 weeks**
Visit Days		**D1**	**D1**	**D1**
Visit Window (days)		**± 7**	**± 14**	**± 14**
Safety Assessments				
Vital signs (temperature, BP, and heart rate/pulse rate)	Section 8.3.1	X		
Physical examination [4]	Section 8.3.1	X		
ECOG performance status	Section 8.3.7.3	X		
12-Lead ECG (local read) [5] /QTcF	Section 8.3.3	X		
Concomitant medications review	Section 8.3.7.1	X		
AE review [6]	Section 8.3.7.2	X	X	X
Efficacy Assessments				
Physical examination of liver, spleen, and lymph nodes [7]	Section 8.2.2		X	
Disease-related constitutional symptoms	Section 8.2.1		X	
CT with intravenous contrast [8]/MRI	Section 8.2.3	X (at the time of suspected CR/CRi or PD based on physical examination findings)		
Bone marrow biopsy [9]	Section 8.2.4	X (at the time of suspected CR/CRi if not confirmed prior, or suspected cytopenic disease progression)		
Disease status/ response assessment [10]	Section 8.2		X	
Survival status data collection [11]	Section 8.2.8			X
PRO questionnaires [12]	Section 8.5	X	X	
Laboratory Assessments				

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Study Procedures	Protocol Section	Safety Follow-up Visit [1]	Post-treatment Follow-up Period [2]	Survival Follow up Period [3]
Treatment Cycles/ Visit		**30 days from the last dose of study drug**	**Every 16 weeks**	**Every 16 weeks**
Visit Days		**D1**	**D1**	**D1**
Visit Window (days)		**± 7**	**± 14**	**± 14**
Hematology [13]	Section 8.3.4	X	X	
Serum chemistry [14]	Section 8.3.4	X		
Serum immunoglobulins [15]	Section 8.3.4	X		
HBV DNA and HCV RNA monitoring by PCR (if applicable) [16]	Section 8.1.3	Every 12 weeks for 12 months starting from the last dose of study drug		
Pregnancy test (if applicable) [17]	Section 8.3.5			
Resistance mechanism biomarker sample [18]	Section 4.2.5.4	At unequivocal disease progression		

1.3.1.6. Footnotes for Table 6: Schedule of Activities (Post-treatment and Survival Follow-up) – Arm B

Abbreviations: AE, adverse event; AESI, adverse event of special interest; BM, bone marrow; BMA, bone marrow aspirate; BP, blood pressure; CIRS, Cumulative Illness Rating Scale; CLL, chronic lymphocytic leukemia; CR, complete response; CRi, complete response with incomplete bone marrow recovery; CT, computed tomography; ECG, electrocardiogram; FISH, fluorescence in situ hybridization; GHPS, gated heart pool scan; HBcAb, hepatitis B core antibody; HBV, hepatitis B virus; HBsAb, hepatitis B surface antibody; HBsAg, hepatitis B surface antigen; HCV, hepatitis C virus; HCVAb, hepatitis C antibody; Ig, immunoglobulin; IRT, Interactive Response Technology; IV, intravenous; LVEF, left ventricular ejection fraction; MRI, magnetic resonance imaging; MRD, minimal residual disease; MUGA, multigated acquisition scan; NGS, next-generation sequencing; PB, peripheral blood; PCR, polymerase chain reaction; PK, pharmacokinetic; SAE, serious adverse event; SZ, sonrotoclax + zanubrutinib; Z, zanubrutinib.

[1] Approximately 30 days after permanent treatment discontinuation or before initiation of a new anticancer therapy, whichever comes first.

[2] Patients with no assessed unequivocal disease progression at permanent treatment discontinuation will enter Post-Treatment Follow-up period the day after the last dose of study drug. PTFU visits will be performed every 16 weeks. Patients remain on PTFU period until unequivocal disease progression has been assessed.

[3] Patients with assessed unequivocal disease progression will enter the Survival Follow-up period until death, withdrawal of patient consent, loss of follow-up, or study closure whichever comes first. Survival Follow-up visits will be performed every 16 weeks.

[4] Assess systems per standard of care at the study site and as clinically indicated by symptoms. Includes weight (height at Screening only).

[5] After C16D1, 12-lead ECG is only required at safety follow up visit, or as clinically indicated.

[6] After initiation of study drug, all AEs and SAEs, and AESI regardless of relationship to study drug, will be collected and reported until 30 days after the last dose of study drug, or before the first day of next line CLL treatment, whichever occurs earlier. After this period, the investigator should report any SAEs that

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are believed to be related to prior study drug treatment. For all patients, any new second primary malignancy, regardless of severity and relationship to study drug, should be recorded until the end of the study.

[7] This procedure can occur at the same time as the physical examination under "Safety Assessment" so long as the visit date adheres to the required visit window for response evaluation.

[8] Imaging studies will be performed for confirmation of suspected occurrence of CR or CRi and also promptly at the time of suspected progressive disease based on physical examination findings. Imaging performed outside of the optimal procedure window specified in the Schedule of Assessments will not necessarily be assessed as a protocol deviation; the sponsor will make the final determination.

[9] A bone marrow biopsy is required under the following conditions during the Post-treatment Follow-up period: if clinical and laboratory results demonstrate a potential CR or CRi, perform a bone marrow biopsy to confirm a CR or CRi if not already confirmed; in cases of progression of cytopenia unrelated to autoimmune cytopenia or study treatment, perform a bone marrow biopsy to confirm progressive disease. Note: Cytopenic disease progression is documented by a decrease of Hb levels > 2 g/dL or to < 10 g/dL or by a decrease of platelet counts ≥ 50% or to < 100,000/µL, which occurs ≥ 3 months after treatment, if the marrow biopsy demonstrates an infiltrate of clonal CLL cells. During therapy, cytopenia cannot be used to define disease progression. The bone marrow biopsy will be centrally tested and reviewed.

[10] Disease status is the baseline assessment of CLL before the first dose of study drug. After the start of study drug: for a timepoint without imaging provide a clinical response evaluation and for a timepoint including imaging provide an overall response evaluation.

[11] Patients in survival follow-up will be followed for survival. Contact may be a visit in person or via phone (with the patient's guardian, if applicable). Additionally, during the survival follow-up period, subsequent new anticancer treatment information, including the reason for starting CLL anticancer treatment, name of each medication along with start date and end date, and date of progression after receiving the subsequent new anticancer treatment will be collected, as well as data regarding second primary malignancy and death.

[12] PROs (EORTC QLQ-C30 and QLQ-CLL17) will be collected at the safety follow-up visit and PTFU period. Complete prior to any health-related procedures or discussion.

[13] Complete blood count will be done at local lab, including hemoglobin, hematocrit, platelet count, red blood cell count, and white blood cell count with differential, which includes neutrophils, lymphocytes, monocytes, eosinophils, and basophils. It is not necessary to duplicate the hematology laboratory tests under "Safety Assessment" and "Efficacy Assessment" so long as the date of laboratory test results adheres to the required visit windows for both safety and efficacy assessments.

[14] Serum chemistry will be evaluated by local laboratory. Serum chemistries include sodium, potassium, chloride, bicarbonate or CO2 (or if neither are available, CO2 combining power), glucose, blood urea nitrogen (or urea), creatinine, calcium, phosphate/phosphorus, magnesium, total bilirubin, total protein, albumin, ALT, AST, lactate dehydrogenase, ALP, and uric acid.

[15] Serum immunoglobulin evaluations by local laboratory include IgG, IgM, and IgA.

[16] After the end of treatment, for patients who are HBsAb negative, HBcAb positive and HBV DNA negative, or patients who are HCV antibody positive and HCV RNA negative, monitoring of HBV DNA/HCV RNA PCR should continue every 12 weeks for 12 months starting from the last dose of study treatment.

[17] For women of childbearing potential (including those who had a tubal ligation), either serum or urine pregnancy tests will continue to be performed every 4 weeks until ≥ 90 days after the last dose of study drug. If a urine pregnancy test is positive, it must be confirmed by a serum pregnancy test. Patients are allowed to take the pregnancy tests locally in the cycles with no scheduled visits, but the investigators should be informed of the test results and should record them in the eCRF timely.

[18] To study the resistance mechanisms of Z, all patients will be asked to provide a peripheral blood sample at C1D1 predose, at the time when unequivocal progressive disease occurs. These peripheral blood samples will be used for exploratory biomarker assessment such as molecular profiling to understand potential resistance mechanisms in patients receiving Z.

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1.3.2. Pharmacokinetic Sampling Schedule

Table 7: Pharmacokinetic Schedules for Arm A (Sonrotoclax + Zanubrutinib)

Day	**Day 1 of C5, C6, and C10**	
Hours	**Predose**	**Postdose**
Visit window	**≤ 30 minutes**	**4-6 hours**
PK blood sampling for sonrotoclax	X	X

1.3.2.1. Footnotes for Table 7: Pharmacokinetic Sampling Schedule

Abbreviations: eCRF, electronic case report form; PK, pharmacokinetic(s); PRO, patient-reported outcome.
Note: Actual drug dosing and PK sampling times must be documented by the sites and will be captured in the database.
General note: It is important that PK sampling occurs as close as possible to the scheduled time. To achieve this, some of the other assessments scheduled at the same timepoint may need to be initiated before or after the scheduling time to allow for completion of these measurements in enough time for the PK sampling to be taken at the designated timepoint. Thus, the sequence at a particular timepoint is: 1) PRO questionnaires;2) vital sign measurements; 3) PK blood sample collection; and 4) any other scheduled or unscheduled measurements occurring at that timepoint.
Sampling window: All efforts will be made to collect the PK samples at the nominal time window relative to dosing. However, samples obtained outside of the sampling window will not be captured as a protocol deviation if the exact time of the sample collection is noted on the source document and eCRF.

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2. INTRODUCTION

2.1. Study Rationale

CLL is characterized by progressive accumulation of leukemic cells in the peripheral blood, bone marrow, and lymphoid tissues with immunophenotypes of CD19+, CD5+ and clonal expression of immunoglobulin light chain (κ or λ), and commonly CD23+. CLL is the most common leukemia in the Western world with an incidence of 4.2 in every 100,000 persons per year (Tsimberidou et al 2007). The incidence increases to > 30 in 100,000 per year in people ≥ 80 years of age. The disease has a median age at diagnosis of 72 years (Eichhorst et al 2015). The incidence of CLL is lower in Asia than in the West, with an age-adjusted incidence rate of 0.1 to 0.2 in every 100,000 persons in East Asia (Yang et al 2021). While no population-based studies of CLL in the mainland of China have been published (Gale 2022), a study in Taiwan found the age-standardized incidence rate of CLL (including SLL) to be 0.54 per 100,000 persons in 2015 with a noted increase from 0.36 per 100,000 persons in 2006 (Ko et al 2021).

CLL is considered a treatable but essentially incurable disease. Historically, treatment options for patients with CLL have included chemotherapy (such as chlorambucil, fludarabine, cyclophosphamide, and bendamustine), and anti-CD20 monoclonal antibodies (such as rituximab and obinutuzumab) (Shadman et al 2023). Currently, first-line treatment for CLL often consists of a regimen including either a covalent BTK inhibitor (eg, zanubrutinib, acalabrutinib, or ibrutinib) or a Bcl-2 inhibitor (eg, venetoclax). Although BTK inhibitor monotherapy is effective at disease control, it does not induce deep remissions in most patients, thus committing patients to continuous use, complicated by toxic effects and the emergence of resistance (Burger et al 2015). In contrast with indefinite therapy until progression or intolerance with the BTK inhibitor, treatment with venetoclax-obinutuzumab provides the opportunity for a fixed duration of treatment (for 1 year) and induces deep remission (CR rate 49.5%). However, resistance and clinical relapse can still occur, especially in high-risk patients (Kristen et al 2019).

Preclinical models have indicated that the combination of a BTK inhibitor and a Bcl-2 inhibitor is synergistic. From a mechanism of action perspective, the BTK inhibitors mobilize CLL cells from protective microenvironmental niches in lymphoid organs, which confer survival advantage to CLL cells through cross talk with accessory cells (Herman et al 2011; Lu et al 2021), to the PB, resulting in an increased sensitivity of CLL cells to the pro-apoptotic activities of Bcl-2 inhibitors (Cervantes-Gomez et al 2015).

The combination of BTK and Bcl-2 inhibition has been explored as time-limited treatment and found to be highly effective, with deep remissions achieved in clinical trials of patients with treatment naïve (TN) CLL (Barr et al 2023; Niemann et al 2023) or Relaxed/Refractory (R/R)CLL (Hillmen et al 2019; Niemann et al 2019; Siddiqi et al 2021).

In the CAPTIVATE study, fit patients ≤ 70 years of age with TN CLL were treated with ibrutinib + venetoclax. After the 12 cycles of I+V, best uMRD4 response rates were 75% (peripheral blood) and 68% (bone marrow). (Wierda et al 2021). At 4 years of follow-up, the best CR rate was 58% and ORR was at 96%. At year 4, the PFS rate was 79% (95% CI: 71% to 84%) and the OS rate was 98% (95% CI: 94% to 99%). Four-year PFS rates by MRD status 3

months after stopping treatment (EOT+3) were significantly higher overall in patients with uMRD versus those with detectable MRD in PB (90% versus 66%）(Ghia et al 2023).

In the Glow study, older (≥ 65 years of age) or unfit younger (18 to 64 years of age) TN patients were randomized to receive ibrutinib + venetoclax (I+V) or chlorambucil + obinutuzumab (C+O). At 3 months after I+V treatment, 54.7% of patients had achieved PB uMRD4 and 49.1% of patients maintained uMRD4 from 3 months post treatment to 12 months posttreatment (Munir et al 2023). With a median follow-up of 46 months, IRC-assessed PFS rates remained superior for the I+V group compared with the C+O group (HR = 0.214 [95% CI: 0.138 to 0.334]; $p < 0.0001$). The estimated 42-month PFS rate was 74.6% (95% CI: 65.0% to 82.0%) for I+V group and 24.8% (95% CI: 16.5% to 34.1%) for the C+O group. In the I+V treatment arm, 2-year PFS rates were 93.0% (95% CI: 82.4% to 97.3%) in patients with uMRD and 79.6% (95% CI: 60.1% to 90.3%) in patients with detectable MRD 3 months after the end of treatment (Niemann et al 2023).

The most important potential risk related to Bcl-2 inhibition is TLS, especially in CLL patients with high tumor burden. A 2- to 3-cycle “lead-in” period with BTK inhibitor monotherapy (ie, before the addition of Bcl-2 inhibitor) can reduce tumor burden and thereby abrogate the risk of TLS (Kater et al 2022; Wierda et al 2021). In the CAPTIVATE study, the proportion of patients at high risk for TLS was reduced from 24% at initiation of the ibrutinib lead-in period to just 2% following its completion and before the initiation of venetoclax (Tam et al 2019). This approach can therefore significantly decrease the risk of TLS and in particular, clinical TLS with attendant end-organ damage.

Although the combination deepened the response and decreased the TLS risk in clinical studies, ibrutinib and venetoclax have demonstrated overlapping toxicities (eg, myelosuppression and gastrointestinal [GI] effects) that need to be managed with supportive care. Frequently reported AEs include arthralgia, diarrhea, nausea, and neutropenia, with diarrhea and neutropenia leading to dose modifications or treatment discontinuation (Kater et al 2022; Hillmen et al 2019). The toxicity profile for I+V suggests the need for a more tolerable BTK/Bcl-2 inhibitor combination.

Zanubrutinib is a more selective covalent BTK inhibitor, having demonstrated a more tolerable AE profile, and better efficacy compared to ibrutinib in CLL. BGB-11417 (sonrotoclax) is a highly potent and selective Bcl-2 protein inhibitor. In biochemical studies, BGB-11417 demonstrated strong inhibition of both Bcl-2 and a putative venetoclax-resistant Bcl-2 mutant BCL2-G101V. BGB-11417 demonstrated better antitumor activity than venetoclax in human acute lymphoblastic leukemia, mantle cell lymphoma, and diffuse large B-cell lymphoma xenograft models.

BGB-11417-101 is the first-in-human clinical study that evaluates the safety, PK, and efficacy of sonrotoclax as monotherapy and in combination with zanubrutinib. In the study, the combination of sonrotoclax + zanubrutinib revealed robust efficacy and safety. The observed uMRD4 rate by Week 48 on the sonrotoclax target dose of 320 mg was 90.9% in TN CLL, where uMRD rate was reported as 52% in patients with TN CLL, after 6 cycles of combination treatment. This data led to the development of a Phase 3 study in patients with TN CLL for registrational purposes (BGB-11417-301).

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To support the registration plan of sonrotoclax + zanubrutinib treatment in patients with previously untreated CLL this Phase 2 study is designed to evaluate the contribution of sonrotoclax to the efficacy outcome of the combination of zanubrutinib and sonrotoclax. In addition, safety information will be collected to confirm the safety of zanubrutinib and sonrotoclax combination.

Zanubrutinib is approved for the treatment of CLL/SLL by the FDA, and CLL by the EMA and the CDE. In the Phase 3 SEQUOIA study, zanubrutinib treatment resulted in a higher ORR and a statistically significant improvement in PFS compared to bendamustine + rituximab treatment in patients with untreated CLL without del(17p)/TP53 mutation (HR = 0.42; $p < 0.0001$) (Tam et al 2022b). The biomarker subgroup analysis from the SEQUOIA study confirmed that a PFS benefit with zanubrutinib treatment was observed in all subgroups, including patients with del (11q) ($p < 0.001$), those with unmutated *IGHV* ($p < 0.0001$), and those with mutated *IGHV* ($p < 0.01$) (Ramakrishnan et al 2023). Zanubrutinib monotherapy is one of the preferred treatment options for frontline treatment of CLL across all clinical guidelines, indicating its suitability as the control arm treatment.

2.2. Background

2.2.1. Chronic Lymphocytic Leukemia

CLL is characterized by the proliferation and accumulation of lymphocytes that appear mature but are immunologically incompetent and malignant (Dameshek 1967). The malignant lymphocytes in CLL are clonal and characterized by coexpression of the B-cell surface antigens CD23 and CD19 with the aberrantly expressed T-cell surface antigen, CD5 (Chiorazzi et al 2005).

CLL is considered a treatable but essentially incurable disease and has a heterogeneous course. Therapy is required at the onset of symptomatic disease (either constitutional symptoms or symptoms of worsening lymphadenopathy or organomegaly), clinically significant cytopenias, or rapidly increasing circulating lymphocyte count; these criteria are formalized in the consensus guidance for CLL management (Hallek et al 2018). Infection is a common complication of the disease, and patients suffering from recurrent infection should receive prophylactic antibiotics and antivirals (Eichhorst et al 2015). Ultimately, patients who succumb to CLL typically die from infectious complications (due to profound suppression of both cellular and humoral immunity) or, less commonly, from refractory cytopenias (Hallek et al 2018).

2.2.2. Current Treatment of Treatment-naïve Chronic Lymphocytic Leukemia

When selecting the initial therapy for CLL, multiple patient factors should be considered, such as prognostic risk, comorbidities, concurrent medications, cytogenetics/molecular features, and disease presentation (eg, bulky disease or cytopenias). Before 2013, front-line treatments for patients with TN CLL relied on chemo-immunotherapy, such as fludarabine/cyclophosphamide/rituximab for fit patients aged ≤ 65 years, bendamustine/rituximab for fit patients aged > 65 years, or chlorambucil/obinutuzumab for unfit patients. Over the last decade, the development and approval of novel agents for treatment of CLL has led to a paradigm shift from chemo-immunotherapies to targeted treatments enabling significant improvements in outcomes and tolerability.

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The BCR signaling pathway is involved in the pathogenesis of CLL, and drugs that inhibit the enzymes involved in the BCR pathway, specifically BTK, is the standard of care to treat CLL. BTK inhibitors, such as ibrutinib, acalabrutinib, and zanubrutinib, are used indefinitely as monotherapy because continuous inhibition of the target enzyme is essential for their antiproliferative effect. More than 90% of patients respond to these treatments. Survival rates are approximately 88% at 4 years for acalabrutinib (Sharman et al 2022), 94% at 2 years for zanubrutinib (Constantine et al 2022), and 78% at 7 years for ibrutinib (Barr et al 2022). BTK inhibitor monotherapy can achieve a high ORR and prolonged duration of response when used continuously; however, the potential resistance associated with continuous treatment, as well as the burden of treatment, have led to the development of therapeutic schemas to further optimize the BTK targeting strategies, with the aim of time-limited therapy and deep response.

Bcl-2 proteins inhibit the mitochondrial pathway of apoptosis. Overexpression of the Bcl-2 family proteins is an essential component of the pathophysiology of CLL (Chong and Davids 2020). Limited duration treatment with venetoclax, the only approved Bcl-2 inhibitor, induces high rate of CRs without detectable measurable disease in the blood or marrow. The CLL13 study showed safety and efficacy benefits of venetoclax plus obinutuzumab (VO) as treatment in younger, fit patients with untreated CLL (Eichhorst et al 2023). At Month 15, the percentage of patients with undetectable minimal residual disease was higher in the VO group (86.5%) and the venetoclax–obinutuzumab–ibrutinib group (92.2%) than in the chemoimmunotherapy group (52.0%). 3-years PFS rates were 87.7%, 90.5%, and 75.5%, respectively. In the CLL14 study of VO compared to obinutuzumab plus chlorambucil (GClb) in patients with untreated CLL with coexisting medical conditions, VO showed a superior PFS (HR = 0.31; 95% CI: 0.22 to 0.44) (Al-Sawaf et al 2020). In particular, fixed-duration combination treatments of VO may confer a longer treatment-free interval without a detrimental effect on subsequent treatment upon progression, offer improved safety, and retain the opportunity to re-treat after disease relapse (Guru Murthy and Atallah 2023).

Currently, the National Comprehensive Cancer Network Guidelines (NCCN 2024) list of preferred front-line treatments for CLL includes zanubrutinib, venetoclax + obinutuzumab, and acalabrutinib ± obinutuzumab. These regimens apply regardless of age, fitness status, or *IGHV* mutation status. Based on the available evidence and in the absence of head-to-head clinical trials comparing BTK inhibitors with venetoclax, continuous treatment with a BTK inhibitor is preferred for patients with del17p/TP53 by most hematologists, but VO remains a reasonable option.

The combination of BTK and Bcl-2 inhibition has been explored as time-limited treatment and found to be highly effective with deep remissions achieved in clinical trials (Barr et al 2023; Niemann et al 2023; Hillmen et al 2019; Niemann et al 2019; Siddiqi et al 2021). With respect to combination therapy involving a 3-month ibrutinib lead-in followed by venetoclax for 12 months, the lead-in eliminates the risk of high TLS, thereby avoiding hospitalization for the venetoclax ramp-up period. The depth of remissions enables fixed duration treatment, which can improve safety and tolerability as well as reduce the development of acquired resistance that may in turn, enable retreatment with the same regimen after disease relapse.

A list of all investigational medicinal products that will be used in this study and their global regulatory approval statuses are provided in Table 8. Further details are included in Section 6.

No auxiliary medicinal products, as defined by Article 2(8) of Regulation (EU) 536/2014, will be used in this study.

Table 8: Investigational Medicinal Products Used in Study BGB-11417-204 and Their Regulatory Approval Status

Country/Region	Approved for Treatment of TN CLL (Yes/No)		
	Sonrotoclax[a]	**Zanubrutinib**	
US	No	Yes[b]	
EU	No	Yes[c]	
China	No	Yes[d]	

Abbreviations: CLL, chronic lymphocytic leukemia; EMA, European Medicines Agency; EU, European union; FDA, Food and Drug Administration; FL, follicular lymphoma; MCL, mantle cell lymphoma; MZL, marginal zone lymphoma; NMPA, National Medical Products Administration; R/R, relapsed or refractory; SLL, small lymphocytic lymphoma; TN, treatment-naïve; US, united states of the America; WM, Waldenstrom macroglobulinemia.
[a] Sonrotoclax, both alone and in combination with other active agents, is currently in clinical development globally and is not approved in any country worldwide.
[b] Zanubrutinib was approved by the FDA for the treatment of MCL, R/R MZL, WM, CLL/SLL and R/R FL.
[c] Zanubrutinib was approved by the EMA for the treatment of CLL, R/R MZL, WM and R/R FL.
[d] As of April 2024, zanubrutinib has been approved by the NMPA for the treatment of R/R MCL, CLL/SLL and WM.

Detailed descriptions of the chemistry, pharmacology, efficacy, and safety of sonrotoclax and zanubrutinib are provided in the respective investigator's brochures.

2.3. Benefit/Risk Assessment

More detailed information about the known and expected benefits and risks of sonrotoclax study drug/treatment is provided in the IB Edition 6.0.

More detailed information about the known and expected benefits and risks of zanubrutinib is provided in IB Edition 12.0 and the package insert.

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2.3.1. Risk Assessment

Table 9: Risk Assessment

Study Drug/Treatment(s): sonrotoclax -	
Important Identified and Potential Risks	No important identified risks associated with sonrotoclax have been identified until now, and 2 important potential risks were identified: 1) tumor lysis syndrome (TLS), and 2) neutropenia and febrile neutropenia
Summary of Data/Rationale for Risk	TLS: TLS occurs when tumor cells release their contents into the bloodstream, leading to electrolyte and metabolic disturbances, which in turn can progress to clinical toxic effects including renal insufficiency, seizures, or death (Howard et al 2018). Based on a thorough clinical and safety review and evaluation of all cases, there appears to be no direct correlation with specific clinical findings or baseline characteristics, and no identifiable pattern or discernible causes was identified for these TLS cases. Neutropenia and febrile neutropenia: Neutropenia is a common toxicity seen in patients with lymphoid and myeloid malignancies who have received multiple prior therapies and can result from bone marrow infiltration of their underlying disease or reduced bone marrow reserves after therapy (Lustberg 2012). Neutropenia may also be induced or exacerbated in these patients due to an on-target pharmacological effect of sonrotoclax. Patients with clinically significant neutropenia may be susceptible to serious infections, and corrective treatments may be required. Febrile neutropenia is a common complication of cancer therapy that can be life-threatening. Preclinical studies of sonrotoclax did not show an increased risk of neutropenia. As of 04 February 2024, in patients with B-cell malignancies treatment with sonrotoclax monotherapy, neutropenia shift from baseline was reported in all indications without an observed dose-relation.
Mitigation Strategy	TLS: It is critically important that all sites strictly follow the safety guidance in the clinical study protocol (Section 6.10.3) and ensure all protocol requirements are fully implemented. Investigators should assess risk factors for TLS and provide prophylactic hydration and anti-hyperuricemic agents to patients as needed to mitigate the risk. TLS-related laboratory results should be monitored pre- and post-dose at all visits to detect the early occurrence of TLS and provide

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	timely intervention. Sonrotoclax should be interrupted for any patient who develops laboratory TLS, clinical TLS, or persistent TLS-related laboratory abnormalities until resolution. Neutropenia and febrile neutropenia: Changes in neutrophils should be closely monitored, and patients should be observed for signs of infections.
Study Drug/Treatment(s): zanubrutinib	
Important Identified and Potential Risks	Important identified risks: Hemorrhage Infections (Including Lower Respiratory Tract Infections and Hepatitis B Reactivation) Important potential risks: Second primary malignancies (other than non-melanoma skin cancer) and second primary non-melanoma skin cancer Cardiac arrhythmia, mainly presenting as atrial fibrillation and flutter
Summary of Data/Rationale for Risk	Hemorrhage: Serious and fatal hemorrhagic events have occurred in patients with hematological malignancies treated with zanubrutinib monotherapy. Grade 3 or higher bleeding events including intracranial and gastrointestinal haemorrhage and haematuria and hemothorax have been reported in patients. Bleeding events of any grade including purpura and petechiae occurred in patients with hematological malignancies. The mechanism for the bleeding events is not well understood. Infection: Fatal and nonfatal infections (including bacterial, viral, or fungal infections or sepsis) and opportunistic infections (eg, herpes viral, cryptococcal, aspergillus and pneumocystis jiroveci infections) have occurred in patients with hematological malignancies treated with zanubrutinib monotherapy. Grade 3 or higher infections occurred in these patients. The most common Grade 3 or higher infection was pneumonia. Infections due to hepatitis B virus (HBV) reactivation have occurred in patients treated with zanubrutinib. Second Primary Malignancies: Second primary malignancies, including non-skin carcinoma have occurred in patients with hematological malignancies treated with zanubrutinib monotherapy. The most frequent second primary malignancy was skin cancers (basal cell carcinoma and squamous cell carcinoma of skin). Cardiac Arrhythmia:

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	Atrial fibrillation and atrial flutter have occurred in patients with hematological malignancies treated with zanubrutinib monotherapy, particularly in patients with cardiac risk factors, hypertension, and acute infections.
Mitigation Strategy	Hemorrhage: Zanubrutinib may increase the risk of hemorrhage in patients receiving antiplatelet or anticoagulant therapies and patients should be monitored for signs of bleeding. Consider the benefit-risk of withholding zanubrutinib for 3-7 days pre- and post-surgery depending upon the type of surgery and the risk of bleeding. Infections: Consider prophylaxis according to the standard of care in patients who are at increased risk for infections. Monitor patients for signs and symptoms of infection and treat appropriately. Second Primary Malignancies: Advise patients to use sun protection. Cardiac Arrhythmia: Monitor signs and symptoms for atrial fibrillation and atrial flutter and manage as appropriate.

2.3.2. Benefit Assessment

The distinct and complementary actions of BTK and Bcl-2 inhibitors can have a powerful synergistic effect resulting in deeper anti-tumor responses. The dynamic between these two treatments involves BTK inhibitor therapy moving CLL cells from the lymph node, where they are well supported and proliferate, to the periphery where, without the supporting tumor microenvironment, the cells become non-proliferating/resting and die by neglect over a relatively long period of time. Once CLL cells are moved to the periphery, Bcl-2 inhibitor treatment can actively eradicate them, thus reducing tumor burden (Lu et al 2021). Thus, the two targeted agents offer complementary functions in the mitochondrial apoptotic pathway and act on distinct CLL subpopulations that have different proliferative capacities; BTK inhibitors preferentially kill the dividing cell subpopulation, and Bcl-2 inhibitors act on the resting cell subpopulation. By acting on both subpopulations of CLL cells, combination therapy has the potential to minimize or even eradicate MRD at all anatomic sites (Zhang et al 2022).

Studies have demonstrated that combination BTK and Bcl-2 inhibitors can be highly effective with deep remissions achieved in clinical studies. In the CAPTIVATE study, fit patients ≤ 70 years of age with TN CLL were treated with ibrutinib + venetoclax. After the 12 cycles of ibrutinib + venetoclax, best uMRD4 response rates were 75% (peripheral blood) and 68% (bone marrow). (Wierda et al 2021). At 4 years posttreatment, PFS rate was 79% (95% CI: 71% to 84%) and OS rate was 98% (95% CI: 94% to 99%) (Ghia et al 2023). In the Glow study, older (≥ 65) or unfit younger (18 to 64) TN patients were randomized to receive I+V or chlorambucil + obinutuzumab. At 3 months post I+V treatment, 54.7% patients achieved peripheral blood uMRD4 and 49.1% patients maintained uMRD4 from 3months post treatment

to 12 months post treatment (Munir et al 2023). With a median follow-up of 46 months, PFS remained superior for the ibrutinib–venetoclax group compared with the chlorambucil–obinutuzumab group (HR = 0.214; p < 0.0001) (Niemann et al 2023).

Zanubrutinib is a more selective covalent BTKi, demonstrated a more tolerable adverse event profile, and better efficacy compared to ibrutinib in CLL. BGB-11417 (sonrotoclax) is a highly potent and selective Bcl-2 protein inhibitor. As of 04 February 2024, 112 TN CLL patients were enrolled and 108 patients were exposed to sonrotoclax in combination with zanubrutinib in the Phase 1/1b Study BGB-11417-101. With a median follow-up of 15.11 months, the ORR rate was 100%, and the CR rate was 40.7%. The uMRD in peripheral blood was 76.8%.

Zanubrutinib is approved for the treatment of CLL/SLL by FDA, and CLL by EMA and CDE. In the phase III SEQUOIA study, zanubrutinib treatment resulted in higher ORR and statistically significant improvement in PFS compared to bendamustine + rituximab treatment in patients with untreated CLL without del(17p)/TP53 mutation (HR = 0.42; p < 0.0001) (Tam et al 2022b). Zanubrutinib monotherapy is one of the preferred treatment options for frontline CLL across all clinical guidelines, which is appropriate as the control arm therapy.

2.3.3. Overall Benefit/Risk Conclusion

Taking into account the measures taken to minimize risk to patients participating in this study, the potential risks identified in association with sonrotoclax and zanubrutinib (SZ) are justified by the anticipated benefits that may be afforded to patients with previously untreated CLL.

There is both a strong scientific rationale as well as significant clinical data to support the initiation of this Phase 2 study comparing the combination of SZ versus Zanubrutnib monotherapy in patients with previously untreated CLL.

A comprehensive plan has been developed to monitor patient safety in both the investigational and comparator treatment arms. The data available for the combination of SZ for the treatment of patients with previously untreated CLL shows this treatment is safe and tolerable and has compelling efficacy. This supports further investigation of this combination in this randomized study.

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3. OBJECTIVES, ENDPOINTS, AND ESTIMANDS

3.1. Objectives and Endpoints

Table 10: Objectives and Endpoints

Objectives	Endpoints
Primary	
To compare efficacy between Arm A (sonrotoclax plus zanubrutinib) versus Arm B (zanubrutinib) measured by the proportion of patients that achieve a complete response (CR) or complete response with incomplete bone marrow recovery (CRi)	Best CR/CRi rate up to the first Post-treatment Follow-up (PTFU 1) Visit for Arm A and up to C16D1 Visit for Arm B per Independent Review Committee (IRC) response assessment using the 2018 International Workshop on Chronic Lymphocytic Leukemia (iwCLL) (Hallek et al 2018) guidelines with modification for treatment-related lymphocytosis (Cheson et al 2012) for patients with chronic lymphocytic leukemia (CLL)
Key Secondary	
To compare efficacy between Arm A (sonrotoclax plus zanubrutinib) versus Arm B (zanubrutinib) by rate of undetectable minimal residual disease at < 10^{-4} sensitivity (uMRD4)	uMRD4 rate measured in both peripheral blood and bone marrow aspirate at PTFU 1 Visit for Arm A and at C16D1 for Arm B based on next-generation sequencing (NGS, clonoSEQ®)
Secondary	
To compare the efficacy between Arm A versus Arm B through CR/CRi per investigator response assessment; overall response rate (ORR), duration of response (DOR), and time to response (TTR) per IRC and investigator assessment; progression-free survival (PFS) per investigator assessment, and overall survival (OS).	• CR/CRi rate per investigator response assessment • ORR per IRC and investigator response assessment • Landmark DOR event-free rate at 12 months per IRC response assessment • DOR per investigator response assessmentTTR per IRC and investigator response assessment • Landmark PFS rate at 24 months and overall PFS per investigator response assessment • OS
To compare the safety and tolerability of Arm A (sonrotoclax plus zanubrutinib) versus Arm B (zanubrutinib)	Safety will be assessed by monitoring and recording of all treatment emergent adverse events graded by National Cancer

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Objectives	Endpoints
	Institute-Common Terminology Criteria for Adverse Events (NCI-CTCAE) v5.0 and the iwCLL Grading Scale for Hematologic Toxicities in CLL Studies as appropriate. Tumor lysis syndrome will be monitored per Howard criteria.
Exploratory	
To evaluate the prognostic biomarkers and biomarkers of relapse and correlation with clinical outcomes	To evaluate the correlation between clinical outcomes (CR rate, ORR, DOR, PFS, and OS) and baseline prognostic biomarkers (eg, del11q, del13q, trisomy 12, CLL-related prognosis mutations, immunoglobulin heavy-chain variable region gene [*IGHV*] mutations, complex karyotype), and identification of potential resistance biomarkers and mechanisms
To assess undetectable MRD (uMRD) rate at various timepoints by flow cytometry and delta MRD by ClonoSEQ® in Arm A	• uMRD rate at various timepoints and uMRD duration; time to uMRD assessed by flow cytometry • To evaluate the correlation between PB delta MRD at C7D1 and bone marrow uMRD4 at PTFU 1 Visit, and the correlation between delta MRD at C7D1 and clinical outcomes (CRR, ORR, DOR, PFS and OS)
To evaluate the pharmacokinetics of sonrotoclax when co-administered with zanubrutinib in Arm A	Plasma concertation data, including but not limited to C_{trough}
To evaluate the patient-reported outcomes (PROs)	• PROs will be assessed using European Organisation for Research and Treatment of Cancer (EORTC) QLQ-C30 and EORTC QLQ-CLL17 • The key PRO endpoints include the symptom burden and physical condition/fatigue measured by QLQ-CLL17, and global health status (GHS)/qualify of life (QoL) and physical functioning measured by EORTC QLQ-C30

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3.2. Estimands

3.2.1. Definition of Primary Estimand for Primary Analysis:

The primary clinical question of interest: "Will sonrotoclax + zanubrutinib increase the CR/CRi rate compared to zanubrutinib in previously untreated CLL patients?"

The primary estimand is described by the following attributes:

1. Treatments of interest:

   The experimental treatment is sonrotoclax + zanubrutinib fixed-duration therapy. The control treatment is zanubrutinib continuous therapy.

2. Population:

   Patients aged ≥ 18 years with previously untreated CLL and without del17p or known TP53 mutation.

3. Variable:

   The primary variable is a binary response variable defined as whether or not the patient achieved CR/CRi up to the PTFU 1 for Arm A and C16D1 for Arm B and prior to disease progression or death, as determined by IRC. Response assessment will be assessed using the 2018 International Workshop on Chronic Lymphocytic Leukemia (Hallek et al 2018) guidelines with modification for treatment-related lymphocytosis (Cheson et al 2012) for patients with CLL.

4. Handling of intercurrent events:

   - New anticancer therapy started before PTFU 1 for Arm A or before Day 1 of Cycle 16 for Arm B: Patients starting any new anticancer therapy without achieving a CR or CRi before completing Cycle 15 of treatment will be considered as non-responders (composite strategy).
   - Discontinuation of treatment before PTFU 1 for Arm A or before Day 1 of Cycle 16 for Arm B: Response assessment after discontinuation of treatment before completion of Cycle 15 will be counted and used for analysis (treatment policy strategy).

5. Population-level summary:

   CR/CRi rate difference.

3.2.2. Definition of Primary Estimand for Key Secondary Analysis:

The primary clinical question of interest: "Will sonrotoclax + zanubrutinib increase the uMRD4 rate compared to zanubrutinib in previously untreated CLL patients?"

The primary estimand is described by the following attributes:

1. Treatments of interest:

   The experimental treatment is sonrotoclax + zanubrutinib fixed duration therapy. The control treatment is zanubrutinib continuous therapy.

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2. Population:

   Patients aged ≥ 18 years with previously untreated CLL and without del17p or known TP53 mutation.

3. Variable:

   The primary variable is a binary response variable defined as whether or not the patient achieved uMRD4 measured both in PB and bone marrow aspirate at the first Post-Treatment Follow-up (PTFU 1) Visit for Arm A and at Day 1 of Cycle 16 for Arm B based on NGS (clonoSEQ®).

4. Handling of intercurrent events:

   - New anticancer therapy started before MRD assessment at PTFU 1 for Arm A or at Day 1 of Cycle 16 for Arm B: Patients starting any new anticancer therapy before the MRD assessment at PTFU 1 for Arm A or at Day 1 of Cycle 16 for Arm B will be considered as not achieving uMRD4 (composite strategy).
   - Discontinuation of treatment before MRD assessment at PTFU 1 for Arm A or at Day 1 of Cycle 16 for Arm B: MRD data collected after discontinuation of treatment will be counted and used for analysis (treatment policy strategy).
   - Disease progression or death before MRD assessment at PTFU 1 for Arm A or at Day 1 of Cycle 16 for Arm B: Patients with disease progression or who die before the MRD assessment will be considered as not achieving uMRD4 (composite strategy).

5. Population-level summary:

   uMRD4 rate difference.

For full details, refer to the SAP.

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4. STUDY DESIGN

4.1. Overall Design

This study is a multicenter, global Phase 2, open-label, randomized study to evaluate the efficacy and safety of a fixed duration of sonrotoclax in combination with zanubrutinib compared to continuous zanubrutinib monotherapy in patients with previously untreated CLL without del17p or known *TP53* mutation. Eligible patients will be randomized 2:1 to receive either 3 cycles of lead-in zanubrutinib monotherapy followed by sonrotoclax in combination with zanubrutinib for 12 cycles (Arm A), or to receive zanubrutinib monotherapy continuously until disease progression or intolerance (Arm B). Refer to the study schema (Section 1.2) for further details.

The sample size of the study will be approximately 87 patients enrolled at approximately 40 sites. Stratification factors include age (age < 65 years versus ≥ 65 years) and *IGHV* status (mutated versus unmutated). The study is composed of a screening period, a treatment period, and a follow-up period (see Section 4.1.1).

The primary efficacy endpoint is the best CR/CRi rate per the IRC response assessment up to PTFU 1 Visit for Arm A and up to C16D1 Visit for Arm B. Response assessment will be performed using the 2018 iwCLL guidelines (Hallek et al 2018) with modification for treatment-related lymphocytosis (Cheson et al 2012) for patients with CLL. The primary analysis will be conducted at the time when all randomized patients have had the opportunity to complete 15 cycles of treatment or follow-up. All patients in both arms will be followed until disease progression and for overall survival, with the final analysis planned to occur approximately 4 years after the last patient is randomized.

4.1.1. Study Description

Each patient will participate in a screening period of ≤ 35 days (at most, from Day -35 to day of enrollment). Patients who agree to participate will sign the informed consent form (ICF) before any study specific screening evaluations. Screening procedures are outlined in the Schedules of Assessments (Section 1.3.1). Any screening assessments that are repeated must fall within the 35-day window. Patients who screen fail may be rescreened once.

The first dose of study drug is to be administered ≤ 5 calendar days after date of randomization.

Patients will be randomized in a 2:1 ratio to receive open-label study treatment with 1 of the following:

- Arm A: sonrotoclax + zanubrutinib
- Arm B: zanubrutinib

Crossover between treatment arms is not allowed.

Treatment Period:

Patients in Arm A will receive a combination treatment for a fixed duration of 15 cycles. Each study cycle is 28 days. Zanubrutinib will be taken orally at a 320 mg total daily dose (320 mg once per day or 160 mg twice per day) starting on Day 1 of Cycle 1, alone for the first 3 cycles, then along with sonrotoclax for 12 cycles. Zanubrutinib may be taken with or without food. On Day 1

of Cycle 4, patients will initiate once-daily sonrotoclax administration with food using the following twice a week ramp-up schedule until the target dose of 320 mg is reached. The twice a week ramp-up schedule will include incremental doubling doses of 1 mg, 2 mg, 5 mg, 10 mg, 20 mg, 40 mg, 80 mg, and 160 mg, increasing biweekly over 1 month, to reach the sonrotoclax target dose of 320 mg. The ramp-up doses and visits may be adjusted per the available data in conjunction with SMC review. Thereafter, patients will continue sonrotoclax 320 mg through Day 28 of Cycle 15.

Patients in Arm B will receive a continuous zanubrutinib monotherapy. Zanubrutinib will be taken orally at a 320 mg total daily dose (320 mg once per day or 160 mg twice per day), with or without food, until disease progression or unacceptable toxicity, whichever occurs first.

All patients who permanently discontinue study treatment either due to a disease progression/intolerant toxicity or following completion of protocol-specified treatment will undergo a Safety Follow-up Visit 30 days (± 7 days) after the last dose of study treatment or before initiation of a new anticancer therapy, whichever occurs first.

Post-treatment Follow-up Period:

For Arm A, the PTFU period will start the day after the last dose of the study drug is taken (whichever is the latest administered in the combination), upon either completing the protocol-specified treatment or discontinuing study treatment prematurely. PTFU visits will be conducted every 16 weeks (± 14 days). All patients in arm A will be followed in the PTFU period for response until unequivocal disease progression has been assessed by the investigator, or the start of next-line anti-CLL treatment, whichever occurs first.

For Arm B, patients with no assessed unequivocal disease progression at the time of permanent treatment discontinuation will enter PTFU period the day after the last dose of study drug. PTFU visits will be performed every 16 weeks (± 14 days) until unequivocal disease progression has been assessed by the investigator, or the start of next-line anti-CLL treatment, whichever occurs first.

Patients enter the survival follow-up period after unequivocal disease progression has been assessed or the start of next line anti-CLL therapy. Survival follow-up may be performed via telephone calls, patient medical records review, a public records search if permitted by local agencies, and/or clinic visits. Survival follow-up will be conducted every 16 weeks (± 14 days), and continues until death, withdrawal of consent, loss to follow-up, or the end of the study. Survival information will be collected including: the date and cause of death, any post-treatment CLL therapies; date of disease progression after each post-treatment CLL therapies, especially the second progression for PFS2; and second primary malignancies.

Study Assessments:

The efficacy assessment will be conducted at screening, C1D1, every 12 weeks (from C1D1) up to PTFU 1 or C16D1, then every 16 weeks thereafter, until unequivocal disease progression, or start of next line anti-CLL therapy, which occurs first.

Assessments of CLL status will include confirmation of CLL diagnosis and disease-related constitutional symptoms; physical examination for detection of lymphadenopathy, hepatomegaly, and splenomegaly; complete blood count (CBC); bone marrow examination. Central laboratory

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assessments of genetic alterations in tumor cells (including del17p, del11q, trisomy 12, del13q, *TP53*, karyotype, and *IGHV* mutation analyses); imaging studies of the neck, chest, abdomen, and pelvis; and patient-reported outcomes assessed by EORTC QLQ-C30 and EORTC QLQ-CLL17.

All patients must have baseline imaging within 35 days before randomization. A CT scan with intravenous contrast/magnetic resonance imaging (MRI) of the neck, chest, abdomen, pelvis, and any other disease sites should be performed at Screening, Cycle 4 Day 1, Cycle 10 Day 1, and at the PTFU 1(Arm A)/C16D1 (Arm B) Visit. Imaging studies will also be performed for confirmation of suspected occurrence of CR or CRi and also promptly at the time of suspected progressive disease based on physical examination findings. For each of the patient, copies of all scans up to PTFU 1 (Arm A)/C16D1 (Arm B) Visit will be sent to the IRC for potential response assessment.

For those patients with suspected CR or CRi based on clinical, laboratory, and imaging criteria, a bone marrow biopsy must also be performed to confirm a CR.

At PTFU 1 (Arm A)/C16D1 (Arm B) Visit, a critical sample of bone marrow aspirate in addition to peripheral blood for MRD by ClonoSEQ® will be collected from all patients. All patients will undergo screening peripheral blood sample collection for MRD by ClonoSEQ®. Patients from Arm A will undergo serial peripheral blood sample collection, bone marrow aspirate sample collection at CR/CRi for assessment of MRD by flow cytometry and peripheral blood sample collection at C7D1 for assessment of MRD by ClonoSEQ®.

To measure potential resistance mechanisms for zanubrutinib plus sonrotoclax and zanubrutinib, blood samples would be collected at C1D1 predose and the time of disease progression for all patients in Arm A and Arm B. Additional biomarker samples collected at PTFU 1 Visit are also required for patients in Arm A.

All patients will be closely monitored for adverse events throughout the study and for 30 days after the last dose of the study treatment or before the start of next line anti CLL therapy, whichever occurs earlier. After this period, the investigator should report any SAEs that are believed to be related to prior study drug treatment. For all patients, any new second primary malignancy, regardless of severity and relationship to study drug, should be recorded until the end of the study. Adverse events will be graded according to the NCI-CTCAE v5.0 and the iwCLL Grading Scale for Hematologic Toxicities in CLL Studies as appropriate (Hallek et al 2018). Refer to Section 8.4 for additional and specific information regarding adverse event monitoring and reporting.

PK analysis will be performed for all patients in Arm A who have received the combination of sonrotoclax and zanubrutinib and have provided ≥ 1 valid postbaseline blood sample for PK analysis of sonrotoclax.

Biomarker analyses include associations between prognostic, predictive biomarkers as well as MRD and the clinical outcomes.

4.1.2. Duration of the Study

The total duration of this study is expected to be approximately 60 to 68 months.

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4.2. Scientific Rationale for Study Design

4.2.1. Rationale for Treatment With Sonrotoclax and Zanubrutinib in the Untreated Chronic Lymphocytic Leukemia Patient Population

Rationale for the combination of sontotoclax and zanubrutinib is provided in Section 2.1.

4.2.2. Rationale for Zanubrutinib Monotherapy as an active comparator

Zanubrutinib monotherapy is considered an effective treatment for previously untreated CLL, and is planned to be the active comparator to the combination of zanubrutinib and sonrotoclax in this study. Rationale for Zanubrutinib monotherapy as an active comparator is provided in Section 2.1.

4.2.3. Rationale for the Ramp-up Schedule

The potency of Bcl-2 inhibitors results in a risk of TLS in MCL, CLL, and other sensitive disease types. A careful ramp-up dosing strategy is therefore required to safely reach the optimal dose.

When patients with CLL or other Bcl-2 inhibitor-sensitive diseases are treated with venetoclax, a 5-step ramp-up schedule is used, starting at 20 mg and increasing weekly to 50 mg, 100 mg, 200 mg, and 400 mg. This has been shown to be safe for monotherapies. During the initial Phase 1 study of venetoclax in patients with CLL, several cases of clinical TLS (including fatal cases) were seen when higher starting doses were used; however, since adopting the 5-week ramp-up, no further cases of clinical TLS were reported in a review of study patients with CLL/SLL (Roberts et al 2016).

In the sonrotoclax clinical studies, several ramp up schedules are being investigated, considering the TLS risk level of each indication. A weekly ramp-up schedule was investigated in B cell malignancies (including CLL/SLL) in several studies (Study BGB-11417-101, Study BGB-11417-102 and BGB-11417-202). The dose is started from 1 mg and roughly doubled each week, until the target dose is reached. As of 04 February 2024, no clinical TLS was observed in patients with CLL/SLL who used the weekly ramp-up schedule for sonrotoclax combined with zanubrutinib therapy. The weekly ramp-up schedule is safe and tolerable for the patients, but considering a ramp-up schedule that could reach a therapeutic dose more quickly may benefit the target population, another daily ramp-up schedule, whereby the dose is increased in small amounts each day until the target dose is reached, was tested in a Phase 1 study of zanubrutinib combined with sonrotoclax in CLL/SLL, as well as a Phase 2 study of sonrotoclax monotherapy in MCL. However, the daily ramp-up schedule is complex, with 24 dose increases during ramp up period, and a simpler administration scheme may increase treatment compliance and safety. As of 4 Feb 2024, 30 patients with R/R CLL/SLL in study BGB-11417-202 were treated with sonrotoclax monotherapy and used the twice a week ramp-up schedule to reach to the target dose of 320mg/day with no reported clinical TLS events. Therefore, the implementation of a BIW ramp up schedule in this study, combined with targeted TLS monitoring, may offer a better way to ensure patient safety and convenience.

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4.2.4. Rationale for Pharmacokinetic Sampling

The sparse PK sampling schedule that follows the multiple doses of study drug/treatment is designed to capture data at different visits to provide a detailed profile of the population PK parameters.

4.2.5. Rationale for Biomarker Strategy

Biomarker analyses will be performed to exclude or stratify patients, as well as the association of biomarkers with patient prognosis, response, and potential resistance to the treatment (Section 4.2.5.1, Section 4.2.5.2, Section 4.2.5.3, and Section 4.2.5.4).

4.2.5.1. Biomarker as Exclusion Criteria

The heterogeneity of the patient population together with the small sample size raises concerns about interpretability and robustness of the treatment effect. Given the small sample size and the two stratification factors already planned, sponsor restrict eligibility to patients without del (17p) and/or TP53 mutation.

Peripheral blood will be collected at screening from patients to assess del17p and other CLL cytogenetics (eg, del(11q), del(13q), and +12, karyotype), and mutation status of TP53.

4.2.5.2. Biomarkers for Stratification

The somatic hypermutation status of the clonotypic *IGHV* gene is a critical biomarker for assessing the prognosis of patients with CLL. Independent studies have documented that *IGHV* somatic hypermutation status is a predictor of responses to therapy, including both chemoimmunotherapy and novel, targeted agents. A systematic review and meta-analysis based on 31 clinical studies between 1999 and 2015 found newly diagnosed CLL with unmutated *IGHV* had shorter PFS and OS than those with mutated *IGHV* genes (Parikh et al 2016). In addition, unmutated *IGHV* is associated with inferior PFS by targeted therapy in newly diagnosed CLL patients treated with venetoclax or ibrutinib-containing therapy (Al-Sawaf et al 2020; Moreno et al 2023).

Choosing *IGHV* mutation as stratification factors enables balance of the most relevant baseline characteristics between the 2 arms. Peripheral blood will be collected at screening from patients to assess mutational status of *IGHV*.

4.2.5.3. Minimum Residual Disease Assessment

In order to isolate the contribution of sonrotoclax in the combination versus zanubrutinib, the primary endpoint of CR/CRi will be coupled with key secondary endpoints such as rate of uMRD4. Achievement of uMRD4 has previously been shown to be independently associated with extended PFS in TN CLL, and therefore, may enhance the assessment of contribution of components (Wierda et al 2021; Kater et al 2022).

Furthermore, both FDA and EMA recommend to test MRD status in patients with CR/CRi (European Medicine Agency 2014; Food and Drug Administration 2020). And considering uMRD4 is well-correlated with PFS as mentioned above, MRD dynamic could be applied in

time to uMRD4, uMRD4 duration exploration in Arm A, which would provide another data package to demonstrate duration of deep remission.

BoVEN study showed among 21 patients who reached blood delta MRD400 at C5D1 (1 month after venetoclax target dose), all had bone marrow uMRD after a median of 6 cycles, and later met the prespecified MRD treatment-discontinuation criterion to stop therapy. Therefore, blood delta MRD might be a surrogate marker for early disease control in Arm A.

A serial MRD assessment in the current study would demonstrate contribution of component, allow for exploration of the landmark uMRD4 and uMRD kinetics, surrogate endpoint in patients from Arm A.

4.2.5.4. Resistance of Mechanism

BTK and PLCG2 mutations have been reported in zanubrutinib-progressed patients with CLL (Xu et al 2023; Blombery et al 2022) and limited data in Zanubrutinib resistance from gene expression perspective. Although there is no study that demonstrates the mechanism of sonrotoclax resistance, it has been reported that dysregulation or genetic alterations or gene expression dysregulation in Bcl-2 family members were involved in venetoclax resistance in CLL (Thomalla et al 2022; Thijssen et al 2022; Blombery et al 2022). It is worth noting that in patients with progression after fixed-duration treatment with ibrutinib plus venetoclax, BTK, PLCG2, and Bcl-2-resistance mutations were not detected, and these patients achieved best response of PR after re-treating with single-agent ibrutinib (Tam et al 2022a).

Studying the mutational status of BTK, PLCG2, other CLL-related pathogenetic mutations, mutational and expression status Bcl-2 families could help to inform the clinical practice for progressive patients.

For patients treated with Sonrotoclax plus Zanubrutinib, peripheral blood sample will be collected at C1D1 predose, PTFU 1 Visit and disease progression to study the acquired resistant clone and its dynamics, gene expression dysregulation. And for patients treated with Zanubrutinib, peripheral blood sample will also be collected at C1D1 predose and disease progression to study gene expression dysregulation.

4.2.6. Patient Input Into Design

No participant/patient/patient organization input has been obtained for this study.

4.3. Justification for Dose

4.3.1. Rationale for Dose Selection

4.3.1.1. Sonrotoclax

The sonrotoclax target dose for patients with TN CLL/SLL is 320 mg daily, based on the totality of the PK, efficacy, safety, and tolerability data, and exposure-response (E-R) analyses, including R/R CLL/SLL patients who received sonrotoclax as monotherapy in Studies BGB-11417-101 and

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BGB-11417-102, and R/R CLL and TN CLL/SLL patients who received SZ in Study BGB-11417-101.

In Studies BGB-11417-101 and BGB-11417-102, patients with CLL/SLL have been enrolled at the doses of 40 mg, 80 mg, 160 mg, 320 mg, and 640 mg once daily, as monotherapy and in combination with zanubrutinib. Sonrotoclax PK exposures appeared to increase in a dose dependent manner from 40 mg to 640 mg with no significant evidence of saturation in exposure until a dose level of 640 mg.

Sonrotoclax was evaluated in patients with R/R CLL as monotherapy (target doses of 80 mg, 160 mg, 320 mg, and 640 mg once daily) and in combination with zanubrutinib (target doses of 40 mg, 80 mg, 160 mg, 320 mg, and 640 mg once daily). Data in patients with R/R CLL exhibited a trend towards improved clinical benefit with increasing sonrotoclax dose as measured by objective responses, CRs, and MRD, without compromising safety and tolerability. In patients with R/R CLL receiving combination therapy, no maximum tolerated dose has been reached up to doses of 640 mg sonrotoclax, and no dose-limiting toxicity (DLT) was observed in any of the dose escalation cohorts. The safety and tolerability of the combination has been consistent across dose levels of 40 mg to 640 mg and consistent with the expected safety profile of both drugs and the underlying diseases. The E-R analysis for safety suggests that there is no apparent relationship between sonrotoclax exposures ($AUC_{0\text{-}24,ss}$, $C_{max,ss}$, or $C_{min,ss}$) and selected adverse events of interest in CLL/SLL patients treated with combination therapy.

Two target doses of sonrotoclax (160 mg and 320 mg once daily) were selected for further evaluation of safety and efficacy in combination with zanubrutinib in TN CLL patients in Study BGB-11417-101. As of the cutoff date of 15 August 2023, efficacy data suggest deeper response associated with the 320 mg dose compared to the 160 mg dose, as is reflected by an increase in CR/CRi and uMRD rate, which could potentially translate to longer survival benefit with various treatment modalities (Seymour et al 2018; Wierda et al 2021; Kater et al 2022).

An E-R analysis for uMRD was performed by including patients with both R/R CLL and TN CLL/SLL receiving combination therapy. Increases of sonrotoclax exposures are associated with an increase in the probability of achieving uMRD, indicating that deeper responses can be achieved at higher exposures. Given an approximately dose-proportional increase from 160 mg to 320 mg, a daily dose of 320 mg is expected to deliver higher sonrotoclax exposures relative to those associated with 160 mg once per day, potentially leading to deeper responses. In light of no safety data collected beyond 640 mg and minimum gain on PK exposure from 320 mg to 640 mg, the 640 mg dose level was not considered for further exploration in TN CLL patients with combination therapy.

In summary, available PK, efficacy, safety and tolerability data, dose- and exposure-response analyses support a 320 mg daily dose of sonrotoclax in combination with a 320 mg daily dose of zanubrutinib in treatment to patients with TN CLL.

4.3.1.2. Zanubrutinib

In the FIH Phase 1 study, BGB-3111-AU-003, the PK of zanubrutinib was linear between 40 mg and 320 mg once a day administered orally. The absorption of zanubrutinib is rapid with t_{max} of 2 hours. The terminal elimination half-life is approximately 4 hours at 320 mg once a day. Results from a food effect study showed that zanubrutinib exposure was not altered by high-fat breakfast,

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and mean AUC and C_{max} were increased by 12% and 51%, respectively, with standard breakfast when compared to fasting. The magnitude of increase in exposure with food was well within doubling of exposure associated with 320 mg administered once a day in the ongoing Phase 1 study and was not associated with any new safety findings; therefore, zanubrutinib can be administered with or without food.

Full occupancy of BTK in peripheral blood mononuclear cells was achieved in all patients in Study BGB-3111-AU-003, while occupancy in lymph node tissue was assessed only at 160 mg twice a day and 320 mg once a day (Tam et al 2015). At the 160 mg twice-a-day dose, full BTK occupancy was observed at trough, suggesting that sustained target occupancy could be achieved in disease-originating tissues, thus more efficiently inhibiting BTK on a continuous basis, further preventing breakthrough signaling despite cycles of new BTK synthesis. Activity has been observed across various B-cell malignancies (including CLL, MCL, WM, and FL) at all tested dose levels; thus, a minimum effective dose cannot be established at this time. Conversely, there is now extensive experience at the 160 mg twice a day and 320 mg once-a-day doses; both schedules show a high level of activity without compromise of the tolerability profile as compared to lower doses of zanubrutinib. Therefore, the dose approved for zanubrutinib monotherapy in CLL/SLL (Brukinsa [zanubrutinib] US prescribing information 2023), 160 mg administered orally twice a day or 320 mg administered orally once a day, has been selected as the recommended dose for this study based on sustained target occupancy, high rates of objective response in multiple types of B-cell malignancies, and a favorable safety and tolerability profile.

The zanubrutinib dose currently under investigation in combination with sonrotoclax and planned for use in this Phase 2 study is also the dose approved for zanubrutinib monotherapy in CLL/SLL. Abundant clinical data was collected in CLL patients treated with zanubrutinib at a total daily dose of 320 mg in combination with doses of sonrotoclax from 40 mg to 640 mg daily, and its safety profile is well characterized with no new safety signals identified.

4.4. Start of Study and End of Study Definitions

The start of the study is defined as the date that the first patient is randomized.

The end of the study is defined as the date of the last visit of the last patient in the study or last scheduled procedure shown in the SoA (Section 1.3) for the last patient in the study globally.

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5. STUDY POPULATION

BeiGene promotes diversity, equity, and inclusion in all clinical studies. Potential patients who cannot fulfill all requirements of this protocol should not be automatically excluded where the reason(s) for noncompliance is due to personal, physical, or social characteristics. This may include, but is not limited to age, sex, gender identity, ethnicity, religious beliefs, sexual orientation, and disability.

Where noncompliance is identified prospectively in potential patients, written approval of the medical monitor must be obtained before informed consent.

Prospective approval of protocol deviations to recruitment and enrollment criteria, also known as protocol waivers or exemptions, is not permitted unless the diversity, equity, and inclusion conditions have been satisfied.

Patients who do not meet the eligibility criteria requirements will be considered screen failures (Section 5.4).

5.1. Inclusion Criteria

Patients are eligible to be included in the study only if they meet all the following criteria:

1. Previously untreated adult patient ≥ 18 years with a confirmed diagnosis of CLL that meets the iwCLL criteria (Hallek et al 2018). For those patients with a screening lymphocyte count < 5000 cells/µL, historical data confirming a lymphocyte count ≥ 5000 cells/µL at time of CLL diagnosis is required.
2. CLL requiring treatment as defined by ≥ 1 of the following criteria:
   a. Evidence of progressive marrow failure as manifested by the development of, or worsening of, anemia and/or thrombocytopenia. Hemoglobin concentrations < 10 g/dL or platelet counts < 100 x 10^9 cells/L are generally regarded as indications for treatment.
   b. Massive (ie, ≥ 6 cm below the left costal margin), progressive, or symptomatic splenomegaly.
   c. Massive (ie, ≥ 10 cm in longest diameter [LDi]), progressive or symptomatic lymphadenopathy.
   d. Progressive lymphocytosis with an increase of ≥ 50% over a 2-month period, or lymphocyte doubling time (LDT) < 6 months.
   Note: LDT can be obtained by linear regression extrapolation of absolute lymphocyte counts obtained at intervals of 2 weeks over an observation period of 2 to 3 months; patients with initial blood lymphocyte counts < 30 x 10^9 cells/L may require a longer observation period to determine the LDT. Factors contributing to lymphocytosis other than CLL (eg, infections and steroid administration) should be excluded.
   e. Symptomatic or functional extranodal involvement (eg, skin, kidney, lung, and spine).
   f. Disease-related symptoms as defined by any of the following:
      – Unintentional weight loss ≥ 10% within the previous 6 months.
      – Fevers ≥ 100.5°F or ≥ 38.0°C for 2 or more weeks without evidence of infection.

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- Night sweats for ≥ 1 month without evidence of infection.
- Significant fatigue (ie, ECOG Performance Status score of 2 or worse; cannot work or unable to perform usual activities).

  Note: Patients with significant fatigue cannot have an ECOG Performance Status score of 0.

3. ECOG Performance Status score of 0, 1, or 2.
4. Measurable disease by CT/MRI. Measurable disease is defined as ≥ 1 lymph node > 1.5 cm in LDi and measurable in 2 perpendicular diameters.
5. Adequate marrow function as defined by:
   a. Absolute neutrophil count ≥ 1.0 x 10^9 cells/L with an exception for patients with bone marrow involvement, in which case absolute neutrophil count must be ≥ 0.75 x 10^9 cells/L (without growth factor support within past 14 days).
   b. Platelet counts ≥ 75 x 10^9 cells/L; in cases of thrombocytopenia clearly due to marrow involvement of CLL (per the discretion of the investigator), platelet count should be ≥ 50 x 10^9 cells/L (without growth factor support or transfusion within past 14 days).
   c. Hemoglobin > 75 g/L (may be post-transfusion).
6. Adequate liver function as indicated by aspartate aminotransferase (AST) and alanine aminotransferase (ALT) values ≤ 2.5 x the institutional upper limits of normal (ULNs) value; serum total bilirubin < 3.0 x ULN (unless documented Gilbert's syndrome).
7. Adequate renal function as defined as creatinine clearance ≥ 50 mL/min directly measured with a 24-hour urine collection or ≥ 50 mL/min/1.73 m^2 calculated according to the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) calculation (see Appendix 16).
8. Life expectancy > 6 months.
9. Signed informed consent and able to comply with the study protocol in the investigator's judgment.
10. Women of childbearing potential must be willing to use a highly effective method of birth control for the duration of the study and for ≥ 90 days after the last dose of study drug (sonrotoclax and/or zanubrutinib). They must also have a negative serum pregnancy test result ≤ 7 days before randomization.
11. Nonsterile men must be willing to use a highly effective method of birth control and must refrain from donating sperm for the duration of the study and for ≥ 90 days after the last dose of study drug (sonrotoclax and/or zanubrutinib). Sterile male is defined as one for whom azoospermia has been previously demonstrated in a semen sample examination as definitive evidence of infertility.

    Males with known "low sperm counts" (consistent with "subfertility") are not to be considered sterile for purposes of this study.

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5.2. Exclusion Criteria

Patients are excluded from the study if they meet any of the following criteria:

1. Presence of del17p or known TP53 mutation.
2. Known prolymphocytic leukemia or history of, or currently suspected, Richter's transformation (biopsy based on clinical suspicion may be needed to rule out transformation).
3. Known central nervous system involvement.
4. Received previous systemic treatment for CLL. Note: Up to 4 doses of anti-CD20 antibody specifically for autoimmune cytopenia is allowed; the last dose should have been given ≥ 6 months before screening.
5. Clinically significant cardiovascular disease including the following:
   a. Myocardial infarction within 6 months before screening.
   b. Unstable angina within 3 months before screening.
   c. New York Heart Association Class III or IV congestive heart failure (see Appendix 6).
   d. History of clinically significant arrhythmias (eg, sustained ventricular tachycardia, ventricular fibrillation, or torsades de pointes).
   e. QTcF > 480 milliseconds based on Fridericia's formula.
      – Note: QTcF value may be calculated as the numerical average of ≤ 3 separate readings for eligibility.
   f. History of Mobitz II second-degree or third-degree heart block without a permanent pacemaker in place.
   g. Uncontrolled hypertension as indicated by a minimum of 2 consecutive blood pressure measurements showing systolic blood pressure > 170 mmHg and diastolic blood pressure > 105 mmHg at screening.
6. Severe or debilitating pulmonary disease, defined as chronic supplementation of oxygen and/or respiratory failure requiring assisted ventilation.
7. History of prior malignancy, except for conditions as listed below and as long as patient has recovered from the acute side effects incurred because of previous therapy:
   a. Malignancies surgically treated with curative intent and with no known active disease present for ≥ 3 years before randomization.
   b. Adequately treated nonmelanoma skin cancer or lentigo maligna without evidence of disease.
   c. Adequately treated cervical carcinoma in situ without evidence of disease.
   d. Localized prostate cancer with Gleason score ≤ 6.
8. Active fungal, bacterial, and/or viral infection requiring systemic therapy.
9. Positive HIV serology (HIVAb) status or serologic status reflecting active hepatitis B or C infection as follows:
   a. Presence of hepatitis B surface antigen (HBsAg).

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b. Patients with presence of hepatitis B core antibody (HBcAb), in the absence of HBsAg, with detectable hepatitis B virus (HBV) DNA.

   – Note: The limit of detection for HBV DNA must have a sensitivity of < 20 IU/mL (see Section 8.1.3). Patients with presence of HBcAb but undetectable HBV DNA are eligible if they are willing to undergo HBV DNA monitoring every 4 weeks for HBV reactivation.

c. Patients with the presence of hepatitis C virus (HCV) antibody and detectable HCV RNA.

   – Note: The limit of detection for HCV RNA must have a sensitivity of < 15 IU/mL (see Section 8.1.3. Patients with presence of HCVAb and undetectable HCV RNA are eligible if willing to undergo HCV RNA monitoring every 4 weeks for HCV reactivation.

10. Uncontrolled autoimmune hemolytic anemia or immune thrombocytopenia requiring treatment.
11. History of severe bleeding disorder such as hemophilia A, hemophilia B, von Willebrand disease, or history of spontaneous bleeding requiring blood transfusion or other medical intervention.
12. History of stroke or intracranial hemorrhage ≤ 6 months before the first dose of study treatment.
13. Unable to swallow capsules or tablets or diseases significantly affecting GI function such as malabsorption syndrome, resection of the stomach or small bowel, bariatric surgery procedures, symptomatic inflammatory bowel disease, or partial or complete bowel obstruction.
14. Hypersensitivity to zanubrutinib, sonrotoclax, or any of its excipients (eg, trehalose).
15. Receiving treatment with any moderate or strong CYP3A4 inhibitor (≤ 7 days or 5 half-lives, whichever is shorter) or strong CYP3A4 inducer (≤ 14 days or 5 half-lives, whichever is shorter) before the first dose of study drug, or requiring ongoing treatment with a moderate or strong CYP3A inhibitor or a strong CYP3A inducer.
16. Consumed grapefruit, grapefruit products, Seville oranges (including marmalade containing Seville oranges), or star fruit ≤ 3 days before the first dose of study drug.
17. At time of enrollment, receiving systemic corticosteroids, unless administered for adrenal replacement.
18. Vaccination with a live vaccine received for a minimum of 4 weeks before enrollment (see also prohibited medications, Section 6.9.2.2).
19. Use of investigational agents within the last 4 weeks before screening.
20. History of illicit drug use or alcohol abuse ≤ 12 months before randomization in the investigator's judgment.
21. Pregnant and lactating females.

22. Any serious medical condition or abnormality in clinical laboratory tests that, in the investigator's judgment, precludes the patient's safe participation in the study.
23. Prisoners or patients who are institutionalized by regulatory or court order or persons who are in dependence to the sponsor or an investigator.

5.3. Lifestyle Considerations

Patients must follow the contraception requirements outlined in Appendix 1.

The most frequent second primary malignancy in zanubrutinib clinical trials was skin cancers (basal cell carcinoma and squamous cell carcinoma of skin). Advise patients to use sun protection.

5.3.1. Meals and Dietary Restrictions

- Each sonrotoclax dose will be administered with food once a day. Zanubrutinib may be administered with or without food.
- Refrain from consumption of grapefruit, grapefruit products, Seville oranges (including marmalade containing Seville oranges) and star fruit from 3 days before the start of study drug until after the final dose.

5.3.2. Caffeine, Alcohol, and Tobacco

No caffeine, alcohol and tobacco restrictions are required.

5.3.3. Physical Activity

No activity restrictions are required.

5.3.4. Other Restrictions

No other restrictions are required.

5.4. Screen Failures

A screen failure occurs when a patient who has consented to participate in the clinical study does not meet the criteria required and therefore is not subsequently assigned to study drug/treatment. A minimal set of screen failure information is required to ensure transparent reporting of screen failure patient to meet the Consolidated Standards of Reporting Trials publishing requirements and to respond to queries from regulatory authorities. Minimal information includes demography, screen failure details, eligibility criteria, and any serious adverse event.

Individuals who do not meet the criteria for participation in this study (screen failure) may be rescreened once under limited conditions after consultation with the sponsor. Rescreening may be considered when a patient's laboratory result narrowly misses a laboratory criterion that is correctable and not due to a rapidly deteriorating condition or progressive disease. Rescreening is allowed only once. If a patient is rescreened, the patient should be assigned a new patient number for every screening/rescreening event and must provide new informed consent as described in Section 10.3. Rescreened patients must meet all inclusion/exclusion criteria to participate in the study.

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5.5. Procedures for Handling Incorrectly Randomized Patients

Incorrect randomization of patients is defined as randomization using incorrect baseline information, ineligible patients being randomized, multiple randomizations being performed for the same patient, and patients receiving the incorrect treatment after randomization.

The sponsor should be notified as soon as the randomization error is discovered, and the investigator should ensure clear documentation of the error and provision of relevant information to the sponsor. The sponsor will promptly provide a decision how to manage the patient (eg, continuation of treatment, discontinuation from the study). Until then the investigator should ensure the patient continues to be managed according to the incorrect randomization and the protocol.

The incorrect randomization of patients is deemed a protocol deviation.

5.6. Criteria for Temporarily Delaying Administration of Study Drug

Refer to Section 6.6 for on treatment dose holds.

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6. STUDY DRUG/TREATMENT(S) AND CONCOMITANT THERAPY

Study drug/treatment are all prespecified, investigational, and non-investigational medicinal products and other study drugs/treatments (eg, surgical and behavioral) that are intended to be administered to the study patients during the study conduct.

6.1. Study Drug/Treatment(s) Administered and Dosage

Table 11: Study Drug/Treatment(s) Administered

Drug/Treatment Label	**Sonrotoclax**	**BRUKINSA**
Drug/Treatment Name	Sonrotoclax	Zanubrutinib
Drug/Treatment Description	Formulated as film-coated, immediate-release tablets for oral administration	Formulated as capsule for oral administration
Type	Drug	Drug
Dose Formulation	Tablet	Capsule
Unit Dose Strength(s)	1, 5, 20, 80 mg/tablet	80 mg/capsule
Dosage Level(s)	Once daily following the twice a week ramp-up schedule in Table 4 until the target dose of 320 mg is reached	320 mg total daily dose (320 mg QD or 160 mg BID)
Route of Administration	Oral	Oral
Use	Experimental	Combination therapy with sonrotoclax: experimental Monotherapy: active comparator
IMP and NIMP/AxMP	IMP	IMP
Sourcing	Provided centrally by the sponsor	Provided centrally by the sponsor

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Drug/Treatment Label	Sonrotoclax	BRUKINSA
Packaging and Labeling	Sonrotoclax will be provided in a child-resistant, high-density polyethylene bottle with a desiccant, induction seal, and bottle label. The contents of the label will be in accordance with all applicable local regulatory requirements.	Zanubrutinib will be provided in a child-resistant, high-density polyethylene bottle with an induction seal and bottle label. The contents of the label will be in accordance with all applicable local regulatory requirements.

Abbreviations: AxMP, auxiliary medicinal product; BID, twice per day; QD, once per day; IMP, investigational medicinal product.

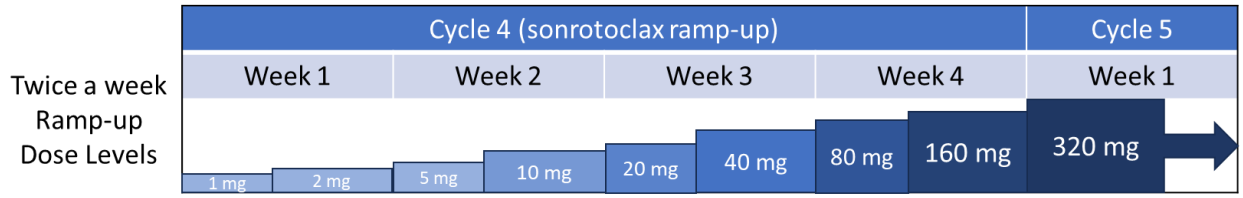
Table 12: Study Arm(s)

Arm Title	Arm A	Arm B
Arm Type	Experimental	Active comparator
Arm Description	Patients will receive zanubrutinib orally at 320 mg daily dose (320 mg QD or 160 mg BID) starting on Day 1 of Cycle 1. Each study cycle is 28 days. Zanubrutinib is taken with or without food for a total of 15 cycles. On Day 1 of Cycle 4, patients will initiate sonrotoclax once daily with food using the following twice a week ramp-up schedule (Table 4) until the target dose of 320 mg is reached. Thereafter, patients will continue both zanubrutinib and sonrotoclax through Day 28 of Cycle 15. The first dose of study drug is to be administered ≤ 5 calendar days after date of randomization.	Patients will receive zanubrutinib monotherapy orally at 320 mg total daily dose (320 mg QD or 160 mg BID) until disease progression, unacceptable toxicity, withdrawal of consent, death, loss to follow-up, investigator's decision, or study termination by the sponsor, whichever occurs first. The first dose of study drug is to be administered ≤ 5 calendar days after date of randomization.

Abbreviations: BID, twice per day; QD, once per day.

As previously noted, patients will employ twice a week ramp-up schedule until the target dose is reached (Table 4 and Figure 2).

Figure 2: Sonrotoclax Twice a Week Ramp-up Schedule



A total of 2 laboratory tests (pre-dose and 4 to 6 hours post-dose) will be conducted and evaluated at Day 1 of each week during the ramp-up period and C5D1. Visits and lab tests of Day 4 of each ramp-up week are not mandatory but could be considered based on the investigator's discretion.

Sonrotoclax should be administered with food in the morning at the same time as zanubrutinib in any order. Zanubrutinib capsules should not be opened, broken, or chewed at any time. Patients will receive zanubrutinib 160 mg twice a day or 320 mg once a day. However, if the patient's preference is 320 mg once a day, it is recommended that the patient continues to use the same dosing schedule throughout the treatment cycles to support better treatment adherence (ie., recommend that patients take the medication at approximately the same time each day). This dosing preference will be recorded in the eCRF.

Patients should take the prescribed dose with water at approximately the same time(s) daily.

Patients will be advised that if a dose of Sonrotoclax is not taken at the scheduled time, they should take the missed dose as soon as they remember, ensuring that the dose is taken within 8 hours of the missed dose with food. Otherwise, patients should skip the missed dose. An extra dose of Sonrotoclax should not be taken to make up for the missed dose.

If a dose of zanubrutinib is missed, it should be taken as soon as possible on the same day with a return to the normal schedule the following day.

Patients will complete a daily diary for as long as they are receiving study treatment. Patients must bring their diary cards to the site at each study visit so that they can be checked by study site personnel for compliance.

Patients will be requested to bring their unused medication and all empty containers/packaging to the center at each visit. All doses of study treatments and all dose changes, including reasons for dose changes, must be recorded on the appropriate eCRF.

6.1.1. Duration of Treatment

The treatment period starts with the first day of assigned study treatment and continues until the last dose of study treatment has been administered.

Patients in Arm A will receive a fixed duration treatment composed of 3 cycles of zanubrutinib monotherapy followed by 12 cycles of zanubrutinib combined with sonrotoclax treatment. Patients in Arm B will receive zanubrutinib continuous therapy until disease progression or intolerance.

Crossover between the study arms will not be allowed.

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6.2. Preparation, Handling, Storage, and Accountability

1. The investigator or designee must confirm appropriate conditions (eg, temperature) have been maintained during transit for all sonrotoclax and/or zanubrutinib received, and any discrepancies are reported and resolved before use of the sonrotoclax and/or zanubrutinib.
2. Only patients enrolled in the study may receive sonrotoclax and/or zanubrutinib, and only authorized site staff may supply, prepare, or administer sonrotoclax and/or zanubrutinib.
3. All sonrotoclax and/or zanubrutinib must be stored in a secure, environmentally controlled, and monitored (manual or automated) area in accordance with the labeled storage conditions with access limited to the investigator and authorized site staff.
4. The investigator, institution, the head of the medical institution (where applicable), or authorized site staff is responsible for study drug/treatment accountability, reconciliation, and record maintenance (ie, receipt, reconciliation, and final disposition records).
5. The investigator or designee (ie, pharmacist) is responsible for ensuring adequate accountability of all used and unused sonrotoclax and/or zanubrutinib. This includes acknowledgment of receipt of each shipment of sonrotoclax and/or zanubrutinib (quantity and condition), patient drug dispensation records, and returned or destroyed sonrotoclax and/or zanubrutinib. Dispensation records will document quantities received from the sponsor's designated depot or its designee and quantities dispensed to patients, including batch/lot number, date dispensed, patient identifier number, and the initials of the person dispensing the medication.
6. At study initiation, the monitor will evaluate the site's SOP for sonrotoclax and/or zanubrutinib disposal/destruction to ensure that it complies with the sponsor's requirements specified in the Pharmacy Manual.
7. At appropriate times during the conduct of the study or at the end of the study after final sonrotoclax and/or zanubrutinib inventory reconciliation by the monitor, the study site will dispose of and/or destroy all unused sonrotoclax and/or zanubrutinib supplies, including empty containers, according to these procedures. If the site cannot meet the sponsor's requirements specified in the Pharmacy Manual for disposal, arrangements will be made between the site and the sponsor or its representative for destruction or return of unused sonrotoclax and/or zanubrutinib supplies.
8. All sonrotoclax and/or zanubrutinib supplies and associated documentation will be periodically reviewed and verified by the study monitor over the course of the study.
9. Further guidance and information for the final disposition of unused sonrotoclax and/or zanubrutinib are provided in the study reference manual or other specified location.

Management (ie, labeling, handling, storage, administration, and disposal) of standard of care products provided by the site will be in accordance with the relevant local guidelines and/or prescribing information. For further details, see the manufacturer's prescribing information for sonrotoclax and/or zanubrutinib.

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6.3. Assignment to Study Treatment

All patients will be centrally assigned to randomized treatment using an IRT system by permuted block stratified randomization. Details of the randomization procedure will be approved by study statistician before the study is initiated.

6.4. Blinding

No blinding or masking is required.

6.5. Study Drug/Treatment Compliance

The dose of sonrotoclax and/or zanubrutinib and study patient identification will be confirmed at the time of dosing by a member of the study site staff other than the person administering the sonrotoclax and/or zanubrutinib.

When patients self-administer sonrotoclax and/or zanubrutinib at home, compliance with sonrotoclax and/or zanubrutinib will be assessed at each visit. Compliance will be assessed by the investigator and/or study personnel and based on information provided by the patient and/or guardian [direct questioning, counting returned tablets/capsules, patient diary, etc.] during the site visits and documented in the source documents and relevant form. Deviation(s) from the prescribed dosing regimen should be recorded.

A record of the quantity of sonrotoclax and/or zanubrutinib dispensed to and administered by each patient must be maintained and reconciled with sonrotoclax and/or zanubrutinib and compliance records by the investigator. Sonrotoclax and/or zanubrutinib start and stop dates, including dates of sonrotoclax and/or zanubrutinib delays and/or dose reductions will also be recorded.

In accordance with all applicable regulatory requirements, the investigator or designated study center personnel must maintain study treatment accountability records throughout the course of the study. The investigator or designated study center personnel will document the amount of study treatment received from the sponsor as well as the amount supplied and/or administered to and returned by the patients, if applicable.

6.6. Dose Modification

Every effort should be made to administer the study drugs according to the planned dose and schedule. In the event of significant toxicities, actions should be taken based on the guidelines below and provided in Section 6.10. Reasons for dose reductions or delays, the supportive measures taken, and the outcome will be documented in the patient's chart and recorded in the eCRF.

The dose modification guidelines in this section are not intended to be a substitute for clinical judgment. Investigators may delay or reduce doses for other reasons (eg, adverse events or laboratory findings) as appropriate.

Dose modification is defined as any of the following: dose reduction, dose delay, and missed doses. A dose delay is a deviation from the prescribed dosing schedule (ie, the study drug/treatment is withheld beyond visit window).

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Study drug/treatment dosing may be interrupted if the patient experiences a toxicity that requires a dose to be withheld. Study drug/treatment should resume as soon as possible and ≤ 28 days at the same dose or at a reduced dose, depending on the toxicity, grade, prior events, and number of dose reductions as per the safety guidance provided in Section 6.10. If the patient is unable to resume treatment within 28 days after the last dose, then the patient should be discontinued from study drug/treatment. Resumption of study drug/treatment after dose interruption for >28 days due to study drug/treatment-related toxicity requires consultation with the medical monitor. If, in the opinion of the investigator, a patient is deriving benefit from the study drug/treatment while meeting the discontinuation criteria, study drug/treatment may be resumed only after discussion with the medical monitor and approval by the sponsor.

For Arm A, if the toxicity is highly likely to be attributed to a particular study drug based on the established toxicity profile of the study drug/class, the investigator may continue another study drugs as deemed clinically appropriate. In this case, discussion with sponsor medical monitor is encouraged.

6.6.1. Sonrotoclax

Dose reduction levels for sonrotoclax are presented in Table 13 and should be followed for hematologic (Section 6.10.1.1) or non-hematologic (Section 6.10.2.1) toxicities.

Dose holds and modifications during ramp-up and after reaching target dose of sonrotoclax are described in Section 6.6.1.1.

For Arm A, if zanubrutinib is discontinued permanently before sonrotoclax dosing has been initiated, then no sonrotoclax will be administered to the patient at any time during this study.

Table 13: Dose Reduction Levels for Sonrotoclax

Toxicity Occurrence	Target Dose Level	Sonrotoclax Dose Upon Resumption of Dosing [a]
First	0 = target dose	Restart at 320 mg
Second [b]	-1 dose level	Restart at 160 mg
Third	-2 dose level	Restart at 80 mg
Fourth	Discontinue sonrotoclax	Discontinue sonrotoclax

[a] Resumption of dosing may occur following resolution of toxicity to ≤ Grade 1 or baseline.
[b] The first instance of Grade 4 non-hematologic toxicity should begin with the guidance given for "Second" toxicity occurrence and study treatment should be discontinued upon recurrence of the Grade 4 toxicity (see Section 6.10.2.1).

Sonrotoclax may be restarted upon resolution of toxicity to ≤ Grade 1 or baseline if held for up to 28 consecutive days. Written approval must be obtained from the sponsor medical monitor for restarting treatment after interruptions > 28 days if it is thought to be in the patient's interest.

The mitigation, monitoring, and management guidelines for TLS are described in Section 6.10.3.1.1.

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6.6.1.1. Sonrotoclax Dose Holds

During Ramp-up

During ramp-up period, the reasons for sonrotoclax dose holds are described in Section 6.10.1.1 and Section 6.10.2.1; management of ramp-up dose levels are described in this section. If a dose hold occurs during the ramp-up period, the ramp-up may continue after resolution of the toxicity to Grade ≤ 1 or baseline.

If the sonrotoclax dose is held for 7 days or less, the patient may resume sonrotoclax at the same ramp-up dose level or at one lower ramp-up dose level as determined by the investigator, based on a risk assessment (including tumor burden status).

If the dose hold lasts more than 7 days, at least one lower ramp-up dose level must be used, with the exception of the initial dose of 1 mg. At each intended ramp-up dose level, all patients must receive sonrotoclax for at least 3 days before starting the next ramp-up dose level (Table 4 and Figure 2). The Investigator or designee must consult with the medical monitor, in writing, regarding the restarting dose and ramp-up prior to restarting the treatment.

Repeated dose holds during ramp-up should be discussed with the sponsor medical monitor inclusive of potential reduction in target dose level.

After reaching target dose

For patients who have reached sonrotoclax target dose and sonrotoclax dose is held for 7 days or less, the patient may resume sonrotoclax at the same dose level or at least one lower dose level as determined by the investigator, based on a risk assessment (including tumor burden status).

If dosing interruption lasting more than 7 days, the investigator must discuss with the medical monitor and have a written approval before patient restarts sonrotoclax dosing. For patients with significant active disease present, the restart dose must be at least one dose level lower than the last dose given prior to interruption, and at each intended dose level the patient must receive sonrotoclax for at least 3 days before starting the next dose level (Table 4 and Figure 2).

All appropriate TLS prophylaxis, monitoring and management guidelines for TLS should be resumed before restarting dosing and during ramp-up (per Section 6.10.3.1).

6.6.2. Zanubrutinib

The treatment modification guidelines in Table 14 below should be followed for hematologic (Section 6.10.1.2) and non-hematologic (Section 6.10.2.2) toxicities (other than hypertension adequately controlled with oral medication or asymptomatic laboratory abnormalities).

Laboratory events indicating liver or renal dysfunction will not be considered asymptomatic laboratory events (Section 6.10.2.2).

Study drug may be held for a maximum of 28 days and restarted upon resolution of toxicity. Written approval must be obtained from the sponsor medical monitor for restarting treatment after interruptions lasting > 28 days if it is thought to be in the patient's interest.

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Table 14: Dose Reduction for Zanubrutinib

Toxicity occurrence	Dose level	Zanubrutinib dosage (starting dose = 160 mg twice a day)	Zanubrutinib dosage (starting dose = 320 mg once a day)
First	0 = starting dose	Restart at 160 mg twice a day	Restart at 320 mg once a day
Second	-1 dose level	Restart at 80 mg twice a day	Restart at 160 mg once a day
Third	-2 dose level	Restart at 80 mg once a day	Restart at 80 mg once a day
Fourth	Permanently discontinue zanubrutinib	Permanently discontinue zanubrutinib	Permanently discontinue zanubrutinib

Consider the benefit-risk of withholding zanubrutinib for 3-7 days pre- and post-surgery depending upon the type of surgery and the risk of bleeding.

6.6.3. Retreatment Criteria

Not applicable.

6.7. Continued Access to Study Drug/Treatment After the End of the Study

Study drugs in Arm A will be administered for up to 15 cycles, and zanubrutinib in Arm B will be administered continuously until PD or intolerance or until the end of the study as determined by the sponsor.

6.8. Treatment of Overdose

Any dose of study drug in excess of that specified in this protocol, whether accidental or intentional, is considered to be an overdose. Any overdose of study drug should be noted in the patient's chart and on the appropriate eCRF. Adverse events associated with an overdose or incorrect administration of study drug will be recorded on the Adverse Event eCRF. Any serious adverse events associated with an overdose or incorrect administration must be reported ≤ 24 hours after awareness via the serious adverse event reporting process described in Appendix 2. There is no specific antidote for overdose for any of the study drugs. In the event of an overdose, patients should be closely monitored and given appropriate supportive treatment.

6.9. Prior and Concomitant Therapy

6.9.1. Prior Therapy

The exclusion criteria (Section 5.2) specify that patients will not have received prior systemic therapy for CLL, with the exception of up to 4 doses of anti-CD-20 antibody specifically for autoimmune cytopenia; the last dose should be given ≥ 6 months prior screening.

6.9.2. Concomitant Therapy

All medication or vaccine (including over the counter or prescription medicines, recreational drugs, vitamins, and/or herbal supplements, intravenous medications and fluids) or other specific

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categories of interest that the patient is receiving within 35 days before the first dose of study drug or receives during the study must be recorded in the eCRF until 30 days after the last dose of study drug along with:

- Reason for use.
- Dates of administration including start and end dates.

Patients must be instructed not to take any medication, including over the counter products without first consulting with the investigator.

Unless noted otherwise, most concomitant medications and therapies deemed necessary, and/or in keeping with local standards of medical care at the discretion of the investigator for supportive care (eg, antiemetics, antidiarrheals, hematopoietic growth factors, red blood cell/platelet transfusions) and in a patient's interest are allowed. Patients must notify the investigator of all concurrent medications used after the start of the study. The investigator should take concomitant medications into account when evaluating adverse events.

The medical monitor should be contacted if there are any questions regarding concomitant therapy. Patients with high tumor burden should be monitored closely for the occurrence of TLS; prophylactic measures, including allopurinol or rasburicase, should be instituted in the setting of high tumor burden per institutional care standards. Guidelines for the prevention and management of TLS are discussed in Section 6.10.3.1.1 and Appendix 11.

Antimicrobial prophylaxis and/or for the prevention of HBV reactivation may be used per institutional care standards.

In patients with a high risk of opportunistic infections, including *Pneumocystis jirovecii* pneumonia, prophylaxis should be considered per institutional standards.

6.9.2.1. Permitted Medications

The following treatments are allowed:

- Blood transfusions and growth factor support per institutional standards of care.
- Corticosteroids for non-CLL indications: Patients should not receive treatment with systemic corticosteroids other than for short durations (< 2 weeks) to treat non-CLL-related conditions (eg, to treat a flare of chronic obstructive pulmonary disease). For patients requiring systemic corticosteroids for longer durations (eg, to treat patients with COVID-19 who require oxygen support), consult the sponsor medical monitor. Long-term chronic systemic corticosteroid use is not permitted, except for adrenal replacement therapy.
- Therapy to manage constitutional or GI symptoms (nausea, vomiting, diarrhea, constipation), pain, infectious complications, etc, per institutional standards of care.

Patients who develop a second primary malignancy while on study treatment should be discussed with the sponsor medical monitor before initiating any nonsurgical treatment.

Adjuvant antihormonal therapy for prostate or breast cancer is allowed. All localized treatments (such as radiation or surgery) expected to be completed within a 6-week period during the

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Treatment Period are allowed, provided the study drug may be held during the period of localized anticancer treatment. Patients with a second primary malignancy who require systemic chemotherapy or immunotherapy must permanently discontinue study treatment before starting non-CLL cancer treatment. Patients who require lifesaving therapy without available alternative that may interact with study treatment or significantly obscure the interpretation of CLL response should permanently discontinue study drug.

The sponsor medical monitor should be consulted to discuss risks and benefits before restarting study drug after treatment interruption due to treatment for second primary malignancy.

6.9.2.2. Prohibited Medications

Patients should not receive another therapy for CLL (cytotoxic, biologic, or immunotherapy) while in this study. Non-protocol therapy for CLL should not be administered until documented disease progression.

Concomitant use of strong CYP3A inhibitors and inducers, or moderate CYP3A inhibitors should be avoided during the ramp-up phase for sonrotoclax (Section 6.9.3.1).

Live vaccines are prohibited throughout study treatment until > 30 days after the last dose of study treatment.

For a full list of restricted and prohibited medications, refer to Appendix 4.

6.9.3. Potential Interactions Between the Study Drugs and Concomitant Medications

6.9.3.1. Sonrotoclax

Based on nonclinical data, the primary metabolic pathway for sonrotoclax is via CYP3A. Administration of sonrotoclax with strong and moderate CYP3A inhibitors or strong CYP3A inducers (see Appendix 4 for a list of examples of these medications) are prohibited during the ramp-up. Moderate CYP3A inhibitors or inducers can be used with caution at other times. Strong CYP3A inhibitors or inducers should be avoided if at all possible; however, if no alternative treatment is available, use may be permitted outside of the ramp-up. Patients taking CYP3A inhibitors or inducers should be closely monitored for potential toxicities of sonrotoclax.

Table 15: Dose Modification Table for Sonrotoclax when Coadministered with Strong/Moderate CYP3A Inhibitors or Inducers

Coadministered Drug	Initiation and Ramp-Up Phase	Steady Daily Dose (after target dose reached)
Strong CYP3A inhibitors	Prohibited during initiation and ramp-up	Avoid if at all possible; however, if no alternative treatment is available, use may be permitted and the dose needs to be reduced to 1/8 of the target dose. Monitor closely for

		potential toxicities of sonrotoclax.
Moderate CYP3A inhibitors	Prohibited during initiation and ramp-up	Use with caution and the dose needs to be reduced to 1/4 of the target dose
Strong CYP3A inducers	Prohibited during initiation and ramp-up	Avoid if at all possible; however, if no alternative treatment is available, use may be permitted. Monitor closely for potential toxicities of sonrotoclax.
Moderate CYP3A inducers	Use with caution	Use with caution

Sonrotoclax exhibited IC_{50} values of 2.3 µM for CYP2C8 in the reversible CYP inhibition study and 3.3 µM for CYP3A4 in the time-dependent inhibition study, which are much higher than targeted plasma levels expected in humans at therapeutic doses. Therefore, it is unlikely that the concomitant administration of sonrotoclax would interfere with the metabolism of compounds that are CYP2C8 or CYP3A4 substrates.

Based on the in vitro transporter data, sonrotoclax was identified as a P-gp and BCRP substrate and has the potential to inhibit P-gp or BCRP intestinal transporters when administered orally. Therefore, any medications known to be strong inhibitors of P-gp or BCRP should be avoided during the study. Moderate P-gp or BCRP inhibitors should be used with caution, unless reviewed and approved by the principal investigator and the medical monitor (refer to Appendix 15 for examples of these medications). Patients taking P-gp or BCRP substrates should be closely monitored for potential toxicities of sonrotoclax.

The concomitant use of sensitive, narrow therapeutic index P-gp or BCRP substrates (eg, tacrolimus, cyclosporine, digoxin, etc) should be avoided if at all possible because sonrotoclax may increase their concentrations. However, if no alternative treatment is available, use with sonrotoclax may be permitted per the investigator's judgement and following the local country prescriber information for these medications (P-gp or BCRP substrates). Patients should be closely monitored for potential toxicity of the P-gp or BCRP substrate.

6.9.3.2. Zanubrutinib

6.9.3.2.1. Cytochrome P450-Inhibiting/Inducing Drugs for Zanubrutinib

Administration of zanubrutinib with strong/moderate CYP3A inhibitors or CYP3A inducers (see Appendix 4 for a list of these medications) and grapefruit juice and Seville oranges should be used with caution as they may affect the metabolism of zanubrutinib. If at all possible, patients are encouraged not to use strong/moderate CYP3A inhibitors and inducers and should consider using alternative agents. If these agents will be used, follow the dose modification guidelines in Table 16 or according to local country zanubrutinib prescriber information. Please refer to http://medicine.iupui.edu/clinpharm/ddis/main-table/ for a more complete list.

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Table 16: Dose Modification Table for Zanubrutinib When Coadministered With Strong/Moderate CYP3A Inhibitors or Inducers

CYP3A	Coadministered Drug	Recommended Use
Inhibition	Strong CYP3A inhibitor (eg, ketoconazole, conivaptan, clarithromycin, indinavir, itraconazole, lopinavir, ritonavir, telaprevir, posaconazole, voriconazole)	80 mg once daily
	Moderate CYP3A inhibitor (eg, erythromycin, ciprofloxacin, diltiazem, dronedarone, fluconazole, verapamil, aprepitant, imatinib, grapefruit products, Seville oranges)	80 mg twice daily
Induction	Strong CYP3A inducer (eg, carbamazepine, phenytoin, rifampin, St. John's wort)	Avoid concomitant use.
	Moderate CYP3A inducer (eg, bosentan, efavirenz, etravirine, modafinil, nafcillin)	Avoid concomitant use. If these inducers cannot be avoided, increase zanubrutinib dose to 320 mg twice daily.

Abbreviations: CYP, cytochrome P450.

6.9.3.2.2. Effects of Zanubrutinib on Exposure of Other Concomitant Medications

A clinical drug-drug interaction study (BGB-3111-108) indicated that zanubrutinib is a mild inducer of CYP3A and CYP2C19. Narrow therapeutic index drugs that are metabolized by CYP3A4 (eg, alfentanil, cyclosporine, dihydroergotamine, ergotamine, fentanyl, pimozide, quinidine, sirolimus, or tacrolimus) and CYP2C19 (eg, S-mephenytoin) should be used with caution because zanubrutinib may decrease the plasma exposure of these drugs.

Because ethinylestradiol (a key ingredient in a variety of combined oral contraceptives) is partly metabolized by CYP3A4, patients using hormonal contraceptives (ie, birth control pills or devices) must use a barrier method of contraception (eg, condoms) as well (see Appendix 1).

Repeated dosing of zanubrutinib increased exposure of digoxin (a P-gp substrate) with a mean increase of 11% for AUC_{0-t} and 34% for C_{max}. Exercise caution when coadministering oral P-gp substrates with a narrow therapeutic index (eg, digoxin) because zanubrutinib may increase their exposures.

6.9.4. Rescue Medication

Not applicable.

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6.10. Safety Management for Adverse Events

Following discussion with the sponsor medical monitor, if the toxicity is highly likely to be attributed to a particular study drug based on the established toxicity profile of the study drug/class, the investigator may continue other study drugs as deemed clinically appropriate.

6.10.1. Hematological Toxicities

Study drug should be held for treatment-related hematological toxicity as described in Table 17 and Table 18 and may require dose reduction or permanent discontinuation.

6.10.1.1. Sonrotoclax

Table 17: Sonrotoclax Dose Modification and Management for Hematological Toxicities

Toxicity	Sonrotoclax Management	Other Actions
·≥ Grade 3 febrile neutropenia ·≥ Grade 3 thrombocytopenia that results in clinically significant bleeding · Any Grade 4 toxicity with the following exceptions: - Grade 4 neutropenia lasting ≤ 7 days in the presence or absence of growth factor support - Grade 4 leukopenia	Sonrotoclax will be held for individual patients under any of the conditions, based on investigator assessment of study-drug relatedness. Resumption of dosing may occur following resolution of toxicity to Grade ≤ 1 or baseline. If any of the conditions listed on the left occur before initiation of sonrotoclax, the first dose must be held until resolution of toxicity to Grade ≤ 1 or baseline. A maximum of 2 dose reductions will be allowed. Patients who experience ≥ Grade 3 thrombocytopenia associated with significant bleeding (ie, requiring medical intervention) will be discussed with the sponsor medical monitor. Re-escalation to the original (ie, target) dose level for patients with 1 or more dose reductions may be allowed under certain circumstances after discussion	Closely monitor hematology labs for worsening signs or symptoms Proper medical management should be instituted Record/report AE/SAE according to Section 8.4

Toxicity	Sonrotoclax Management	Other Actions
	with the sponsor medical monitor	

Abbreviations: AE, adverse event; SAE, serious adverse event.

6.10.1.2. Zanubrutinib

Table 18: Zanubrutinib Dose Modification and Management for Hematological Toxicities

Toxicity	Zanubrutinib Management	Other Actions
·Grade 4 neutropenia that is persistent for ≥ 10 consecutive days ·Grade 4 thrombocytopenia that is persistent for ≥ 10 consecutive days ·Grade 3 thrombocytopenia associated with significant bleeding ·≥ Grade 3 febrile neutropenia	Treatment will be interrupted under any of the conditions not clearly related to the underlying disease. Resumption of dosing may occur following resolution of toxicity to Grade ≤ 1 or baseline. The dose hold should start on the day that one of the conditions is met. A maximum of 2 dose reductions will be allowed. Patients who experience ≥ Grade 3 thrombocytopenia associated with significant bleeding (ie, requiring medical intervention) should be discussed with the sponsor medical monitor before resuming treatment. Asymptomatic, treatment-related lymphocytosis should not be considered a hematological toxicity that requires dose reduction. Patients with treatment related lymphocytosis should remain on study treatment and	Closely monitor hematology labs for worsening signs or symptoms Proper medical management should be instituted Record/report AE/SAE according to Section 8.4

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Toxicity	Zanubrutinib Management	Other Actions
	continue with all study related procedures.	

Abbreviations: AE, adverse event; SAE, serious adverse event.

6.10.2. Non-Hematological Toxicities

Study drug should be held for treatment-related non-hematological toxicity as described in Table 19 and Table 20 and may require dose reduction or permanent discontinuation.

6.10.2.1. Sonrotoclax

Table 19: Sonrotoclax Dose Modification and Management for Non-Hematological Toxicities

Toxicity	Study Treatment Management	Other Actions
Liver function abnormalities meeting Hy's law criteria defined as follows: - ALT or AST > 3 x the ULN and total bilirubin > 2 x ULN without initial findings of cholestasis (elevated serum alkaline phosphatase [ALP]) and after excluding other potential causes. - Patients who experience liver function abnormalities meeting Hy's Law criteria may only restart study treatment once, upon resolution. If the abnormalities recur, study treatment must be permanently discontinued.	Sonrotoclax will be held for individual patients under any of the conditions, based on investigator assessment of study-drug relatedness. Resumption of dosing may occur following resolution of toxicity to Grade ≤ 1 or baseline. If any of the conditions occur before initiation of sonrotoclax, the first dose must be held until resolution of toxicity to Grade ≤ 1 or baseline. For the first instance of a Grade 4 non-hematologic toxicity that is not clearly related to the underlying disease, follow the "Second Occurrence" guidance in Table 13 upon resolution of toxicity to Grade ≤ 1 or baseline (ie, restart sonrotoclax at Dose Level-1). If the Grade 4 toxicity	Closely monitor for worsening signs or symptoms Proper medical management should be instituted as medically indicated Record/report AE/SAE according to Section 8.4

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Toxicity	Study Treatment Management	Other Actions
Any≥ Grade 3 toxicity with the following exceptions: - TLS-related laboratory analyte abnormalities, that do not meet Howard criteria of Laboratory TLS (see Appendix 12), (hyperuricemia, hyperphosphatemia, hyperkalemia, and/or hypocalcemia) that resolve before the next dose, with or without treatment. - Grade 3 GI toxicity (ie, nausea, vomiting, diarrhea) responsive to supportive care measures administered over a period of ≥ 3 days. - Asymptomatic laboratory abnormalities (eg, hypophosphatemia, hypomagnesemia) that resolve in ≤ 7 days either with (eg, mineral supplementation) or without intervention.	reoccurs, study treatment should be discontinued. Re-escalation to the original (ie, target) dose level for patients with 1 or more dose reductions may be allowed under certain circumstances after discussion with the sponsor medical monitor.	

6.10.2.2. Zanubrutinib

Table 20: Zanubrutinib Dose Modification and Management for Non-Hematological Toxicities

Toxicity	Action for Zanubrutinib	Restart Dose
≥ Grade 3 bleeding not considered related to study drug	Hold until recovery to ≤ Grade 1	Restart at either the original dose or dose level

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Toxicity	Action for Zanubrutinib	Restart Dose
		(-1), at the discretion of the treating investigator
≥ Grade 3 bleeding considered related to study drug	Hold until underlying condition has fully resolved. If underlying condition cannot be treated to full resolution, permanently discontinue zanubrutinib. The drug should be permanently discontinued for any related ≥ Grade 3 hemorrhage with the exception of those where the underlying condition or reason for the bleeding can be fully treated (eg, gastric ulcer resulting in GI bleed, use of anticoagulation) and the risk of a re-bleed is deemed acceptable by the sponsor medical monitor and investigator.	Restart at dose level (-1)
Any grade intracranial hemorrhage	Permanently discontinue	Not applicable
AF that is symptomatic and/or incompletely controlled	Hold until AF is controlled	Restart at either the original dose or dose level (-1), at the discretion of the treating investigator
Other ≥ Grade 3 toxicity considered related to study drug, including inadequately controlled hypertension and/or liver or renal laboratory value abnormalities	Hold until recovery to ≤ 1 or baseline	Restart at the original dose level
≥ Grade 3 opportunistic infection	Permanently discontinue	Not applicable

Abbreviations: AF, atrial fibrillation; GI, gastrointestinal

6.10.3. Risks Associated With Study Drugs

6.10.3.1. Tumor Lysis Syndrome Associated With Sonrotoclax

For sonrotoclax, the risk of TLS will affect the hospitalization and monitoring requirements applicable to a given patient on this study. Patients will be categorized for TLS risk, on the basis

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of tumor burden (Table 21). Each tumor burden category defines the corresponding risk for TLS for sonrotoclax. Renal dysfunction does not affect this classification but should be considered as a factor and may warrant additional prophylaxis and monitoring for a given patient.

Table 21: Tumor Lysis Syndrome Risk Stratification for Sonrotoclax

	Sonrotoclax
Low tumor burden	Patients with low tumor burden are defined as those with an absence of high tumor burden (see criteria below). A patient with low tumor burden may follow the increased monitoring and hospitalization guidelines for those with high tumor burden/renal dysfunction cited below, at the discretion of the investigator. Lymph node size used to classify the tumor burden category must be determined by imaging.
High tumor burden	Patients considered to have high tumor burden include those meeting any of the following criteria: • Any lymph node ≥ 10 cm • Both of the following: – Any lymph node ≥ 5 cm – ALC ≥ 25 x 10^9/L Lymph node size used to classify the tumor burden category must be determined by imaging.
Renal dysfunction	Renal dysfunction (creatinine clearance < 80 mL/min) may increase the risk of clinically relevant TLS independent of tumor burden. Regardless of tumor burden, patients with a creatinine clearance ≤ 80 mL/min should be monitored closely when the dose of sonrotoclax is increased. Additional prophylaxis and monitoring of these patients should be considered.

Abbreviations: ALC, absolute lymphocyte count.

6.10.3.1.1 Mitigation and Monitoring of Tumor Lysis Syndrome

Any patient who develops clinically relevant TLS-related laboratory abnormalities or fulfills the Howard Criteria (Howard et al 2018) for laboratory or clinical TLS must interrupt sonrotoclax administration until the abnormalities have resolved. Patient should be immediately treated with uricolytic agents and hydration until resolution of these abnormalities.

To mitigate and detect the occurrence of TLS during this study, TLS prophylaxis, patient hospitalization, and regular laboratory monitoring are employed. The need for such measures will vary depending on a patient's tumor burden and renal function.

Management recommendations of TLS are provided in Appendix 11, but patients should be treated according to institutional guidelines.

6.10.3.1.1.1 Hydration

For patients in arm A, oral hydration consisting of 1.5 to 2 L/day should be administered 2 to 3 days before until the 24 hours after the first dose of sonrotoclax, coincident with

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subsequent dose increases. As applicable, intravenous hydration is encouraged at first dose of sonrotoclax and at subsequent dose increases for patients unable to maintain such oral hydration. Intravenous hydration in the outpatient setting on the day of dosing during the clinic stay is recommended to assure this full amount of hydration is achieved. For patients for whom volume overload is considered a significant risk, hospitalization should be considered.

6.10.3.1.1.2 Prophylaxis With Uric Acid Reducing Agents

For patients in Arm A, treatment for the prevention of hyperuricemia should be given 2 to 3 days before the first dose of sonrotoclax and continue until ≥ 7 days after reaching the target dose. Daily allopurinol is generally considered the standard of care for prevention of TLS.

Rasburicase is an option for TLS prophylaxis. If rasburicase will be administered, it must be given before the initial dose or dose increment of sonrotoclax, unless there is a medical contraindication. This should be strongly considered if baseline uric acid levels are elevated.

For patients with a contraindication to rasburicase (ie, glucose 6 phosphate dehydrogenase deficiency), the TLS risk-mitigation plan must be reviewed with the investigator. Uric acid levels following treatment with rasburicase must be analyzed using specific guidelines described below.

For samples from patients treated with rasburicase: rasburicase causes enzymatic degradation of the uric acid in blood/plasma/serum samples left at room temperature, potentially resulting in spuriously low plasma uric acid assay readings. The following special sample handling procedure must be followed to avoid ex vivo uric acid degradation. Uric acid must be analyzed in plasma. Blood must be collected into prechilled tubes containing heparin anticoagulant. Immediately immerse plasma samples for uric acid measurement in an ice water bath. Plasma samples must be prepared by centrifugation in a precooled centrifuge (4°C). Finally, the plasma must be maintained in an ice water bath and analyzed for uric acid within 4 hours of collection.

6.10.3.1.1.3 Hospitalization

Arm A: For patients with high tumor burden, hospitalization should be considered based on the investigator’s discretion during ramp-up.

A nephrology consultation should be considered upon admission to ensure emergency dialysis is available and hospital staff are prepared to deal with TLS as necessary.

During the ramp-up schedules, all patients are not required to be hospitalized and may be hospitalized per investigator’s discretion. Elective hospitalization is not considered to be an SAE. If there is a limited reduction of circulating tumor cells or nodal disease burden at the lower dose levels, patients may should be considered for hospitalization per the investigator’s decision. Also additional TLS laboratory assessments may be performed per investigator’s discretion.

6.10.3.1.1.4 Tumor Lysis Syndrome-Related Monitoring

Tumor Lysis Syndrome-Related Monitoring for Sonrotoclax

TLS-related laboratory assessments (Section 8.4.8) will be monitored pre and post-dose when initiating sonrotoclax, when increasing the dose level, and when sonrotoclax is restarted at a

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lower dose level after dose interruption during sonrotoclax ramp-up. This schedule may be adjusted at the request of the sponsor or per SMC recommendations, based on data gathered during the study.

Patients will undergo the following assessments before dosing and up to 4-6 hours after dosing:

- Vital signs: temperature, blood pressure, and heart rate or pulse rate
- Hematology: hemoglobin, hematocrit, platelet count, red blood cell count, and white blood cell count with differential, which includes counts neutrophils, lymphocytes, monocytes, eosinophils, and basophils
- Serum chemistry: uric acid, creatinine, phosphorus, potassium, and calcium

Laboratory samples collected for hematology and serum chemistry should be sent immediately to the laboratory, and the results reviewed promptly by the investigator or sub investigator.

Predose TLS laboratory results before the first dose do not need to be reviewed before dosing, if prior test results from ≤ 7 days before dosing have been reviewed and assessed as acceptable.

For all required visits during the ramp-up period, ≥ 1 post-dose set of TLS laboratory results must be obtained and reviewed before the patient leaves the clinic. Additional TLS monitoring may be required before the patient leaves the clinic.

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7. DISCONTINUATION OF STUDY DRUG/TREATMENT AND PATIENT WITHDRAWAL

Discontinuation of specific sites or of the study as a whole are detailed in Section 10.10.

7.1. Discontinuation of Study Drug/Treatment

It may be necessary for a patient to permanently discontinue study drug/treatment. If study drug/treatment is permanently discontinued, the patient should, if possible, remain in the study to be evaluated for safety/long-term and survival follow-up. Patients who discontinue study drug/treatment before progressive disease will continue to undergo tumor assessments as outlined in the SoA (Section 1.3).

See the SoA (Section 1.3) for data to be collected at the time of discontinuation of study drug/treatment and follow-up and for any further evaluations that need to be completed.

The primary reason for study drug/treatment discontinuation should be documented on the appropriate eCRF. Patients may discontinue study drug/treatment for reasons including but not limited to the following:

- Adverse event.
- Patient decision. Patients have the right to discontinue study drug/treatment at any time for any reason without prejudice to further treatment.
- Investigator decision. The investigator has the right to discontinue a patient from study drug/treatment at any time.
- Pregnancy.
- Any medical condition that the investigator or sponsor determines may jeopardize the patient's safety if they were to continue the study drug/treatment.
- Patient noncompliance. Study site staff should first counsel patients who are significantly noncompliant on the importance of study drug/treatment compliance and accountability. The investigator may, in consultation with the medical monitor, discontinue patients from treatment who are consistently noncompliant.
- Progressive disease.
- Use of any concurrent systemic anticancer therapy, including chemotherapy, standard or investigational agents (including Chinese patent medicines) for the treatment of cancer.

7.2. Patient Withdrawal From the Study

- A patient may withdraw from the study at any time at the patient's own request for any reason (or without providing any reason). Note: Withdrawal of consent occurs only when a patient does not want to participate in the study anymore or no longer wants their personal data to be collected or used in the study.

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- A patient may be withdrawn at any time at the discretion of the investigator for safety, behavioral, or compliance reasons.
- If the patient withdraws consent, the sponsor may retain and continue to use any information (including data and biological samples) collected before such a withdrawal of consent. In the US, all biological samples not yet analyzed at the time of withdrawal may still be used for further testing/analysis in accordance with the terms of this protocol and of the ICF. In the EU and Rest of World all biological samples not yet analyzed at the time of the withdrawal will no longer be used, unless permitted by applicable law. These samples will be stored according to applicable legal requirements.
- If a patient withdraws from the study, the patient may request destruction of any samples taken and not tested, and the investigator must document this in the site study records.

7.3. Loss to Follow-up

A patient will be considered lost to follow-up if the patient repeatedly fails to return for scheduled visits and is unable to be contacted by the study site.

The following actions must be taken if a patient fails to return to the clinic for a required study visit:

- The site must attempt to contact the patient and reschedule the missed visit as soon as possible, counsel the patient on the importance of maintaining the assigned visit schedule and ascertain whether the patient wishes to and/or should continue in the study.
- Before a patient is deemed lost to follow-up, the investigator or designee must make every effort to regain contact with the patient (where possible, 3 telephone calls, and if necessary, a certified letter to the patient's last known mailing address or local equivalent methods). These contact attempts should be documented in the patient's medical record.
- Should the patient continue to be unreachable, the patient will be considered to have withdrawn from the study.
- Site personnel, or an independent third party, will attempt to collect the vital status of the patient within legal and ethical boundaries for all patients randomized, including those who did not receive study drug/treatment. Hospital records and public sources may be searched for vital status information where local laws allow. If vital status is determined as deceased, this will be documented, and the patient will not be considered lost to follow-up. Sponsor personnel will not be involved in any attempts to collect vital status information.

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8. STUDY ASSESSMENTS AND PROCEDURES

Investigators are responsible for recruitment of patients in this study. The study is or will be publicly registered in national or international clinical study databases as required by regulation. Potential patients will be identified via medical record review or by contact from other referring providers or patients. The first act for recruitment is defined as the first interaction between the potential patient and the investigator or site designee in relation to the study.

- Study procedures and their timing are summarized in the SoA (Section 1.3). Protocol waivers or exemptions are not allowed.
- Adherence to the study design requirements, including those specified in the SoA (Section 1.3), is essential and required for study conduct.
- All screening evaluations must be completed and reviewed to confirm that potential patients meet all eligibility criteria. The investigator will maintain a screening log to record details of all patients screened and to confirm eligibility or record reasons for screening failure, as applicable.
- Procedures conducted as part of the patient's routine clinical management [eg, blood count] and obtained before signing of the ICF may be utilized for screening or baseline purposes provided the procedures met the protocol-specified criteria and were performed within the timeframe defined in the SoA (Section 1.3).
- In the event of a significant study-continuity issue (eg, caused by a pandemic), alternative strategies for patient visits, assessments, medication distribution and monitoring may be implemented by the sponsor or the investigator, per local health authority/ethics requirements.

Repeat or unscheduled samples may be taken for safety reasons or for technical issues with the samples.

8.1. Administrative and Screening Assessments and Procedures

Screening evaluations will be performed ≤ 35 days before the first dose of study drug/treatment (refer to the SoA (Section 1.3) for details). Patients who agree to participate must sign the ICF before undergoing any study-specific screening assessment. The screening period begins on the first day that a screening assessment is conducted. Screening evaluations may be repeated as needed within the screening period. The investigator will assess patient eligibility according to the latest screening assessment results. Results of standard of care tests or examinations performed before informed consent has been obtained and ≤ 35 days before the first dose of study drug/treatment may be used for the purposes of screening rather than repeating the standard of care tests unless otherwise indicated. Procedures conducted during the Screening Visit only, are described in this section.

Rescreening requirements are presented in Section 5.4.

Any abnormality identified during screening will be graded according to the CTCAE v5.0 and the iwCLL Grading Scale for Hemotologic Toxicities in CLL, and recorded in the source data, and

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the appropriate disease/condition terms should be recorded in the source data and as medical history in eCRF.

8.1.1. Demographics

Demographic factors such as age, sex, race, and ethnicity could influence the effects (safety and efficacy) of medicines and the risk/benefit assessment in different populations. Race and ethnicity data are collected in accordance with ICH guidance (ICH E5 1998, ICH E17 2017) adopted by the EMA and US FDA, to understand whether race/ethnicity could influence the PK, safety, and/or efficacy of the study drug/treatment. For example, population PK analysis is a well-established, quantitative method that can quantify and explain the variability in drug concentrations among patients. Such variability can be attributed to intrinsic factors (eg, body weight, age, sex, race/ethnicity), or to extrinsic factors (eg, concomitant medications), and can lead to clinically relevant changes in drug concentrations that require a change in the dose or dosing regimen. Results from race/ethnicity and other demographic analyses will be incorporated into study drug/treatment product labeling to provide guidance on safety and efficacy variations (if any) linked to certain populations (eg, race or ethnic group) as well as any potential dose adjustment needed for those populations. Therefore, collecting race/ethnicity data in the study is essential to understand whether race/ethnicity could influence the PK, safety, and/or efficacy.

8.1.2. Medical and Cancer History

Review medical and cancer history after obtaining informed consent, including the presence or absence of disease-related constitutional symptoms. Clinically significant medical history findings (ie, previous diagnoses, diseases, or surgeries) not pertaining to the indication under study that started before signing the informed consent, and considered relevant for the patient's eligibility will be captured in the eCRF. Clinically significant is defined as any events, diagnoses, or laboratory values requiring treatment, follow-up or the presence of signs or symptoms that require medical intervention. Concurrent medical signs and symptoms must be documented to establish baseline severities.

Other baseline information to be collected includes history of CLL (including the date of initial diagnosis and current disease status), Binet staging, Cumulative Illness Rating Scale, sites of disease, and presence or absence of disease-related constitutional symptoms. Prior and concomitant medications/significant nondrug therapies will also be collected.

8.1.3. Hepatitis B and C Testing

Hepatitis B and C serologies (includes HBsAg, hepatitis B surface antibody [HBsAb], HBcAb and HCVAb) and viral load (HBV DNA will be assessed by polymerase chain reaction [PCR] and HCV RNA by PCR) will be tested at screening.

Hepatitis B and C serologic testing will be performed locally. PCR tests are required to have a sensitivity < 20 IU/mL and < 15 IU/mL for HBV and HCV, respectively.

Patients positive at screening for HBsAg, HBV DNA, or HCV RNA are not eligible. Table 22 describes how the results for HBV and HCV testing at screening relate to study eligibility.

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Table 22: Eligibility and Monitoring of Patients With HBV and/or HCV Infection

Screening Assessment	Meets Inclusion Criteria	To Be Excluded
HBV	HbsAg (-) and HbcAb (-)	HbsAg (+)
	HbsAg (-) and HbcAb (+) HBV DNA "Not detected" Assess HBV DNA by PCR every ≤ 4 weeks (or every ≤ 12 weeks for patients receiving prophylactic antiviral therapy)	HBV DNA detected
HCV	Antibody (-) or antibody (+) HCV RNA "Not detected" Assess HCV RNA by PCR every ≤ 4 weeks	HCV RNA detected

Abbreviations: HbcAb, hepatitis B core antibody; HbsAg, hepatitis B surface antigen; HBV, hepatitis B virus; HCV, hepatitis C virus; PCR, polymerase chain reaction.

Patients who are HBsAg negative, HBcAb positive, and HBV DNA negative and not on prophylactic antiviral treatment must have HBV DNA assessed every cycle (cycle length 28 days) during the study treatment period. These patients should be considered for prophylactic antiviral treatment in consultation with a local HBV expert. Patients treated prophylactically with antivirals should undergo testing for HBV DNA every third cycle. After the end of treatment, monitoring should continue every 12 weeks for 12 months starting from the last dose of study treatment. If HBV DNA is found to be ≥ 100 IU/mL, study treatment should be held and antiviral therapy initiated. If HBV DNA is ≥ 20 IU/mL but < 100 IU/mL, HBV DNA should be rechecked in ≤ 2 weeks. If the repeat level is again ≥ 20 IU/mL, study treatment should be held and antiviral therapy initiated. Resumption of study treatment in patients whose HBV reactivation resolves must be discussed with and approved by the sponsor medical monitor in consultation with a physician with expertise in managing hepatitis B.

Patients positive for HCV antibody but negative for HCV RNA must have HCV RNA testing every 4 weeks during the study treatment period. After the end of treatment, monitoring should continue every 12 weeks for 12 months starting from the last dose of study treatment. If HCV RNA is detected (≥ 15 IU/mL), study treatment should be held and antiviral therapy initiated. Resumption of study treatment in patients whose HCV reactivation resolves must be discussed with and approved by the sponsor medical monitor in consultation with a physician with expertise in managing hepatitis C.

The sponsor medical monitor or designee should be informed of any suspected HBV or HCV reactivation.

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8.1.4. Patient Numbering

After obtaining informed consent, study site personnel will access the Interactive Response Technology (IRT) system to obtain a unique patient number for a potential study participant.

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Patient number will be assigned in chronological order starting with the lowest number. Once a patient number has been assigned, it cannot be reassigned to any other patient.

8.1.5. Diagnostic Flow Cytometry for CLL Markers

A peripheral blood sample will be collected and sent to the Central Lab to evaluate for CLL markers to document CLL diagnosis according to the iwCLL criteria (see Appendix 9).

8.1.6. HIV Serology Test

An HIV serology test will be conducted at screening unless previous HIV test results from ≤ 4 weeks before screening are available. Patients with HIV infection are excluded from the study.

8.1.7. Coagulation tests

The coagulation profile includes prothrombin time, which will also be reported as international normalized ratio, and activated partial thromboplastin time. The coagulation profile will be performed at screening only and thereafter as medically indicated by local laboratory.

8.1.8. Coombs test

Direct antiglobulin test (or direct Coombs test) will only be done at screening and will be performed locally.

8.1.9. β2-microglobulin

β2-microglobulin will only be done at screening and will be performed locally.

8.1.10. Urinalysis

Urinalysis is required to be performed as indicated in Section 1.3. Urinalysis will include pH, glucose, proteins, ketones, and specific gravity. A 24-hour urine collection for creatinine clearance test may be performed per investigator's judgment to determine creatinine clearance at screening.

Refer to Section 8.7 for biomarker testing.

8.2. Efficacy Assessments

Response assessment will be conducted at screening, C1D1, every 12 weeks from C1D1 (at C4D1, C7D1, C10D1, C13D1) up to PTFU 1 (Arm A)/C16D1 (Arm B), then every 16 weeks thereafter, until unequivocal disease progression, or the start of next line anti-CLL therapy, whichever occurs first. Response assessment will also be performed promptly at the time of suspected progressive disease as clinically indicated.

Disease assessments are conducted as described above, and the assessment schedule will not be adjusted regardless of any changes in treatment (Section 1.3).

Responses will be assessed in accordance with the iwCLL criteria (Hallek et al 2018) with modifications for treatment-related lymphocytosis (Cheson et al 2012; Appendix 7) for patients with CLL. Disease status at baseline and iwCLL response at each response assessment timepoint must be documented in the EDC system. Response parameters will include assessments of lymphadenopathy, hepatomegaly, and splenomegaly by both imaging and physical examination,

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as well as blood lymphocyte count, platelet count, hemoglobin concentration, neutrophil count, and bone marrow aspirate/biopsy. For efficacy assessments, physical examination findings and CBC results obtained at Cycle 1 Day 1 before the first study drug dose should be used as baseline. If predose values at C1D1 are not available, the most recent prior values will be used for baseline. All abnormal lesions assessed as CLL related will be recorded at baseline. Up to 6 target lesions will be identified by imaging and physical examination and recorded with bi-dimensional measurements. Additional abnormal lymph nodes and non-nodal lesions/sites of disease will be recorded as non-target lesions.

Responses will be assessed by the study site investigator based on the above-cited efficacy parameters. Response at C4D1, C10D1, PTFU 1 (Arm A) / C16D1 (Arm B), at the occurrence of CR or CRi, and also at the time of suspected progressive disease before the PTFU 1 (Arm A) / C16D1 (Arm B) visit also will be assessed by IRC. All disease assessments after PTFU 1 (Arm A) / C16D1 (Arm B) will be done only by investigator.

The treatment decisions of drug discontinuation due to disease progression will be based on the investigators' assessments.

8.2.1. Disease-Related Constitutional Symptoms

The presence or absence of disease-related constitutional symptoms including unexplained fever of ≥ 38°C; unexplained, recurrent drenching night sweats; significant fatigue (ie, ECOG performance scale 2 or worse; cannot work or unable to perform usual activities); and/or unexplained loss of > 10% body weight within the previous 6 months and since last examination will be documented at screening, C1D1, and the following post-baseline response evaluations:

- Arm A: at C4D1, C7D1, C10D1, C13D1, and during each PTFU Visit.
- Arm B: at C4D1, C7D1, C10D1, C13D1, C16D1, and then D1 of every 4 cycles until disease progression.

8.2.2. Examination of Liver, Spleen, and Lymph Nodes by Physical Examination

The same method of assessment and the same technique should be used to characterize the liver, the spleen, and each identified and reported abnormal lymph node at screening, C1D1 and the following post-baseline response evaluations:

- Arm A: at C4D1, C7D1, C10D1, C13D1, and during each PTFU visit.
- Arm B: at C4D1, C7D1, C10D1, C13D1, C16D1, and then D1 of every 4 cycles until disease progression.

A full disease assessment physical examination should be performed to assess the extent of disease involvement. Physical examinations should include the evaluation of the presence and measurements of enlarged lymph nodes in 2 dimensions (lymph node regions: cervical, supraclavicular, axillary, inguinal, and femoral nodes). For each abnormal lymph node identified at baseline, these bi-dimensional measurements should be recorded at all physical examinations, with documentation of lymph node being palpable or non-palpable (follow CRF instructions for entering measurements when lymph node is not palpable). During physical examination, lymph nodes will only be considered measurable when they are superficial (eg, palpable lymph nodes).

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All measurements should be taken and recorded in metric notation using a ruler or calipers. The perpendicular diameters, in 2 planes, of the largest palpable nodes in each lymph node region should be measured.

Splenomegaly and hepatomegaly are assessed based on whether they are palpable and the extent below the costal margin will be recorded.

Clinical response evaluation will be based on blood lymphocyte count, platelet count, hemoglobin concentration, and neutrophil count, as well as physical examination of lymph nodes, spleen and liver, and constitutional symptoms.

Disease progression assessed by physical examination must be promptly confirmed by imaging (CT scan or MRI) within 28 days of the physical examination.

8.2.3. Tumor Imaging

CT with intravenous contrast is the preferred imaging modality. For patients having a contraindication to CT scans with contrast, the reason should be documented in the patient's medical record.

All patients must have baseline imaging within 35 days before randomization. Intravenous contrast-enhanced CT scans of the neck, chest, abdomen, and pelvis and any other disease sites should be performed.

For patients who cannot receive CT intravenous contrast, or who have other documented contraindication to CT imaging, the following imaging studies should be performed at Screening and during the study: contrast-enhanced MRI of the neck, abdomen, and pelvis, and non-contrast CT of the chest. For patients who develop a contraindication to CT scan with intravenous contrast while on study, see the Imaging Manual for details about imaging requirements. All efforts will be made to ensure that the imaging equipment, contrast agent, and person (investigator or radiologist) performing the evaluation are kept constant throughout a patient's course of study; additional details can be found in the Imaging Manual.

After screening, imaging studies should be performed for patients in Arm A at Cycle 4 Day 1, Cycle 10 Day 1, and the PTFU 1 Visit; for patients in Arm B, they should be performed at Cycle 4 Day 1, Cycle 10 Day 1, and Cycle 16 Day 1. Imaging studies will also be performed for confirmation of suspected occurrence of CR or CRi and also promptly at the time of suspected progressive disease based on physical examination findings.

Overall response evaluation will include imaging, blood lymphocyte count, platelet count, hemoglobin concentration, and neutrophil count, as well as physical examination of lymph nodes, spleen and liver, and constitutional symptoms. For these critical overall response evaluations, all required laboratory blood count results must be dated within 28 days before or within 28 days after the date of acquisition of the first set of imaging (if more than one imaging date). At the time of scheduling the imaging, a laboratory blood draw for the required blood count results should also be scheduled unless it is confirmed by the investigator that the patient meets the 28-day window requirement for timing of the laboratory test results with the imaging acquisition date.

For selection and evaluation of target and non-target lesions, see Appendix 8.

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All CT scans and MRIs obtained up to PTFU 1 (Arm A) / C16D1 (Arm B) visit will be collected and reviewed by an IRC. Deidentified copies of all scans and radiology reports (including those from screening) must be provided to the sponsor or its designee (eg, IRC).

In rare instances, the timing of a patient's scans may fall outside the imaging procedure window specified in the Schedule of Assessments due to out-of-town travel or other unforeseen circumstances. Imaging performed outside of the optimal procedure window specified in the Schedule of Assessments will not necessarily be assessed as a protocol deviation; the sponsor will make the final determination.

8.2.4. Bone Marrow Biopsy for Response Assessment

At baseline, a bone marrow aspirate and biopsy are not required; however, results should be recorded in the eCRF if available. During the treatment period and PTFU period bone marrow biopsy and aspirate are required if clinical and laboratory results demonstrate a potential CR/CRi, to confirm a CR/CRi in bone marrow biopsy (bone marrow biopsy is required regardless of minimal residual disease status).

- Patients who are otherwise complete responders but show bone marrow involvement preventing CR/CRi classification should recheck bone marrow biopsy within the next 12 months until CR/CRi is confirmed as long as the patient is still showing evidence of being CR/CRi outside of bone marrow results. Recheck may be done earlier than 12 months as clinically indicated.
- Once CR or CRi in the bone marrow biopsy has been confirmed, no repeated bone marrow biopsies are required for the purpose of CR/CRi response assessment.

At PTFU 1 Visit/C16D1 all patients who have achieved absolute lymphocyte count (ALC) $< 4.0 \times 10^9$/L should have a bone marrow biopsy unless already obtained previously to confirm CR/CRi. This will allow the IRC to perform a thorough and independent assessment of response at end of treatment including CR/CRi.

Bone marrow biopsy is required during the PTFU Period to confirm if clinical and laboratory results demonstrate a suspected cytopenic disease progression that is unrelated to autoimmune cytopenia or study treatment at the discretion of the investigator. Note: The progression of any cytopenia defines disease progression (unrelated to autoimmune cytopenia). This is documented by a decrease of hemoglobin levels > 2 g/dL or to < 10 g/dL, or by a decrease of platelet counts ≥ 50% or to < 100,000/μL, which occurs ≥ 3 months after treatment, if the marrow biopsy demonstrates an infiltrate of clonal CLL cells. During therapy, cytopenias cannot be used to define disease progression.

All the bone marrow biopsies will be collected and reviewed by a pathologist from the central pathology laboratory. For bone marrow biopsy sample to the central laboratory, submit either an archival block or 15 unstained slides (for details on bone marrow sample collection and handling, refer to the Laboratory Manual). If a bone marrow examination was not submitted to the central pathology laboratory per protocol, was submitted centrally and was NE, or a procedure was performed as part of standard of care at any time, please document the results in the eCRF if local pathology results are available.

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8.2.5. Lymph Node or Tissue Biopsy

A lymph or tissue node biopsy is generally not required, unless it is necessary in rare cases in which the distinguish between new malignant lesion and inflammatory disease is difficult.

A lymph node or tissue biopsy is performed to establish the diagnosis of a transformation into an aggressive lymphoma (Richter transformation). A PET scan can be useful to identify the best lymphoid area for establishing the diagnosis by biopsy.

8.2.6. Minimal Residual Disease Assessments - Peripheral Blood and Bone Marrow Aspirate

Peripheral blood and bone marrow aspirate samples for minimal residual disease assessment will be collected as stated in Table 23. The first draw of bone marrow aspirate should be sent for MRD assessment. Additional instructions for sample handling are outlined in the Laboratory Manual. MRD will be assessed by NGS and flow cytometry method using an antibody panel recommended by the European Research Initiative on CLL, both in the central laboratory.

Peripheral blood and bone marrow aspirate samples for MRD assessment by NGS (clonoSEQ®) from Chinese patients will be collected from clinical research centers and temporarily stored at Labcorp Pharmaceutical Research and Development (Shanghai) Co., Ltd. The samples will be transported to ADAPTIVE BIOTECHNOLOGIES CORPORATION for clonoSEQ® testing through Beijing Bio-Platform International Logistics Service Co., Ltd. The remaining samples will be stored at ADAPTIVE BIOTECHNOLOGIES CORPORATION after completion of testing and will be destroyed by third-party destruction company, SteriCycle of Washington Inc., after project completion.

Table 23: Schedule of MRD Sample Collection for Arm A and Arm B Patients

	Screening	**C4D1**	**C7D1**	**C10D1**	**C13D1**	**PTFU 1 Visit (Arm A) or C16D1 (Arm B)**	**Every 16 weeks from PTFU 1 Visit and until PD (Arm A) or every 4 cycles from C16D1 and until PD (Arm B)**	**At CR/CRi**
PB from Arm A and B (NGS)	X		X (Arm A only)			X		
BMA from Arm						X		

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	Screening	**C4D1**	**C7D1**	**C10D1**	**C13D1**	**PTFU 1 Visit (Arm A) or C16D1 (Arm B)**	**Every 16 weeks from PTFU 1 Visit and until PD (Arm A) or every 4 cycles from C16D1 and until PD (Arm B)**	**At CR/CRi**
A and B (NGS)								
PB from Arm A (flow)	X	X	X	X	X	X	X	
BMA from Arm A (flow)								X

8.2.7. Post-treatment Assessments

Post-treatment study visits will occur every 16 weeks after completion of the study treatment and continue until unequivocal disease progression has been assessed, or the start of next line anti-CLL treatment, whichever occurs first.

Response assessments will be performed during each post-treatment study visit. As noted for the study treatment period, CR or CRi suspected on the basis of blood counts and physical findings must be confirmed with contrast-enhanced imaging studies and bone marrow examination.

8.2.8. Survival Assessments

After confirmation of unequivocal disease progression or after the start of next line anti-CLL treatment, survival follow-up calls will be performed every 16 weeks until end of study. Survival follow-up may be performed via telephone calls, patient medical records review, a public records search if permitted by local agencies, and/or clinic visits. Survival information will be collected, including the date and cause of death; any post-treatment CLL therapies; date of disease progression after each post-treatment CLL therapies, especially the second progression for PFS2; and second primary malignancies (see Section 1.3).

8.3. Safety Assessments

Planned timepoints for all safety assessments are provided in the SoA (Section 1.3).

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8.3.1. Physical Examinations and Vital Signs

Physical examination, vital signs (sitting blood pressure, heart rate or pulse rate, and body temperature), and weight will be performed at screening and at each study visit during study treatment; at safety follow-up and PTFU 1 for Arm A and safety follow-up for Arm B. Height (cm) is determined at screening only.

A complete physical examination includes an assessment of all body systems per standard of care at each study site. A targeted physical examination is focused on evaluation of treatment-emergent signs or symptoms.

Assessment of vital signs and physical examination on Cycle 1 Day 1 may be skipped if performed within 7 prior days.

8.3.2. Cardiac Function

An echocardiogram, multigated acquisition scan, or gated heart pool scan will be performed within 90 days before enrollment (ie, a standard-of-care procedure performed before consent was received for this study may be used if within the 90-day window) and as medically necessary.

8.3.3. Electrocardiograms

A 12-lead ECG will be performed in triplicate (≥ 1 minute apart) at screening. The QTcF value may be calculated using Fridericia's Formula as the numerical average of up to 3 separate readings for eligibility. During the treatment period visits, a single 12-lead ECG read will be performed prior to any study drug(s) administration on that day. For safety monitoring purposes, the investigator must review, sign, and date all ECG tracings. Paper or electronic copies of ECG tracings will be kept as part of the patient's permanent study file at the site.

All ECGs are to be obtained before other assessments scheduled at that same time (eg, vital sign measurements, blood draws). The patient should rest in a semi recumbent or supine position for approximately 3 minutes in the absence of environmental distractions that may induce changes in heart rate (eg, television, radio, conversation) before each ECG collection.

New or worsened clinically significant abnormalities are to be recorded as adverse events in the eCRF.

8.3.4. Clinical Safety Laboratory Tests

Table 24: Protocol-Required Safety Laboratory Tests

Laboratory Tests	Parameters
Hematology	Hematocrit Hemoglobin Platelet count Red blood cell count White blood cell count with differential: neutrophil count, lymphocyte count, monocyte count, eosinophil count, basophil count (The CBC result obtained before first study drug administration on Cycle 1 Day 1will be used for response assessment purposes. If no values are

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Laboratory Tests	Parameters
	available predose on the first dose date, then the most recent CBC values before the first dose date should be assigned as baseline for response assessments.)
Clinical Chemistry [a]	Sodium Potassium Chloride Bicarbonate or CO_2 (or if neither are available, CO_2 combining power) Glucose Blood urea nitrogen (or urea) Creatinine Calcium Phosphate/phosphorus Magnesium Total bilirubin Total protein Albumin ALT AST Lactate dehydrogenase ALP Uric acid (An increase of AST or ALT levels to > 3 x ULN should be followed by repeat testing of liver function studies [AST, ALT, as well as ALP and total bilirubin] within 48 to 72 hours to confirm the abnormalities and determine if levels are increasing or decreasing.)
Urinalysis by Dipstick	pH, Glucose, Proteins, Ketones, Specific gravity
Tumor Lysis Syndrome-Relevant Laboratory Assessments	Potassium Phosphorus Uric acid Corrected calcium Serum creatinine
Coagulation	Prothrombin time Activated partial thromboplastin time INR

Laboratory Tests	Parameters
Quantitative Serum Immunoglobulins	IgG, IgM, and IgA
Other Screening Tests	Highly sensitive urine or serum pregnancy test. See Section 8.3.5. Serology (HIV antibody, hepatitis B surface antigen, and HCV antibody). See Section 8.1.3 and Section 8.1.6.

Abbreviations: ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate aminotransferase; HCV, hepatitis C virus; Ig, immunoglobulin; INR, international normalized ratio; TSH, thyroid stimulating hormone; ULN, upper limit of normal.
[a] Total calcium values will be corrected for participants/patients with hypoproteinemia.

- Clinical samples will be tested at timepoints specified in the Schedule of Activities and at unscheduled timepoints as medically necessary. Samples for baseline laboratory assessments should be collected before the first study drug administration on Cycle 1 Day 1 unless otherwise specified. Specified screening blood tests performed ≤ 72 hours before the first study drug administration do not need to be repeated on the first day of study drug administration.
- The investigator must review the laboratory results, document this review, and record any clinically significant changes occurring during the study as an adverse event. The laboratory results must be retained with source documents.
- Refer to Appendix 2 for additional information regarding clinical assessment and management of clinical laboratory abnormalities.
- A detailed description of the procedures for collection, handling, storage, and shipment of local laboratory samples are provided in the Laboratory Manual.

8.3.5. Pregnancy Testing

All women of childbearing potential (including those who have had a tubal ligation) will have a serum pregnancy test ≤ 7 days before the first dose of study drug as part of the screening process. A laboratory-based, highly sensitive urine or serum pregnancy tests will also be performed with the results reviewed by the investigators on Cycle 1 Day 1 prior to dosing. Either serum or urine pregnancy tests will be performed every 4 weeks during the treatment period at specified visits. Patients are allowed to take the pregnancy tests locally in the cycles with no scheduled visits, but the investigators should be informed of the test results and should record them in the eCRF timely. Either serum or urine pregnancy tests will continue to be performed every 4 weeks until ≥ 90 days after the last dose of study drug. This will require additional visits after the Safety Follow-up Visit. If a urine pregnancy test is positive, it must be confirmed by a serum pregnancy test. Any patient with a positive serum pregnancy test result will immediately discontinue study treatment.

8.3.6. Suicidal Ideation and Behavior Risk Monitoring

No SIB risk monitoring is required.

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8.3.7. Other Safety Assessments

8.3.7.1. Prior and Concomitant Medications Review

Record any new medications, changes in ongoing medications, and medications discontinued within 35 days before the first visit in Cycle 1 and since the prior study visit, thereafter. All prior and concomitant medications should be documented at Screening and during the Treatment Period up to and including the Safety Follow-up Visit. During the post-treatment and survival follow-up phase, information on subsequent CLL therapy will be collected (Section 8.2.7 and Section 8.2.8).

8.3.7.2. Adverse Events Review

After informed consent has been signed, but before the administration of study treatment, only serious adverse events should be recorded.

After initiation of study treatment, all adverse events and serious adverse events regardless of relationship to study drug will be reported. The adverse events reporting period is defined in Section 8.4.1.

All treatment-related adverse events and serious adverse events will be followed until resolution or stabilization. The accepted regulatory definition for an adverse events is provided in Appendix 2. Additional requirements for reporting serious adverse events are explained in Section 8.4.4.

8.3.7.3. Eastern Cooperative Oncology Group Performance Status>

ECOG Performance Status (Appendix 5) will be assessed during the study.

8.4. Adverse Event, Serious Adverse Event, and Other Safety Reporting

The definitions of an adverse event and a serious adverse event can be found in Appendix 2.

Adverse events of special interest for this study are presented in Section 8.4.8.

The investigator and any qualified designees are responsible for detecting, documenting, and recording events that meet the definition of an adverse event, serious adverse event and adverse event of special interest and remain responsible for following up of all adverse events or adverse events that are serious, considered related to the study drug/treatment or study procedures, or that caused the patient to discontinue the study drug/treatment or study (see Section 7). This includes events reported by the patient (or, when appropriate, by a caregiver, surrogate, or the patient's legally authorized representative).

The method of recording, evaluating, and assessing causality of adverse events and serious adverse events and Adverse events of special interest and the procedures for completing and transmitting serious adverse event reports are provided in Appendix 2.

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8.4.1. Time Period for Collecting and Recording Adverse Events and Serious Adverse Event Information

After informed consent has been signed but before the administration of the study drug/treatment, only serious adverse events should be recorded by the investigator/investigational site to the sponsor.

After the first dose of study drug, all nonserious adverse events, serious adverse events, and adverse events of special interest (see Section 8.4.8), regardless of relationship to study drug/treatment, will be reported until 30 days after the last dose of study drug/treatment (including other IMP[s]): or initiation of a new anticancer therapy, whichever occurs first. After this period, the investigator should report any SAEs that are believed to be related to prior study drug treatment. For all patients, any new second primary malignancy, regardless of severity and relationship to study drug(s), should be recorded until the end of the study.

All serious adverse events will be recorded and reported to the sponsor or designee immediately and under no circumstance should this exceed 24 hours, as indicated in Appendix 2. The investigator will submit any updated serious adverse event data to the sponsor within 24 hours of them being available.

After the safety reporting period as described above, investigators are not obliged to actively seek information on adverse events or serious adverse events after the conclusion of study participation. However, if the investigator learns of any serious adverse event, including a death, at any time after a patient has been discharged from the study, and the investigator considers the event to be reasonably related to the study drug/treatment or study participation, the investigator must promptly notify the sponsor regardless of time since the last dose of study drug/treatment.

Refer to Table 25 for guidance on duration of recording new or worsening adverse events in Arm A and Arm B. For the follow-up period for adverse events and serious adverse events, see Section 8.4.3.

Table 25: Guidance for Duration of Recording New or Worsening Adverse Events in Arm A and Arm B

Event Type	Record New or Worsening Events That Occur During This Period	
	Begin	**End**
SAEs	Signing of informed consent.	First dose of study drug/treatment.
Nonserious AEs, SAEs[a], and AESIs	First dose of study drug/treatment.	Up to 30 days after last dose of study drug/treatment or initiation of new anticancer therapy, whichever occurs first.

Abbreviations: AE, adverse event; AESI, adverse event of special interest; SAE, serious adverse event.
[a] All SAEs considered related to the study drug/treatment that are brought to the attention of the investigator should be reported regardless of time since the last dose of treatment.

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8.4.2. Method of Detecting Adverse Events

Care will be taken not to introduce bias when detecting adverse events and/or serious adverse events. Open-ended and nonleading verbal questioning of the patient is the preferred method to inquire about adverse event occurrences.

The investigator or designee may ask patients about adverse events. Questions include, but are not limited to, the following:

- How are you feeling?
- Have you had any medical problems since your last visit?
- Have you taken any new medicines since your last visit?

8.4.3. Follow-up of Adverse Events, Serious Adverse Events and Adverse Events of Special Interest

After the initial adverse event, serious adverse event or adverse event of special interest (as defined in Section 8.4.8) report, the investigator is required to proactively follow each patient at subsequent visits/contacts and provide further information to the sponsor on the patient's condition.

All adverse events, serious adverse events or adverse events of special interest will be followed until event is resolved, stabilized, or otherwise explained, or the patient is lost to follow-up (as defined in Section 7.3). The investigator will ensure that follow-up includes any supplemental investigations as may be indicated to elucidate the nature and/or causality of the adverse event, serious adverse event or adverse event of special interest. This may include additional laboratory tests or investigations, histopathological examinations, radiographic imaging, or consultation with other health care professionals.

The sponsor may request that the investigator perform or arrange for the conduct of supplemental measurements and/or evaluations to elucidate as fully as possible the nature and/or causality of the adverse event, serious adverse event and adverse event of special interest. The investigator is obligated to assist. If a patient dies during participation in the study or during a recognized follow-up period, the sponsor will be provided with a copy of any postmortem findings, including histopathology.

New or updated information should be reported to the sponsor according to the serious adverse event adverse event of special interest instructions provided by the sponsor within the timeframes outlined in Appendix 2.

8.4.4. Regulatory Reporting Requirements for Serious Adverse Events

- Prompt notification by the investigator to the sponsor of an SAE is essential so that legal obligations and ethical responsibilities towards the safety of patients and the safety of the study drug/treatment under clinical investigation are met.
- The sponsor has a legal responsibility to notify both the local regulatory authority and other regulatory agencies about the safety of a study drug/treatment under clinical investigation. The sponsor will comply with country-specific regulatory requirements relating to safety reporting to the regulatory authority, IRBs/IECs, and investigators.

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- An investigator who receives an investigator safety report describing a serious adverse event or other specific safety information (eg, summary or listing of serious adverse events) from the sponsor will review and then file it along with the IB/IDFU/prescribing information or state other documents and will notify the IRB/IEC, if appropriate according to local requirements.
- Investigator safety reports must be prepared for SUSARs according to local regulatory requirements and sponsor policy and forwarded to investigators, as necessary.

8.4.5. Pregnancy

- Details of all pregnancies in female patients and, if indicated, female partners of male patients will be collected after the start of study drug/treatment and until time period for reporting pregnancies should align with the time period for post study drug/treatment contraception determined in Section 5.1.
- If a pregnancy is reported, the investigator will record the pregnancy information on the appropriate form and submit it to the sponsor within 24 hours of learning of the pregnancy of the female patient or female partner of male patient (after obtaining the necessary signed informed consent from the female partner).
- While pregnancy itself is not considered to be an adverse event or serious adverse event, any pregnancy complication or elective termination of a pregnancy for medical reasons will be reported as an adverse event or serious adverse event.
- Abnormal pregnancy outcomes (eg, spontaneous abortion, fetal death, stillbirth, congenital anomalies, ectopic pregnancy) are considered serious adverse events and will be reported as such.
- The patient/pregnant female partner will be followed to determine the outcome of the pregnancy. The investigator will collect follow-up information on the patient/pregnant female partner and the neonate and the information will be forwarded to the sponsor.
- Any poststudy pregnancy-related serious adverse event considered reasonably related to the study drug/treatment by the investigator will be reported to the sponsor as described in Section 8.4.4. While the investigator is not obligated to actively seek this information in former study patients/pregnant female partners, they may learn of a serious adverse event through spontaneous reporting.
- Any female patient who becomes pregnant while participating in the study will discontinue study drug/treatment or be withdrawn from the study or may request continuation of study drug/treatment.

Before continuation of study drug/treatment after pregnancy, the following must occur:

- The sponsor and the relevant IRB/IEC give written approval.
- The patient gives signed informed consent.

- The investigator agrees to monitor the outcome of the pregnancy and the status of the patient and her offspring.

8.4.6. Death on Study

For this study, mortality is an efficacy endpoint. Deaths that occur during the protocol-specified adverse event reporting period (see Section 8.4.1) that are attributed by the investigator solely to progression of underlying disease should be recorded on the Death eCRF.

For all other deaths that occur during the study safety reporting period (refer to Section 8.4.1), regardless of relationship to study drug/treatment, the primary cause of death must be recorded as a fatal serious adverse event on the Adverse Event eCRF and be immediately reported to the sponsor (see Appendix 2).

Death should be considered an outcome and not a distinct event. The event or condition that caused or contributed to the fatal outcome should be recorded as the single medical concept on the Adverse Event eCRF. Generally, only one such event should be reported. If the cause of death is unknown or if it is a sudden death, then the death could be recorded as an adverse event as "unknown cause of death" or "sudden death."

8.4.7. Disease Progression

Progression of an underlying malignancy and related symptoms (including fatal outcomes) is not reported as an adverse event if it is clearly consistent with the expected pattern of progression of the underlying cancer. In most cases, the expected pattern of progression will be based on Hallek's and Cheson's guidelines. In rare cases, the determination of clinical progression will be based on symptomatic deterioration. However, every effort should be made to document progression through the use of objective criteria. If there is any uncertainty as to whether an event is due to disease progression, it should be reported as an adverse event during the adverse event recording period on the Adverse Event eCRF, and immediately reported to the sponsor if assessed as a serious adverse event (refer to Appendix 2).

The term "disease progression" should not be used when reporting adverse events or serious adverse events; instead, associated symptoms should be reported. However, in cases of fatal events clearly related to the progression of disease under study, the terms "malignant neoplasm progression" or "disease progression" are acceptable when the immediate cause of death is not known.

8.4.8. Adverse Events of Special Interest

An adverse event of special interest (serious or nonserious) is an adverse event of scientific and medical concern specific to the sponsor's product or program, for which ongoing monitoring and rapid communication by the investigator to the sponsor could be appropriate. Such an event might require further investigation to characterize and understand it. Depending on the nature of the event, rapid communication by the study sponsor to other parties (eg, regulators) might also be warranted.

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Adverse events of special interest may warrant collection of additional information across the entire study population to better characterize these events (eg, particular laboratory parameters, vital signs, risk factors, concomitant therapies, and/or concomitant illnesses).

Adverse events of special interest (either serious or nonserious) are required to be reported by the investigator to the sponsor immediately (ie, no more than 24 hours after learning of the event; see Appendix 2 for reporting instructions). Adverse events of special interest for this study are as follows:

1. **Potential Hy's law cases**, as defined below in the absence of cholestasis or other causes of hyperbilirubinemia:
   - Treatment-emergent ALT or AST > 3 x ULN in combination with total bilirubin > 2 x ULN, or
   - Treatment-emergent ALT or AST > 3 x ULN in combination with clinical jaundice
2. **TLS** based on Howard criteria. TLS is a disorder characterized by metabolic abnormalities that result from a spontaneous or therapy-related cytolysis of tumor cells.
   - **Laboratory TLS** is an abnormality in ≥ 2 of the following, present simultaneously: uric acid > 8 mg/dL, potassium > 6 meq/L, phosphate > 4.5 mg/dL, or calcium< 7 mg/dL occurring within 3 days before or 7 days after study treatment.
   - **Clinical TLS** is defined as laboratory TLS plus ≥ 1 of the following:
     – Increased serum creatinine (1.5 x ULN)
     – Cardiac arrhythmia or sudden death
     – Any symptomatic hypocalcaemia
     – Seizure

8.5. Patient-Reported Outcomes

PROs will continue to be assessed until death, withdrawal of consent, and study discontinuation. Patients will be asked to complete the EORTC QLQ-C30 and QLQ-CLL17 questionnaires per the Schedules of Assessments (Section 1.3.1) before the study drug is administered, any clinical procedures and health-related discussions to ensure objectivity of patients' answers.

EORTC QLQ-C30

The EORTC QLQ-C30 is a standardized PRO questionnaire developed to assess the quality of life (QoL) of cancer patients, which has been translated and validated in over 100 languages and is used in more than 3000 studies worldwide including CLL clinical studies. The EORTC QLQ-C30 contains 30 items that include 5 functional scales (Physical Functioning, Role Functioning, Emotional Functioning, Cognitive Functioning, and Social Functioning), 1 Global Health Status (GHS)/QoL GHS, 3-symptom scales (Fatigue, Nausea/Vomiting, and Pain), and

6-single item scales (Dyspnea, Insomnia, Appetite Loss, Constipation, Diarrhea, and Financial Difficulties; Fayers et al 2001). The recall period is 1 week (the past 7 days). A sample questionnaire is provided in Appendix 3. Higher scores in GHS and functional scales and lower scores in symptom scales indicate better health-related quality of life (HRQoL).

EORTC QLQ-CLL17

The EORTC QLQ-CLL17 is the CLL module of QLQ-C30 consisting of 17 items and comprising 3 scales: symptom burden due to disease and/or treatment (6 items), physical condition/fatigue (4 items), and worries/fears about health and functioning (7 items of which 2 are conditional). The 2 conditional items are about work/education and patients can rate them if they are applicable to them. Items are rated using a 4-point response scale ("not at all," "a little," "quite a bit," and "very much") and the recall period for all items is the past 7 days. Lower scores indicate better HRQoL outcomes. A sample of QLQ-CLL17 is provided in Appendix 3.

8.6. Pharmacokinetics

Blood samples will be collected to characterize the PK of sonrotoclax. Additionally, blood samples may also be used to explore the presence of metabolites.

Blood samples for PK will be collected at the timepoints specified in the SoA (Section 1.3). The samples will be sent to the sponsor or designee for storage and/or analysis.

On all days of PK collection, the time(s) of study drug/treatment administration must be recorded within the eCRF (including the administration before predose PK collection and any other situations where dosing occurred outside the clinic). On the days of PK collection, administration of study drug/treatment must occur under the supervision of the investigator (or designee) after the predose PK sample is obtained.

PK samples will only be collected from sites that are able to adequately follow the sampling, handling, and processing procedures outlined in the laboratory manual. For each sample, the actual time of collection will be captured to the nearest minute in the eCRF and recorded in the database.

Refer to the laboratory manual for instructions regarding PK sample collection, handling, labeling, storage, and shipping of laboratory samples to the central laboratory. Samples will be shipped to the designated bioanalytical laboratory for quantification of the plasma concentration of study drug/treatment using a validated method.

Genetic analyses will not be performed on these plasma/serum/whole blood samples unless consent for this was included in the informed consent.

8.7. Biomarkers

Shipping, storage, and handling of blood for the assessment of biomarkers will be managed through a central laboratory. Refer to the laboratory manual for details of sample handling and the SoA Section 1.3) for sample collection timepoints.

BeiGene may store some samples for the shorter period of up to 10 years after the end of the study or as allowed by the IRB/IEC or applicable law to achieve the study objectives. Additionally, with

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separate written informed consent, samples may be used for further research by the sponsor or other institutions such as universities or other companies to contribute to the understanding of CLL, the investigational agents, the development of related or new treatments, or research methods.

8.7.1. Del17p, TP53 mutation for exclusion and IGHV mutation for stratification

CLL cytogenetics (del17p, del11q, del13q, 12+) determined by Vysis FISH assay and *TP53* mutation determined by NGS in central lab will be used to exclusion criteria and *IGHV* mutation determined by NGS will be used to guide patient stratification.

8.7.2. Blood Biomarkers

Blood samples will be collected at specified times as described in the SoA (Section 1.3).

Blood-based biomarkers may include but are not limited to CLL cytogenetics (del17p, del11q, del13q, 12+), genetic alteration profile (mutational status of relevant genes such as IGHV, TP53 and genes related to CLL prognosis and resistance to BTK or Bcl-2 inhibitor), gene expression profile, complex karyotype. Other assessments may be conducted as allowed by local regulations. Analyses may include the association between these biomarkers and clinically relevant outcomes including response, resistance and prognosis.

8.8. Pharmacodynamics

Pharmacodynamics are not evaluated in this study.

8.9. Genetics

Refer to Section 8.7 biomarker for details.

8.10. Immunogenicity Analyses

Immunogenicity is not evaluated in this study.

8.11. Health Economics OR Medical Resource Utilization and Health Economics

Not applicable

8.12. Visit Windows

All visits must occur within the visit window indicated in the Schedules of Assessments tables (Section 1.3). All assessments will be performed on the day of the specified visit, unless an acceptable time window is specified. Assessments scheduled on the day of study treatment administration (Day 1) of each cycle should be performed before study treatment administration, unless otherwise noted.

If the timing of a protocol-mandated study visit coincides with a holiday, weekend, or other event, the visit should be scheduled on the nearest feasible date (the visit window is provided in the SoA; Section 1.3). The rescheduling due to a weekend or holiday is not considered a protocol deviation.

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A cycle is defined as every 28 days. Subsequent visits should be conducted according to the planned schedule every 28 days from Day 1 of Cycle 1.

8.13. Unscheduled Visit

Unscheduled visits may be performed at any time at the patient's or the investigator's request and may include vital signs/focused physical examination; ECOG Performance Status; adverse event review; concomitant medications and procedures review; radiographic assessments; physical examination of liver, spleen, and lymph nodes; disease-related constitutional symptoms; and safety laboratory assessments, eg, hematology and chemistry laboratory assessments etc. The date and reason for the unscheduled visit must be recorded in the source documentation.

If an unscheduled visit is necessary to assess toxicity or for suspected progressive disease, then diagnostic tests may be performed based on the investigator assessment as appropriate, and the results of diagnostic tests and images should be entered on the unscheduled visit eCRF.

8.14. Safety Follow-up Visit

All patients who permanently discontinue study drug will have a Safety Follow-up Visit approximately 30 days (± 7 days) after the last dose of study drug or before initiation of a new anticancer therapy, whichever comes first to collect adverse events, including adverse events that may have occurred or been ongoing after the patient discontinued study treatment. The investigator or his/her designee will continue to collect information on new anticancer therapy given after the last dose of study drug. Refer to the Schedules of Assessments (Section 1.3) for the assessments to be performed at the Safety Follow-up Visit.

8.15. Post-treatment Follow-up Period

Patients will be evaluated for response until unequivocal disease progression has been assessed, or before the start of next line anti-CLL treatment, whichever occurs first.

The schedule of visits and procedures for PTFU Period are outlined in the Schedules of Assessments (Section 1.3).

8.16. Survival Follow-up Period

Patients in Survival Follow-up will be followed for survival after unequivocal disease progression has been assessed or start of next line anti-CLL treatment. Contact may be a visit or in person or via phone (with the patient's guardian, if applicable). During the survival follow-up period, subsequent new anticancer treatment information, including the reason for starting CLL anticancer treatment, name of each medication along with start date and end date, and date of progression after receiving the subsequent new anticancer treatment will be collected. In addition, data regarding second primary malignancy and death will be collected.

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9. STATISTICAL CONSIDERATIONS

Methodology for summary and statistical analyses of the data collected in this study is outlined here and further detailed in an SAP. The SAP will be developed and finalized before any analyses are performed and will describe the analyses and procedures for accounting for missing, unused, and spurious data.

9.1. Statistical Hypotheses

Hypothesis of Primary Estimand for Primary Analysis:

The null and alternative hypotheses corresponding to the primary estimand for primary analysis are as follows:

- Null hypothesis: CR/CRi rate of Arm A ≤ CR/CRi rate of Arm B
  versus
- Alternative hypothesis: CR/CRi rate of Arm A > CR/CRi rate of Arm B.

Hypothesis of Primary Estimand for Key Secondary Analysis:

The null and alternative hypotheses corresponding to the primary estimand for key secondary analysis are as follows:

- Null hypothesis: uMRD4 rate of Arm A ≤ uMRD4 rate of Arm B
  versus
- Alternative hypothesis: uMRD4 rate of Arm A > uMRD4 rate of Arm B

9.1.1. Multiplicity Adjustment

To maintain the overall Type I error rate at a one-sided 2.5% level, the multiplicity arising from testing primary and key secondary endpoints will be controlled using the fixed-sequence procedure (Dmitrienko and D'Agostino 2013). The multiplicity adjustment will be as follows:

- The primary endpoint of CR/CRi will be tested at one-sided alpha of 0.025.
- The key secondary endpoint of uMRD4 will be tested at one-sided alpha of 0.025 only if the primary endpoint, CR/CRi, reaches statistical significance. The analysis timing for the primary endpoint of CR/CRi and the key secondary endpoint of uMRD4 will be the same.

9.2. Analysis Sets

For the purposes of analysis, the following analysis sets are defined:

Patient Analysis Set	Description
Intent-to-Treat Analysis Set	All randomized patients.
Safety Analysis Set	All patients who are exposed to investigational study drug(s).
PK Analysis Set	All patients in Arm A who received ≥ 1 dose of sonrotoclax and provided ≥ 1 valid postbaseline PK sample.

Abbreviation: PK, pharmacokinetics.

The Intent-to-Treat Analysis Set will be used to analyze endpoints related to the efficacy objectives and the safety analysis set will be used to analyze the endpoints and assessments related to safety.

For the efficacy analyses, patients will be included in the analyses according to the planned investigational treatment, whereas for safety analyses, patients will be included in the analyses according to the investigational treatment they actually received.

The PK Analysis Set will be used for PK analyses.

9.3. Statistical Analyses

9.3.1. General Considerations

The data analysis general considerations include general definitions and analysis strategies, more details about analysis methods, conventions and handling of missing data will be provided in the SAP.

Data from all patients who signed the main informed consent will be used in the analysis. All statistical analyses will be performed using all data collected in the database up to the data cutoff date.

The patient disposition will be summarized from the ITT analysis set. The reason for discontinuation will be summarized by treatment arm.

Data will be summarized by the following treatment arms:

- Arm A: Sonrotoclax+Zanubrutinib fixed duration therapy
- Arm B: Zanubrutinib continuous therapy

Quantitative data will be described by standard descriptive statistics (eg, mean, standard deviation, minimum, and maximum), and qualitative data will be summarized by frequency tables with number and proportion in each category. A missing category will be included as applicable.

Time-to-event will be summarized graphically using Kaplan-Meier plots for each of the treatment groups. The median and other quartiles will be estimated using the Kaplan-Meier method. The 2-sided 95% CIs will be constructed with the generalized Brookmeyer and Crowley method (Brookmeyer and Crowley 1982). Event-free rates at selected timepoints will be estimated using the Kaplan-Meier method with corresponding 95% CI constructed using Greenwood's formula (Greenwood 1926).

All p-values will be one-sided and CIs will be 2-sided 95% CIs.

9.3.2. Patient Demographics and Other Baseline Characteristics

Demographics and other baseline data including disease characteristics will be listed and summarized descriptively by treatment arm.

Relevant medical histories and current medical conditions at baseline will be summarized by System Organ Class and PT.

For further details, refer to the SAP.

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9.3.3. Primary Estimand Analysis

9.3.3.1. Definition of Endpoint

The primary objective of the study is to compare efficacy between Arm A (sonrotoclax plus zanubrutinib) versus Arm B (zanubrutinib) measured by the proportion of patients that achieve a complete response (CR) or complete response with incomplete bone marrow recovery (CRi) per the IRC assessment. The primary estimand is defined in Section 3.

9.3.3.2. Main Analytical Approach

The null and alternative hypotheses corresponding to the primary estimand for primary analysis are as follows:

- Null hypothesis: CR/CRi rate of Arm A ≤ CR/CRi rate of Arm B
  versus
- Alternative hypothesis: CR/CRi rate of Arm A > CR/CRi rate of Arm B.

The CR/CRi rates will be compared based on a Cochran-Mantel-Haenszel test adjusted for the randomization factors (age [age < 65 years vs age ≥ 65 years] and IGHV status [mutated vs unmutated]) per IRT in the ITT analysis set. If the one-sided p value is < 0.025, it will be concluded that sonrotoclax plus zanubrutinib results in a statistically significant increase in CR/CRi rate versus zanubrutinib monotherapy in patients with previously untreated CLL.

A Mantel-Haenszel common risk difference (Mantel and Haenszel 1959) in CR/CRi rates between arms will be constructed, along with its 95% confidence interval constructed using a normal approximation and Sato's standard error (Sato 1989) stratified by the same randomization factors.

The primary analysis will be performed at the time when all randomized patients have the opportunity to complete 15 cycles of treatment or follow-up. All efficacy and safety endpoints will be analyzed at the time of the primary analysis.

9.3.3.3. Handling of Intercurrent Events of Primary Estimand

Handling of ICEs for primary analysis of CR/CRi:

- New anticancer therapy started before PTFU 1 for Arm A or C16D1 for Arm B:
  Patients starting any new anticancer therapy without achieving a CR or CRi before completing Cycle 15 of treatment will be considered as non-responders (composite strategy).
- Discontinuation of treatment before PTFU 1 for Arm A or C16D1 for Arm B:
  Response assessment after discontinuation of treatment before completion of Cycle 15 will be counted and used for analysis (treatment policy strategy).

9.3.3.4. Handling of Missing Values Not Related to Intercurrent Events

Patients who had no postbaseline response assessment by PTFU 1 for Arm A and C16D1 for Arm B, will be considered as nonresponders.

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9.3.3.5. Sensitivity Analysis

Sensitivity analyses will be detailed in the SAP.

9.3.3.6. Supplementary Analysis

Supplementary analyses if any will be detailed in the SAP.

9.3.3.7. Subgroup Analysis

Subgroup analyses will be detailed in the SAP.

9.3.4. Key Secondary Estimand Analysis

The key secondary objective is to compare efficacy between Arm A (sonrotoclax + zanubrutinib) versus Arm B (zanubrutinib) by rate of undetectable measurable residual disease at $< 10^{-4}$ sensitivity (uMRD4). The estimand for the key secondary objective is defined in Section 3.

The key secondary endpoint will be tested only if the primary endpoint of CR/CRi reaches statistical significance. The null and alternative hypotheses corresponding to the key secondary endpoint are as follows:

- Null hypothesis: uMRD4 rate of Arm A ≤ uMRD4 rate of Arm B
  versus
- Alternative hypothesis: uMRD4 rate of Arm A > uMRD4 rate of Arm B

The uMRD4 rate will be compared based on a Cochran-Mantel-Haenszel test adjusted for the randomization factors (age [age < 65 years vs age ≥ 65 years] and IGHV status [mutated vs unmutated]) per IRT in the ITT analysis set. If the one-sided p value is < 0.025, it will be concluded that sonrotoclax plus zanubrutinib results in a statistically significant increase in uMRD4 rate compared to zanubrutinib monotherapy in patients with previously untreated CLL.

A Mantel-Haenszel common risk difference (Mantel and Haenszel 1959) in uMRD4 rate will be estimated, along with its 95% confidence interval constructed by a normal approximation and Sato's variance estimator (Sato 1989) with the same stratification factors.

The analysis of uMRD4 in all patients will account for the following intercurrent events:

- New anticancer therapy started before MRD assessment at PTFU 1 for Arm A or C16D1 for Arm B: Patients starting any new anticancer therapy before the MRD assessment at PTFU 1 for Arm A or C16D1 for Arm B will be considered as not achieving uMRD4 (composite strategy).
- Discontinuation of treatment before MRD assessment at PTFU 1 for Arm A or C16D1 for Arm B: MRD data collected after discontinuation of treatment will be counted and used for analysis (treatment policy strategy).
- Disease progression or death before MRD assessment at PTFU 1 for Arm A or C16D1 for Arm B: Patients with disease progression or die before the MRD assessment will be considered as not achieving uMRD4 (composite strategy).

Patients who do not have the MRD assessment will be considered as not achieving uMRD4.

For multiplicity adjustment, refer to Section 9.1.1.

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9.3.5. Secondary Endpoints Analyses

9.3.5.1. Secondary Efficacy Endpoint Analyses

CR/CRi Rate per Investigator Response Assessment

CR/CRi rate (CRR) is defined as the proportion of patients with best overall response of CR or CRi up to the PTFU 1 for Arm A and C16D1 for Arm B. CRR per the investigator assessment will be summarized in the ITT analysis set. A 2-sided Clopper-Pearson 95% CI of CRR will be constructed to assess the precision of the point estimate of CRR.

Overall Response Rate per IRC and Investigator Response Assessment

Overall response rate (ORR) is defined as the proportion of patients achieving overall response (CR+CRi+PR+nodular PR). ORR per the IRC and the investigator response assessment will be summarized in the ITT analysis set. A 2-sided Clopper-Pearson 95% CI of ORR will be constructed to assess the precision of the point estimate of ORR.

Landmark Duration of Response Event-free Rate at 12 Months per IRC Response Assessment

DOR is defined as the time from the first determination of response until first documentation of progression or death, whichever occurs first. The 12-month landmark DOR event-free rate is defined as the proportion of responders who remain alive and progression-free at 12 months since the first determination of response. Landmark DOR event-free rate at 12 months per the IRC response assessment will be summarized among the responders in the ITT analysis set. The Kaplan-Meier estimate of the 12-month landmark DOR event-free rate and its 2-sided 95% CIs will be provided. The DOR censoring rule will follow US FDA Guidance for Industry Clinical Trial Endpoints for the Approval of Cancer Drugs and Biologics (US FDA 2018).

Duration of Response per Investigator Response Assessment

DOR is defined as the time from the first determination of response until first documentation of progression or death, whichever occurs first. DOR per the investigator response assessment will be summarized among the responders in the ITT analysis set. The Kaplan-Meier estimates of median DOR time among responders and the landmark estimates of DOR event-free rate with 2-sided 95% CIs will be provided. The DOR censoring rule will follow US FDA Guidance for Industry Clinical Trial Endpoints for the Approval of Cancer Drugs and Biologics (US FDA 2018).

Time to Response per IRC and Investigator Response Assessment

Time to Response (TTR) is defined as the time from treatment initiation to the first documentation of response. TTR per the IRC and the investigator response assessment will be summarized among the responders in the ITT analysis set. It will be summarized descriptively with sample statistics (mean, median, range) for responders only.

Landmark Progression-Free Survival and Overall Progression-Free Survival per Investigator Response Assessment

Progression-free survival (PFS) is defined as the time from the start of treatment to the first documentation of disease progression or death, whichever occurs first. The 24-month landmark PFS rate is defined as the proportion of patients who remain alive and progression-free at 24

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months since the start of treatment. PFS per the investigator response assessment will be analyzed in the ITT analysis set. Kaplan-Meier estimates of median PFS time and the landmark estimates with 2-sided 95% CIs will be provided. The PFS censoring rule will follow US FDA Guidance for Industry Clinical Trial Endpoints for the Approval of Cancer Drugs and Biologics (US FDA 2018).

Overall Survival

Overall survival (OS) is defined as the time from treatment initiation to death due to any cause. OS will be analyzed in the ITT analysis set using similar methods to those described for PFS similarly, except for censoring rules. For OS, patients will be censored either at the date that the patient was last known to be alive or the date of data cutoff, whichever comes earlier, in the absence of death.

More details on the definitions and analyses of secondary endpoints will be provided in the SAP.

9.3.5.2. Analyses of Secondary Safety Endpoints

Safety will be assessed by monitoring and recording of all treatment emergent AEs graded by NCI-CTCAE v5.0 and the iwCLL Grading Scale for Hematologic Toxicities in CLL Studies as appropriate. TLS will be monitored per Howard criteria. Treatment emergent AEs ≥ Grade 3, SAEs, treatment related treatment-emergent AEs, treatment-emergent AEs (including relativeness to the treatments) that led to treatment discontinuation, dose reduction, dose interruption, and death will be summarized. Laboratory values (complete blood count [CBC], serum chemistry, coagulation and urinalysis), vital signs, physical examinations and ECG findings will also be used in assessing safety. Descriptive statistics will be used to analyze all safety data in the Safety Analysis Set.

Extent of Exposure

Extent of exposure to each study drug will be summarized descriptively as the number of cycles received (number and percentage of patients), duration of exposure (days), cumulative total dose received per patient (mg), dose intensity (mg/day), and relative dose intensity.

The number (and percentage) of patients with dose reductions, treatment interruptions, and study drug discontinuation will be summarized with the respective reasons.

Concomitant Medication

Concomitant medications before and after the start of the study treatment will be summarized according to the ATC classification system.

Adverse Events

Adverse events will be graded by the investigators using NCI-CTCAE v5.0 and the iwCLL Grading Scale for Hematologic Toxicities in CLL Studies as appropriate. The adverse event verbatim descriptions (investigator terms from the eCRF) will be classified into standardized medical terminology using MedDRA. Adverse events will be coded to the MedDRA (Version 20.0 or higher) lowest level term closest to the verbatim term, along with the linked MedDRA PT and primary System Organ Class.

A treatment-emergent adverse event is defined as an adverse event that had onset or increase in severity level on or after the date of the first dose of study drug through 30 days after the permanent

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discontinuation of study drug or the initiation of new anticancer therapy for CLL, whichever is earlier. Worsening of a treatment-emergent adverse event to Grade 5 beyond 30 days after the last dose of study drugs is also considered a treatment-emergent adverse event. Summary tables will generally focus on those adverse events that were treatment-emergent. All adverse events, treatment-emergent or otherwise, will be presented in patient data listings.

The incidence of treatment-emergent adverse events will be reported as the number (percentage) of patients with treatment-emergent adverse events by System Organ Class, PT and worst grade. A patient will be counted only once by the highest severity grade per CTCAE v5.0 within a System Organ Class and PT, even if the patient experienced > 1 treatment-emergent adverse event within a specific System Organ Class and PT.

The number (percentage) of patients with treatment-emergent adverse events will also be summarized by relationship to the study drugs. Treatment-related adverse events include those events considered by the investigator to be related to study drug or with missing assessment of the causal relationship.

Serious adverse events, deaths, ≥ Grade 3 treatment-emergent adverse events, treatment-related treatment-emergent adverse events, and treatment-emergent adverse events that led to treatment discontinuation, dose interruption, or dose delay will be summarized. The incidence of clinical TLS events will also be reported.

Laboratory Analyses

Clinical laboratory (eg, hematology, clinical chemistry, coagulation, and urinalysis) values will be evaluated for each laboratory parameter as appropriate. Abnormal laboratory values will be flagged and identified as those outside (above or below) the normal range. Reference (normal) ranges for laboratory parameters will be included in the CSR for this protocol. Descriptive summary statistics (eg, n, mean, standard deviation, median, minimum, and maximum for continuous variables; n [%] for categorical variables) for laboratory parameters and their changes from baseline will be calculated. Laboratory values will be summarized by visit and by worst postbaseline change.

Laboratory parameters that are graded in NCI-CTCAE v5.0 or iwCLL Grading Scale for Hematologic Toxicities in CLL Studies will be summarized by CTCAE grade/iwCLL Grading Scale for Hematologic Toxicities in CLL Studies. In the summary of laboratory parameters by grade, parameters with grading in both high and low directions (eg, calcium, glucose, magnesium, potassium, and sodium) will be summarized separately.

Vital Signs

Descriptive statistics for vital sign parameters (body temperature, pulse rate, respiratory rate, and blood pressure [systolic and diastolic]) and weight as well as their changes from baseline will be presented by visit. Vital signs will be listed by patient and visit.

Electrocardiograms

Actual value and change from baseline for the ECG parameters will be summarized.

Eastern Cooperative Oncology Group (ECOG) Performance Status

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A shift table from baseline to worst postbaseline in ECOG Performance Status will be presented. ECOG status will be summarized by visit.

9.3.5.3. Pharmacokinetics

Sparse PK samples to quantitate sonrotoclax plasma concentrations will be collected from all patients assigned to Arm A. The sonrotoclax PK concentration data collected sparsely at predose and postdose will be tabulated and summarized by visit/cycle. For the sparse PK profile of sonrotoclax, PK parameters such as Ctrough at steady state will be summarized. Additional PK parameters may be calculated if deemed appropriate. PK parameters will be tabulated and summarized using descriptive statistics (eg, n, mean, standard deviation, percent of coefficient of variation [CV%], median, minimum, maximum, geometric mean [GeoMean], and geometric coefficient of variance [GeoCV%]).

Plasma concentration-time data will be summarized by nominal time of collection and displayed in both tabular and graphical form. In addition, PK data from this study, along with data from other studies, may be included in a population PK analysis and reported separately.

Correlation between PK and biomarker endpoints may be explored as appropriate. Results of these analyses, such as exposure-response (efficacy or safety endpoints) analysis, may be carried out if supported by data.

Results from the population PK and exposure-response analyses may be reported separately from the main CSR for this study.

9.3.6. Exploratory Endpoints Analysis

The correlation of clinical outcome (CRR, ORR, DOR, PFS, and OS) and baseline prognostic biomarkers (e.g., del11q, del13q, trisomy 12, CLL-related prognosis mutations) would be analyzed and potential resistance biomarker and mechanism would be identified.

Undetectable MRD at various timepoints assessed by flow cytometry, including duration and time to undetectable MRD, would be summarized.

The correlation between blood delta MRD at C7D1 and bone marrow uMRD4 at PTFU 1 Visit, and the correlation between blood delta MRD at C7D1 and clinical outcomes (CRR, ORR, DOR, PFS and OS) may be reported separately from the CSR as appropriate.

Descriptive statistics will be provided for patient-reported outcomes as measured by EORTC-QLQ-C30 and EORTC-QLQ-CLL17 in the ITT analysis set. A mixed model repeated measures will be performed for key PRO endpoints.

9.3.7. Other Analyses

Not applicable.

9.4. Interim Analysis

No formal interim analysis is planned.

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9.5. Sample Size Determination

Approximately 87 patients will be randomized. It is assumed that the proportion of patients achieving a best response of CR/CRi up to PTFU 1 Visit is 40% in Arm A (Tam et al 2023) and 3% in Arm B up to Day 1 of Cycle 16 (unpublished data from SEQUOIA study). Using the normal approximation method with pooled variance for a one-sided statistical test, a study with an overall sample size of N = 87 patient randomized in a 2:1 ratio (58 patients in Arm A and 29 patients in Arm B) will have over 90% power to detect a treatment difference between the 2 arms at a Type I error level of 2.5%.

Additionally, this sample size will provide over 90% power to demonstrate a difference between Arm A and Arm B for the key secondary endpoint of uMRD4 rate. The power is calculated under the assumptions that the uMRD4 rate is 85% at the PTFU 1 Visit in Arm A (unpublished data from BGB-11417-101 study) and < 1% at Day 1 of Cycle 16 in Arm B (unpublished data from SEQUOIA study).

The sample size is calculated using the EAST software version 6.5.

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10. SUPPORTING DOCUMENTATION AND OPERATIONAL CONSIDERATIONS

10.1. Regulatory and Ethical Considerations

This study will be conducted by the investigator and the study center in accordance with the protocol and the following:

- Consensus ethical principles derived from international guidelines including the Declaration of Helsinki and Council for International Organizations of Medical Sciences international ethical guidelines.
- Applicable ICH GCP guidelines.
- Applicable laws and regulations.

This study will be conducted in compliance with Regulation [EU] No 536/2014.

The sponsor shall notify the member states concerned about a serious breach of Regulation 536/2014 or of the version of the protocol applicable at the time of the breach through the EU portal without undue delay but no later than 7 days after becoming aware of the breach.

Safety reporting via Eudravigilance database will be done in accordance with Article 40 CTR.

10.1.1. Regulatory Authority Review

The sponsor will obtain review of the protocol from the appropriate regulatory agency in accordance with any applicable country-specific regulatory requirements or file the protocol to the appropriate regulatory agency before the study is initiated at a study center in that country.

10.1.2. Institutional Review Board/Independent Ethics Committee

This protocol, the ICFs, any information to be given to the patient, and relevant supporting information must be submitted, reviewed, and approved by the IRB/IEC before the study is initiated. In addition, any patient recruitment materials must be approved by the IRB/IEC. Copies of the IEC/IRB correspondence and approval of the amended ICF/other information and the approved amended ICF/other information must be forwarded to the sponsor promptly.

The investigator/sponsor representative (as applicable) is responsible for providing written summaries of the status of the study to the IRB/IEC annually or more frequently in accordance with the requirements, policies, and procedures established by the IRB/IEC. Investigators/sponsor representative (as applicable) are also responsible for promptly informing the IRB/IEC of any protocol amendments. In addition to the requirements for reporting all adverse events to the sponsor, investigators must comply with requirements for reporting serious adverse events to the local health authority and IRB/IEC. Investigators may receive written IND Safety Reports or other safety-related communications from the sponsor.

The investigator will be responsible for the following, as applicable:

- Providing written summaries of the status of the study to the IRB/IEC annually or more frequently in accordance with the requirements, policies, and procedures established by the IRB/IEC.

- Notifying the IRB/IEC of serious adverse events or other significant safety findings as required by IRB/IEC procedures.

Providing oversight of the conduct of the study at the site and adherence to requirements of 21 Code of Federal Regulations, ICH guidelines, the IRB/IEC, European regulation 536/2014 for clinical studies, European Medical Device Regulation 2017/745 for clinical device research, and all other applicable local regulations.

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10.1.3. Protocol Amendments

Any protocol amendments will be prepared by the sponsor. Protocol modifications must be submitted to competent authorities according to local requirements, when required by the applicable country regulations. All protocol modifications must be submitted to the IRB/IEC together with, if applicable, a revised model ICF in accordance with local requirements. Written documentation from competent authorities (according to local requirements) and from the IRB/IEC and required site approval must be obtained by the sponsor before changes can be implemented, except for changes necessary to eliminate an immediate hazard to patients or changes that involve logistical or administrative aspects only (eg, change in medical monitor or contact information).

Information on any change in risk and/or change in scope must be provided to patients already actively participating in the study, and they must read, understand, and sign each revised ICF confirming their willingness to remain in the study.

10.2. Financial Disclosure

For all clinical investigators, the sponsor is required to disclose the presence or absence of certain financial interests, as required, to the appropriate health authorities. Investigators (including any subinvestigators and coinvestigators) are required to provide the sponsor with complete and accurate financial information in accordance with regulations to allow the sponsor to make these submissions. This is intended to ensure financial interests and arrangements of the clinical investigators with the sponsor that could affect reliability of data submitted to health authorities are identified and disclosed by the sponsor. Investigators (including any subinvestigators and coinvestigators) are responsible for providing information about their financial interests before participation in the study and to update this information if any relevant changes occur during the study and for 1 year after completion of the study (ie, last patient, last visit).

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10.3. Informed Consent Process

Before any data collection under this protocol, written informed consent will be obtained as per local requirements.

The sponsor's sample ICF will be provided to each site. If applicable, it will be provided in a certified translation of the local language. Where the investigator is responsible for submission to the IRB/IEC, the final IRB/IEC-approved ICFs must be provided to the sponsor.

The approved ICFs must be signed and dated by the patient or the patient's legally authorized representative in accordance with the applicable local regulations before their participation in the study. The case history or clinical records for each patient shall document the informed consent

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process and that written informed consent was obtained before participation in the study. The ICF must not be altered without the prior agreement of the relevant IEC and study sponsor.

The ICFs will be revised whenever there are changes to study procedures or when new information becomes available that may affect the willingness of the patient to participate. Where the investigator is responsible for submission to the IRB/IEC, the final revised IRB-/IEC-approved consent forms must be provided to the sponsor.

Patients must be reconsented to the most current approved version of the ICFs (or to a significant new information/findings addendum in accordance with applicable laws and IRB/IEC policy) during their participation in the study. For any updated or revised ICFs, the case history or clinical records for each patient shall document the informed consent process and that written informed consent was obtained using the updated/revised ICFs for continued participation in the study.

A copy of each signed ICF must be provided to the patient or the patient's legally authorized representative in accordance with the applicable local regulations. All signed and dated ICFs must remain in each patient's study file or in the site file and must be available for verification by study monitors at any time.

10.4. Study Committees Structure

Safety Monitoring Committee

An SMC will be established and include both investigators and the sponsor (including the medical monitor and study team members from pharmacovigilance/drug safety, clinical pharmacology, and biostatistics with other members as appropriate). The SMC will review all available safety, efficacy, PK, and exploratory data and make recommendations on the ramp-up schedule dose modification, and further enrollment. The SMC may also be called upon by the sponsor on an ad hoc basis where applicable to the conduct of the study.

Independent Review Committee

An IRC will be established to perform an independent review of all scheduled and relevant unscheduled tumor assessments to determine response and disease progression on the basis of the iwCLL criteria (Hallek et al 2018) with modifications for treatment-related lymphocytosis (Cheson et al 2012; Appendix 7) for patients with CLL. For each of the patient, the IRC review will start from screening up to PTFU 1 (Arm A)/C16D1 Visit (Arm B). The tumor assessment by the IRC will be used for reporting the study results.

Detailed rules and guidelines for radiographic imaging and tumor assessments by the IRC are outlined separately in the Imaging Manual and IRC Charter.

10.5. Confidentiality and Data Protection

Patient Confidentiality and Data Protection

The investigator, institution, sponsor, and site will maintain confidentiality and privacy standards for the collection, storage, transmission, and processing of patients' personal and medical information by following applicable laws and regulations related to the confidentiality, use, and protection of such information, including the ICH GCP Guideline, as implemented locally. Such laws may be more stringent than the requirements in this protocol.

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The investigator and site shall code the personal and medical information obtained during the study with a unique patient identification number assigned to each patient enrolled in the study. The investigator must ensure that patients' confidentiality will be strictly maintained and that their identities are protected from unauthorized parties. Unless required to be provided by laws or regulations or specifically requested in exceptional circumstances by the sponsor or its representatives, the principal investigator and site must ensure that any personal and medical information transmitted to or collected by the sponsor or its service providers is: 1) required by the protocol, and 2) appropriately de-identified (eg, via redaction and/or coding with the patient identification number) to ensure the following information about patients is NOT shared:

- Names or initials (full or partial);
- Full dates of birth;
- Contact information (such as phone numbers or home or email addresses);
- Numerical identifiers (eg, hospital or medical record, government, health insurance, or financial account numbers) other than patient identification numbers assigned as part of this study;
- Geographic identifiers smaller than a state, province, or local equivalent (such as city, county, ZIP code, or other equivalent geographic identifiers); or
- Information about marital status, family, or household members; employment, sex life, sexual orientation, or other sensitive data that is not relevant to the study.

Upon enrollment, patients will be required to provide their names, phone and contact information, and similar information on secondary contacts. This information will be stored separately from the study clinical database. Documents generated by the site that will be shared with the sponsor, such as patient enrollment logs, need to have this information redacted before being shared. This information will only be used to obtain patients vital status and disposition if the patients becomes lost to follow-up or in other specific circumstances specified in the ICF provided to the patient at the time of the collection of informed consent for participation in the study (eg, for purposes of expense reimbursement or mailing the latest version of the ICF to the patients when he or she cannot travel to the site).

Patients' personal and medical information obtained during this study is confidential and may only be disclosed to third parties as permitted by the signed ICF (or a separate authorization for the use and disclosure of personal health information that has been signed by the patient), unless permitted or required by law.

In certain circumstances, such as in connection with insurance purposes or patient support services ancillary to certain study sites (eg, for patient travel or reimbursement), the investigator and site may provide certain uncoded personal information to the sponsor and/or its representatives. Such personal information may not be provided as part of the study protocol (eg, as part of the eCRF, on samples or reports submitted to the central laboratory, on safety reporting forms, or on product dispensing logs provided to the sponsor, etc.).

Investigator and site must use only the specific forms and clinical study systems (eg, the EDC system and any secure file transfer platforms) designated by the sponsor for sharing and transfer

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of personal and medical information. Should a site use its own form, it must be reviewed and redacted in accordance with this section before being shared with the sponsor.

In the event of a breach of the confidentiality of a patient's personal and medical information, the investigator, site, and sponsor, as appropriate, shall fulfill all risk containment and remediation steps and reporting obligations under applicable laws. If the sponsor identifies personal or medical information that was not properly de-identified, it may be required to report the disclosure under local applicable laws.

Medical information may be given to a patient's personal physician or other appropriate medical personnel responsible for the patient's welfare for treatment purposes where allowed by local law or the patient's signed ICF.

Information generated during this study must be available for inspection upon request by representatives of the US FDA, the EMA, the China National Medical Products Administration, and all other national and local health authorities; by sponsor and its designee(s); and by the IRBs/IECs for each study site, as appropriate. Every effort will be made to keep the patients' personal data confidential.

In any presentations or in publications of the results of the study, the participants'/patients' identities will remain anonymous and confidential.

BeiGene Confidential Information

The investigator agrees that all information received from the sponsor, including but not limited to the IB, this protocol, eCRFs, the investigational study drug/treatment, and any other study information, are confidential and remain the sole and exclusive property of the sponsor during the conduct of the study and thereafter. This information is not to be disclosed to any third party (except employees or agents directly involved in the conduct of the study or as required by law) without prior written consent from the sponsor. The investigator further agrees to take all reasonable precautions to prevent the disclosure by any employee or agent of the study site to any third party or otherwise into the public domain.

In cases of discrepancies between the confidentiality or data privacy provisions contained in a written contract for the conduct of the study and this Section 10.5 of the protocol, that contract's provisions shall apply to the extent that they provide for a higher level of protection of patients' rights to confidentiality and data privacy or of BeiGene's confidential information.

10.6. Data Collection and Management Responsibilities

10.6.1. Data Collection and Management Responsibilities

Data required by the protocol will be entered into an EDC system. All study-related data collected or received by the investigator or study team shall be promptly entered into the eCRFs. In no event should the entry of the study data into the eCRF be later than what is stipulated in the site contract after the data are collected or received by the investigator or study team without prior communication with and approval by the sponsor.

Data collection in the eCRF should follow the instructions described in the eCRF Completion Guidelines. The investigator has ultimate responsibility for the collection and reporting of all

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clinical data entered in the eCRF. The e-signature of the investigator or designee must be provided in the EDC system to attest to its accuracy, authenticity, and completeness.

Data contained in the eCRFs are the sole property of the sponsor and should not be made available in any form to third parties without written permission from the sponsor, except for authorized representatives of the sponsor or appropriate regulatory authorities.

10.6.2. Data Management/Coding

All final patient data, both eCRF and external data (eg, laboratory data), collected according to the protocol will be stored by the sponsor at the end of the study.

Standard procedures (including following data review guidelines, computerized validation to produce queries, and maintenance of an audit file that includes all database modifications) will be followed to support accurate data collection. Data will be reviewed for outliers, logic, data inconsistencies, and completeness.

During the study, a study monitor (clinical research associate) will make site visits to review protocol compliance, compare eCRFs against individual patient's medical records, and ensure that the study is being conducted according to pertinent regulatory requirements.

The eCRF entries will be verified with source documentation. The review of medical records will be performed in a manner to ensure that patient confidentiality is maintained. Checking the eCRFs for completeness, clarity, and cross-checking with source documents is required to monitor the progress of the study. Direct access to source data is also required for inspections and audits and will be carried out with due consideration to data protection and medical confidentiality.

Adverse events and concomitant diseases/medical history will be coded using MedDRA. Concomitant medications will be coded using the WHO Drug Dictionary.

10.6.3. Study Records Retention

The investigator must maintain adequate and accurate records to enable the conduct of the study to be fully documented and the study data to be subsequently verified. These documents should be classified into at least 1 of the following 2 categories: 1) investigator's study file and/or 2) patient clinical source documents.

The investigator's study file will contain the protocol/amendments, eCRF and query forms, IRB/IEC and governmental approval with correspondence, ICFs, drug records, staff curriculum vitae and authorization forms, and other appropriate documents and correspondence.

Patient clinical source documents (usually defined by the project in advance to record key efficacy/safety parameters independent of the eCRFs) would include documents such as (although not be limited to) the following: patient hospital/clinic records, physician's and nurse's notes, appointment book, original laboratory reports, ECG, electroencephalogram, x-ray, pathology and special assessment reports, consultant letters, screening and enrollment logs, etc.

After closure of the study, the investigator must maintain all study records in a safe and secure location. The records must be maintained to allow easy and timely retrieval, when needed (eg, audit or inspection), and, whenever feasible, to allow any subsequent review of data in conjunction with assessment of the facility, supporting systems, and personnel. Where permitted

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by local laws/regulations or institutional policy, some or all of these records can be maintained in a format other than hard copy (eg, microfiche, scanned, electronic); however, caution needs to be exercised before such action is taken. The investigator must ensure that all reproductions are legible, are a true and accurate copy of the original, and meet accessibility and retrieval standards, including regenerating a hard copy, if required. Furthermore, the investigator must ensure there is an acceptable backup of these reproductions and that an acceptable quality control process exists for making these reproductions.

The sponsor will inform the investigator of the length of time to retain these documents to comply with all applicable regulatory requirements. The minimum retention time will meet the most rigorous standards applicable to the study by law or by IEC/IRB decisions; otherwise, the retention period will default to at least 25 years, and in any case for at least 15 years.

The investigator must notify the sponsor of any changes in the archival arrangements including but not limited to the following: archival at an off-site facility or transfer of ownership of or responsibility for the records in the event the investigator leaves the study center.

If the investigator cannot guarantee this archiving requirement at the study site for any or all of the documents, special arrangements must be made between the investigator and the sponsor to store these in sealed containers outside of the site so that they can be returned sealed to the investigator in case of a regulatory audit. When source documents are required for the continued care of the patient, appropriate copies should be made for storage outside of the site.

Biological samples processed by the sponsor's designated laboratories may be retained at the conclusion of this study for the shorter of a period of up to 10 years after the end of this study or as allowed by the IRB/IEC or applicable law.

10.7. Protocol Deviations

The investigator is responsible for ensuring that the study is conducted in accordance with this protocol, including carrying out the procedures and evaluations described in this protocol that occur at the sites overseen by the investigator. Investigators assert they will apply due diligence to avoid protocol deviations and shall report all protocol deviations to the sponsor.

The investigator is to document and explain any deviations from the approved protocol. The investigator must promptly report any major deviations that might impact patient safety and/or data integrity to the sponsor and to the IRB/IEC, in accordance with established IRB/IEC policies and procedures.

10.8. Data Quality Assurance

10.8.1. Quality Assurance

To ensure compliance with the GCP and all applicable regulatory requirements, the sponsor may conduct a quality assurance audit. Regulatory agencies may also conduct a regulatory inspection of this study. Such audits and inspections can occur at any time during or after completion of the study. If an audit or inspection occurs, the investigator and institution agree to allow the auditor or inspector direct access to all relevant documents and to allocate his/her time and the time of his/her personnel to the auditor or inspector to discuss findings and any relevant issues.

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10.8.2. Study Site Inspections

This study will be organized, performed, and reported in compliance with the protocol, standard operating procedures, working practice documents, and applicable regulations and guidelines. Site audits may be performed periodically by the sponsor's or the contract research organization's qualified compliance auditing team, which is an independent function from the study team responsible for conducting the study.

Site visits will be conducted by the sponsor or an authorized representative to inspect study data, patients' medical records, and eCRFs. The investigator is to permit national and local health authorities; sponsor study monitors, representatives, and collaborators; and IRB/IEC members to inspect all facilities and records relevant to this study.

10.8.3. Drug Accountability

The investigator or designee (ie, pharmacist) shall take reasonable steps to ensure that it has adequate unexpired supplies of each study drug and is responsible for ensuring adequate accountability of all used and unused study drug. This includes acknowledging receipt of each shipment of study product (quantity and condition), patient drug dispensation records, and returned or destroyed study product. Dispensation records will document quantities received from the sponsor's designated depot or its designee and quantities dispensed to patients, including the batch or lot number, date dispensed, patient identifier number, and the initials of the person dispensing the medication.

At study initiation, the monitor will evaluate the site's standard operating procedure for study drug disposal or destruction to ensure that it complies with the sponsor's requirements specified in the Pharmacy Manual. At appropriate times during the conduct of the study or at the end of the study after final drug inventory reconciliation by the monitor, the study site will dispose of and/or destroy all unused study drug supplies, including empty containers, according to these procedures. If the site cannot meet the sponsor's requirements specified in the Pharmacy Manual for disposal, arrangements will be made between the site and the sponsor or its representative for destruction or return of unused study drug supplies.

All drug supplies and associated documentation will be reviewed periodically and verified by the study monitor over the course of the study.

10.9. Source Documents

The investigator must maintain adequate and accurate records to ensure that the conduct of the study may be fully documented. Such records include but are not limited to the protocol, protocol amendments, ICFs, and documentation of IRB/IEC and governmental approvals. In addition, at the end of the study, the investigator will receive patient data, which will include an audit trail containing a complete record of all changes to such data.

10.9.1. Access to Information for Monitoring

In accordance with ICH GCP guidelines, the study monitor must have direct access to the investigator's source documentation to verify the data recorded in the eCRFs for consistency.

The monitor is responsible for routine review of the eCRFs at regular intervals throughout the study to verify adherence to the protocol and the completeness, consistency, and accuracy of the data being entered on them. The monitor should have access to any patient records needed to verify the entries in the eCRFs. The investigator agrees to cooperate with the monitor to ensure that any problems detected during these monitoring visits are resolved.

10.9.2. Access to Information for Auditing and Inspections

Representatives of regulatory authorities or of the sponsor may conduct inspections or audits any time during or after completion of this clinical study. If the investigator is notified of an inspection by a regulatory authority, the investigator agrees to notify the sponsor or its designee immediately. The investigator agrees to provide to representatives of a regulatory agency or representatives of the sponsor access to records, facilities, and personnel for the effective conduct of any inspection or audit.

10.10. Site Closure

Upon completion of the study, the monitor will conduct the following activities in conjunction with the investigator or study center personnel, as appropriate:

- Return/provide all study data to the sponsor;
- Resolution and closure of all data queries;
- Accountability, reconciliation, and arrangements for unused study drug(s);
- Review of study records for completeness;
- Collection of all study documents for the study master file filing according to GCP and local regulation; and
- Shipment of samples (including but not limited to those for PK, and biomarkers) to the designated laboratory for analysis according to protocol and laboratory manual requirements.

In addition, the sponsor reserves the right to suspend the enrollment or prematurely discontinue this study either at a single study center or at all study centers at any time for any reason. Potential reasons for suspension or discontinuation include but are not limited to: safety or ethical issues or noncompliance with this protocol, GCP, the sponsor's written instructions, the clinical study agreement, or applicable laws and regulations. If the sponsor determines such action is needed, the sponsor will discuss this with the investigator (including the reasons for taking such action) at that time. When feasible, the sponsor will provide advance notification to the investigator of the impending action before it takes effect.

The sponsor will promptly inform all other investigators and/or institutions conducting the study if the study is suspended or terminated for safety reasons. The sponsor will also inform the regulatory authorities of the suspension or termination of the study and the reason(s) for the action. If required by applicable regulations, the investigator must inform the IEC/IRB promptly and provide the reason for the suspension or termination.

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If the study is prematurely discontinued, all study data must still be provided to the sponsor. In addition, arrangements will be made for the return of all unused study drug/treatment in accordance with the applicable sponsor procedures for the study.

Financial compensation to the investigators and/or institutions will be in accordance with the clinical study agreement established between the investigator and/or institutions and the sponsor.

10.11. Publications and Data Presentation Policy

A CSR will be prepared and provided to the regulatory agency(ies). The sponsor will ensure that the report meets the standards set out in the ICH Guideline for Structure and Content of CSRs (ICH E3).

The results of this study will be published or presented at scientific meetings in a timely, objective, and clinically meaningful manner that is consistent with good science, industry and regulatory guidance, and the need to protect the intellectual property of the sponsor, regardless of the outcome of the study. The data generated in this clinical study are the exclusive property of the sponsor and are confidential. For a multicenter study, the first publication or disclosure of study results shall be a complete, joint multicenter publication, or disclosure coordinated by the sponsor. Thereafter, any secondary publications will reference the original publication(s). Authorship will be determined by mutual agreement and all authors must meet the criteria for authorship established by the ICMJE Uniform Requirements for Manuscripts or stricter local criteria.

After the conclusion of the study, investigators may communicate, orally present, or publish in scientific journals or other scholarly media the results of the study only after the following conditions have been met:

- results of the study in their entirety have been publicly disclosed by or with the consent of the sponsor in an abstract, manuscript, or presentation form, including all disclosure requirements for clinical study registries have been met; or
- if there is no such a publication within 2 years of study completion.

Each investigator agrees to submit all manuscripts, abstracts, posters, publications, and presentations (both oral and written) to the sponsor for review before submission or presentation in accordance with the clinical study agreement. This allows the sponsor to protect proprietary information, provide comments based on information from other studies that may not yet be available to the investigator, and ensure scientific and clinical accuracy. For details of the submission and review processes see your clinical study agreement. Each investigator agrees that, in accordance with the terms of the clinical study agreement, a further delay of the publication/presentation may be requested by the sponsor to allow for patent filings and/or protection in advance of the publication/presentation.

Within one year from the end of the clinical study in all member states concerned, the sponsor shall submit to the EU database a summary of the results of the clinical study as outlined in CTR Annex IV.

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10.12. Data Disclosure and Transparency and Data Sharing With External Researchers

BeiGene publicly discloses study drug/treatment clinical studies in compliance with BeiGene policy and applicable regulations. When the study is completed and analysis is conducted, results are published on clinical study databases like https://www.ClinicalTrials.gov and BeiGene's Clinical Studies webpage. The facility name, contact information, and other study information for this study may be searched by entering the protocol ID.

BeiGene may make redacted clinical study documents and de-identified patient-level data from this study available to external researchers for scientific analyses or to conduct further research that may help advance medical science and lead to better research and treatments for all people. "De-identified" means that all direct and indirect identifiers, including the study patient identification numbers, have been removed before the data are made available. "Redacted" documents means that personal data and commercially confidential information are no longer visible.

Information Disclosure and Inventions

All rights, title, and interests in any inventions, knowledge, or other intellectual or industrial property rights that are conceived or reduced to practice by the study center personnel during the course of or because of the study are the sole property of the sponsor and are hereby assigned to the sponsor.

If a written contract for the conduct of the study, which includes ownership provisions inconsistent with this statement, is executed between the sponsor and the study center or investigator, that contract's ownership provisions shall apply rather than this statement.

All information provided by the sponsor and all data and information generated by the study center as part of the study (other than a patient's medical records) are the sole property of the sponsor and will be kept confidential by the investigator and other study center personnel.

This information and data will not be used by the investigator or other study center personnel for any purpose other than conducting the study without the prior written consent of the sponsor.

These restrictions do not apply to:

- Information that becomes publicly available through no fault of the investigator or study center personnel.
- Information that is necessary to disclose in confidence to an IEC/IRB solely for the evaluation of the study.
- Information that is necessary to disclose to provide appropriate medical care to a patient.
- Study results that may be published as described in Section 10.11 and Section 10.12.

If a written contract for the conduct of the study, which includes provisions inconsistent with this statement is executed, that contract's provisions shall apply rather than this statement.

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12. APPENDICES

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APPENDIX 1. CONTRACEPTIVE AND BARRIER GUIDANCE

Contraception Guidelines

All male patients with partners of childbearing potential who take part in this study must use condoms during sexual intercourse in addition to 1 of the highly effective methods of contraception listed below, from the time of taking the first dose of study drug until 90 days after taking the last dose of study drug. If a female partner of a male patient is already pregnant, the male patient must use condoms during sexual intercourse for the duration of the study and for ≥ 90 days after the last dose of study drug. Nonsterile male patient must refrain from sperm donation for the duration of the study and for 90 days after the last dose of study drug/treatment.

Female patients of childbearing potential must use 1 of the highly effective forms of birth control listed below (preferably those with low user dependency) for the duration of the study and for 90 days after the last dose of study drug/treatment. Nonsterile female patient must refrain from egg donation for the duration of the study and for 90 days after the last dose of study drug/treatment.

Participating women of childbearing potential should be placed on effective birth control as soon as medically reasonable and before initiation of study drug.

The CTFG's recommendations related to contraception and pregnancy testing in clinical studies include the use of highly effective forms of birth control. These methods include the following:

- Combined (estrogen- and progestogen-containing) hormonal contraception associated with the inhibition of ovulation:
  - Oral, intravaginal, or transdermal.
- Progestogen-only hormonal contraception associated with the inhibition of ovulation:
  - Oral, injectable, or implantable*
    Note: Oral birth control pills are not considered a highly effective form of birth control, and if they are selected, they must be used with a second, barrier method of contraception such as condoms with or without spermicide.
- Intrauterine device.*
- Intrauterine hormone-releasing system.*
- Bilateral tubal occlusion.*
- Vasectomized male partner.*
  Note: This is only considered a highly effective form of birth control when the vasectomized partner is the sole partner of the study patient and there has been a medical assessment confirming surgical success:
  - A sterile man is defined as having definitive evidence of infertility previously demonstrated with azoospermia in a semen sample examination. Men with known "low sperm counts" (consistent with "sub-fertility") are not to be considered sterile for purposes of this study.

- Sexual abstinence (defined as refraining from heterosexual intercourse during the entire period of exposure associated with the study drug).
  Note: Total sexual abstinence should only be used as a contraceptive method if it is in line with the patient's usual and preferred lifestyle. Periodic abstinence (eg, calendar, ovulation, symptothermal, or postovulation methods), declaration of abstinence for the duration of exposure to study drug, and withdrawal are not acceptable methods of contraception.

Of note, barrier contraception (including male and female condoms with or without spermicide) is not considered a highly effective method of contraception and if used, this method must be combined with another acceptable method listed above.

Patients using hormonal contraceptives (eg, birth control pills or devices) must use a barrier method of contraception (eg, condoms) as well.

Definitions of "Women of Childbearing Potential," "Women of No Childbearing Potential"

As defined in this protocol, "women of childbearing potential" are women who are physiologically capable of becoming pregnant.

Conversely, "women of no childbearing potential" are defined as women meeting any of the following criteria:

- Surgically sterile (ie, through bilateral salpingectomy, bilateral oophorectomy, or hysterectomy).
- Postmenopausal, defined as:
  - ≥ 60 years of age
  - < 60 years of age with no spontaneous menses for ≥ 12 months in the absence of prior chemotherapy, tamoxifen, toremifene, or ovarian suppression AND estradiol and FSH in the postmenopausal range OR
  - < 60 years of age with chemotherapy-induced amenorrhea for ≥ 12 months with FSH and estradiol in the postmenopausal range on serial assessments
  - < 60 years of age on tamoxifen with FSH and estradiol in the postmenopausal range
  - If an FSH measurement is required to confirm postmenopausal state, concomitant use of hormonal contraception or hormonal replacement therapy should be excluded.

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APPENDIX 2. ADVERSE EVENTS AND SERIOUS ADVERSE EVENTS: DEFINITIONS AND PROCEDURES FOR RECORDING, EVALUATING, FOLLOW-UP, AND REPORTING

Definitions of Adverse Event

An adverse event is defined as any unfavorable and unintended sign (including an abnormal laboratory finding), symptom, or disease (new or exacerbated) temporally associated with the use of study drug, whether considered related to study drug or not.

Examples of adverse events include:

- Worsening of a chronic or intermittent pre-existing condition, including an increase in severity, frequency, duration, and/or has an association with a significantly worse outcome.
- Detection or diagnosis of a new condition after study drug administration even though the condition might have been present before the start of the study.
- Signs, symptoms, or the clinical sequelae of a suspected interaction.
- Signs, symptoms, or the clinical sequelae of a suspected overdose of study drug or a concurrent medication (overdose per se should not be reported as an adverse event or serious adverse event).

Definition of a Serious Adverse Event

A serious adverse event is any untoward medical occurrence that, at any dose, meets one or more of the criteria listed in the view of either the investigator or sponsor:

a. Results in death

b. Is life-threatening

- The term *life-threatening* in the definition of *serious* refers to an event in which the patient was at risk of death at the time of the event. It does not refer to an event, which hypothetically might have caused death, if it were more severe.

c. Requires inpatient hospitalization or prolongation of existing hospitalization

- In general, hospitalization signifies that the patient has been admitted (usually involving at least an overnight stay) at the hospital or emergency ward for observation and/or treatment that would not have been appropriate in the physician's office or outpatient setting. Complications that occur during hospitalization are adverse events. If a complication prolongs hospitalization or fulfills any other serious criteria, the event is serious. When in doubt as to whether hospitalization occurred or was necessary, the adverse event should be considered serious.
- Hospitalization for elective treatment of a pre-existing condition that did not worsen from baseline is not considered an adverse event.

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- Hospitalization for social/convenience considerations, or scheduled therapy for the target disease of the study including admissions for transfusion support or convenience is not considered to meeting the seriousness criteria.

d. Results in persistent or significant disability/incapacity

- The term disability means a substantial disruption of a person's ability to conduct normal life functions.
- This definition is not intended to include experiences of relatively minor medical significance such as uncomplicated headache, nausea, vomiting, diarrhea, influenza, and accidental trauma (eg, sprained ankle) that may interfere with or prevent everyday life functions but do not constitute a substantial disruption.

e. Is a congenital anomaly/birth defect

f. Important medical events

- Important medical events that may not result in death, be life-threatening, or require hospitalization may be considered serious when, based upon appropriate medical judgment, they may jeopardize the patient and may require medical or surgical intervention to prevent one of the outcomes listed in this definition.
  - Examples of such medical events include allergic bronchospasm requiring intensive treatment in an emergency room or at home, blood dyscrasias or convulsions that do not result in inpatient hospitalization, or the development of drug dependency or drug abuse.

Definition of a Suspected Unexpected Serious Adverse Reaction

A SUSAR is a serious adverse reaction that is both unexpected (ie, not present in the study drug's reference safety information) and meets the definition of a serious adverse drug reaction, the specificity or severity of which is not consistent with those noted in the section of the sonrotoclax and zanubrutinib IB.

The assessment of SUSARs will be performed by the sponsor.

Recording and Follow-Up of Adverse Events, Serious Adverse Events, and/or Adverse Events of Special Interest

Adverse Event, Serious Adverse Event, and Adverse Event of Special Interest Recording

- When an adverse event/serious adverse event/adverse event of special interest occurs, it is the responsibility of the investigator to review all documentation (eg, hospital progress notes, laboratory reports, and diagnostics reports) related to the adverse event or serious adverse event.
- The investigator will then record all relevant adverse event/serious adverse event/adverse event of special interest information in the eCRF.
- The investigator will make an assessment of severity and seriousness.
- It is **not** acceptable for the investigator to send photocopies of the patient's medical records to the sponsor in lieu of completion of the eCRF.

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- There may be instances when copies of medical records for certain cases are requested by the sponsor. In this case, all patient identifiers, with the exception of the patient number, will be redacted on the copies of the medical records before submission to the sponsor.
- Abnormal laboratory findings (eg, clinical chemistry, complete blood count, coagulation, or urinalysis) or other abnormal assessments (eg, ECGs, imaging, or vital signs) that are judged by the investigator as clinically significant will be recorded as adverse events. This includes clinically significant abnormal laboratory findings or other abnormal assessments that are present at baseline and significantly worsen during the study. The definition of clinically significant is left to the judgment of the investigator. In general, these are the laboratory test abnormalities or other abnormal assessments that:
  - Are associated with clinical signs or symptoms, or
  - Require active medical intervention, or
  - Lead to treatment interruption or discontinuation, or
  - Require close observation, more frequent follow-up assessments, or further diagnostic investigation.
- The investigator will attempt to establish a diagnosis of the event based on signs, symptoms, and/or other clinical information.
- Whenever possible, the diagnosis (not the individual signs/symptoms) will be documented as the adverse event/serious adverse event/adverse event of special interest. If a clinically significant laboratory abnormality is a sign of a disease or syndrome (eg, alkaline phosphatase and bilirubin 5 x ULN associated with cholestasis), only the diagnosis (ie, cholestasis) should be recorded on the Adverse Event eCRF. Clinically associated events can be reported as a single event (eg, simultaneous occurrence of anemia, thrombocytopenia, and leukopenia may be reported as myelosuppression). If the laboratory abnormality can be characterized by a precise clinical term per standard definitions, the clinical term should be recorded as the adverse event. For example, an elevated serum potassium level of 7.0 mEq/L (or mmol/L) should be recorded as “hyperkalemia.”
- Observations of the same clinically significant laboratory abnormality from visit to visit should not be repeatedly recorded on the Adverse Event eCRF, unless the etiology or grade changes.
- The initial grade of the event should be recorded, and the grade and seriousness should be updated any time the event changes.

Assessment of Severity

The investigator will assess the severity for each adverse event/serious adverse event/adverse event of special interest reported during the study. Adverse events/serious adverse events/adverse events of special interest should be assessed and graded based upon the NCI-CTCAE v5.0 and the iwCLL Grading Scale for Hematologic Toxicities (if applicable).

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Toxicities that are not specified in the NCI-CTCAE v5.0 and the iwCLL Grading Scale for Hematologic Toxicities (if applicable) will be defined as follows:

- Grade 1: Mild; asymptomatic or mild symptoms; clinical or diagnostic observations only; intervention not indicated.
- Grade 2: Moderate; minimal, local, or noninvasive intervention indicated; limiting age-appropriate instrumental ADL.
- Grade 3: Severe or medically significant but not immediately life-threatening; hospitalization or prolongation of hospitalization indicated; disabling; limiting self-care ADL
- Grade 4: Life-threatening consequences; urgent intervention indicated
- Grade 5: Death related to adverse event

Note: The terms "severe" and "serious" are not synonymous. Severity is a measure of intensity (eg, grade of a specific adverse event, mild [Grade 1], moderate [Grade 2], severe [Grade 3], or life-threatening [Grade 4]); whereas, seriousness is classified by the criteria based on the regulatory definitions. Seriousness serves as the guide for defining regulatory reporting obligations from the sponsor to applicable regulatory authorities as described in Appendix 2.

Assessment of Causality

- The investigator is obligated to assess the relationship between study drug/treatment and each occurrence of each adverse event/serious adverse event/adverse event of special interest. The investigator will use clinical judgment to determine the relationship.
- A *reasonable possibility* of a relationship conveys that there are facts, evidence, and/or arguments to suggest a causal relationship, rather than a relationship cannot be ruled out.
- Alternative causes, such as underlying disease(s), concomitant therapy, and other risk factors, as well as the temporal relationship of the event to study drug/treatment administration, will be considered and investigated.
- For causality assessment, the investigator will also consult the IB and/or product information, for marketed products.
- The investigator must review and provide an assessment of causality for each adverse event/serious adverse event/adverse event of special interest and document this in the medical notes. There may be situations in which a serious adverse event has occurred and the investigator has minimal information to include in the initial report to the sponsor. However, it is very important that the investigator always make an assessment of causality for every event before the initial transmission of the serious adverse event data to the sponsor.
- The investigator may change their opinion of causality in light of follow-up information and send a serious adverse event/adverse event of special interest follow-up report with the updated causality assessment.

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- The causality assessment is one of the criteria used when determining regulatory reporting requirements.

The causality of each adverse event should be assessed and classified by the investigator as "related" or "not related" based on all information available at the time of reporting. An adverse event should be considered "related" to study treatment if any of the following criteria are met; otherwise, the event should be assessed as "not related":

- There is clear evidence to suggest a causal relationship, and other possible contributing factors can be ruled out.
- There is evidence to suggest a causal relationship, and the influence of other factors is unlikely.
- There is some evidence to suggest a causal relationship (eg, the adverse event occurred within a reasonable time after administration of the study drug. However, the influence of other factors may have contributed to the adverse event (eg, the patient's clinical condition or other concomitant adverse events).

Follow-Up of Adverse Events and Serious Adverse Events

- The investigator is obligated to perform or arrange for the conduct of supplemental measurements and/or evaluations as medically indicated or as requested by the medical monitor or safety operations to elucidate the nature and/or causality of the adverse event or serious adverse event as fully as possible. This may include additional laboratory tests or investigations, histopathological examinations, or consultation with other health care professionals.
- If a participant/patient dies during participation in the study or during a recognized follow-up period, the investigator will provide the sponsor with a copy of any postmortem findings including histopathology, when possible.
- New or updated information will be recorded in the originally submitted documents.
- The investigator will submit any updated serious adverse event data to the sponsor within 24 hours of receipt of the information.

Reporting Serious Adverse Events and Adverse Events of Special Interest

As soon as the investigator determines that an adverse event meets the protocol definition of an serious adverse event or adverse event of special interest (see Section 8.4.8), the event must be reported promptly (≤ 24 hours after determination) to the sponsor or designee as described in Table 26.

Table 26: Timeframes and Documentation Methods for Reporting Serious Adverse Events and Adverse Events of Special Interest to the Sponsor or Designee

	Timeframe for Sending Initial/Follow-up Report [a]	Documentation Method	Reporting Method
All SAEs and AESIs	≤ 24 hours after first knowledge of the event	SAE report	Electronic submission of SAE form to safety portal [b]

Abbreviations: AESI, adverse event of special interest; SAE, serious adverse event.

[a] Report follow-up information that is clinically relevant and pertaining to the SAE/AESI, which includes but is not limited to the following: update to the SAE/AESI, new additional SAE /AESI, outcome, seriousness criteria, investigator causality, event start date/date of onset, date of death, and/or relationship to each study drug. Follow-up information will also be reported per the discretion of the investigator if the new or updated information changes the medical assessment of the case.

[b] SAE reports should be submitted to the sponsor safety database electronically from the EDC system. If the electronic submission is not available for any reason, a paper SAE form should be submitted by email or fax.

The serious adverse event report will always be completed as thoroughly as possible with all available details of the event and forwarded to the sponsor or designee within the designated timeframes.

If the investigator does not have all information regarding a serious adverse event/adverse event of special interest, they must not wait to receive additional information before notifying the sponsor or designee of the serious adverse event/adverse event of special interest and completing the form. The form will be updated when additional information is received.

The investigator must always provide an assessment of causality for each serious adverse event/adverse event of special interest as described in Appendix 2.

The sponsor will provide contact information for serious adverse event receipt.

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APPENDIX 3. EUROPEAN ORGANISATION FOR RESEARCH AND TREATMENT OF CANCER QUALITY OF LIFE CANCER QUESTIONNAIRE QLQ-C30

EORTC QLQ-C30 (version 3)

We are interested in some things about you and your health. Please answer all of the questions yourself by circling the number that best applies to you. There are no "right" or "wrong" answers. The information that you provide will remain strictly confidential.

Please fill in your initials: ____
Your birthdate (Day, Month, Year): __ __ __ __ __ __ __ __
Today's date (Day, Month, Year): 31 __ __ __ __ __ __ __ __

	Not at All	A Little	Quite a Bit	Very Much
1. Do you have any trouble doing strenuous activities, like carrying a heavy shopping bag or a suitcase?	1	2	3	4
2. Do you have any trouble taking a long walk?	1	2	3	4
3. Do you have any trouble taking a short walk outside of the house?	1	2	3	4
4. Do you need to stay in bed or a chair during the day?	1	2	3	4
5. Do you need help with eating, dressing, washing yourself or using the toilet?	1	2	3	4

During the past week:	Not at All	A Little	Quite a Bit	Very Much
6. Were you limited in doing either your work or other daily activities?	1	2	3	4
7. Were you limited in pursuing your hobbies or other leisure time activities?	1	2	3	4
8. Were you short of breath?	1	2	3	4
9. Have you had pain?	1	2	3	4
10. Did you need to rest?	1	2	3	4
11. Have you had trouble sleeping?	1	2	3	4
12. Have you felt weak?	1	2	3	4
13. Have you lacked appetite?	1	2	3	4
14. Have you felt nauseated?	1	2	3	4
15. Have you vomited?	1	2	3	4
16. Have you been constipated?	1	2	3	4

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During the past week:	Not at All	A Little	Quite a Bit	Very Much
17. Have you had diarrhea?	1	2	3	4
18. Were you tired?	1	2	3	4
19. Did pain interfere with your daily activities?	1	2	3	4
20. Have you had difficulty in concentrating on things, like reading a newspaper or watching television?	1	2	3	4
21. Did you feel tense?	1	2	3	4
22. Did you worry?	1	2	3	4
23. Did you feel irritable?	1	2	3	4
24. Did you feel depressed?	1	2	3	4
25. Have you had difficulty remembering things?	1	2	3	4
26. Has your physical condition or medical treatment interfered with your family life?	1	2	3	4
27. Has your physical condition or medical treatment interfered with your social activities?	1	2	3	4
28. Has your physical condition or medical treatment caused you financial difficulties?	1	2	3	4

For the following questions please circle the number between 1 and 7 that best applies to you

29. How would you rate your overall health during the past week?

1 2 3 4 5 6 7

Very poor Excellent

30. How would you rate your overall quality of life during the past week?

1 2 3 4 5 6 7

Very poor Excellent

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Specimen

EUROPEAN ORGANISATION FOR RESEARCH AND TREATMENT OF CANCER QUALITY OF LIFE CANCER QUESTIONNAIRE QLQ-CLL17

ENGLISH

EORTC QLQ-CLL17

Patients sometimes report that they have the following symptoms or problems. Please indicate the extent to which you have experienced these symptoms or problems during the past week. Please answer by circling the number that best applies to you.

During the past week:	Not at All	A Little	Quite a Bit	Very Much
31. Have you had muscle weakness?	1	2	3	4
32. Have you had aches or pains in your muscles or joints?	1	2	3	4
33. Have you had aches or pain in your bones?	1	2	3	4
34. Have you had a dry mouth?	1	2	3	4
35. Have you felt ill or unwell?	1	2	3	4
36. Have you had night sweats?	1	2	3	4

During the past week:	Not at All	A Little	Quite a Bit	Very Much
37. Have you had shortness of breath on exertion?	1	2	3	4
38. Have you had a lack of energy?	1	2	3	4
39. Have you felt drowsy?	1	2	3	4
40. Have you had sudden tiredness?	1	2	3	4

During the past 4 weeks:	Not at All	A Little	Quite a Bit	Very Much
41. Have you worried about your health in the future?	1	2	3	4
42. Have you worried about recurrence of your disease?	1	2	3	4
43. Have you worried about becoming dependent on others?	1	2	3	4
44. Have you worried about getting another type of cancer?	1	2	3	4
45. Have you worried about your treatment causing future health problems?	1	2	3	4
46. If applicable: Have you had problems at your work or place of study due to the disease?	1	2	3	4
47. If applicable: Have you worried about not being able to continue working or your education?	1	2	3	4

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APPENDIX 4. LIST OF RESTRICTED AND PROHIBITED MEDICATIONS

Table 27: CYP3A Inhibitors and Inducers

Strong CYP3A Inhibitors (Avoid)
Antibiotics: clarithromycin, telithromycin, troleandomycin
Antifungals: itraconazole, ketoconazole, posaconazole, voriconazole
Antivirals: boceprevir, telaprevir
Other: cobicistat, conivaptan, elvitegravir [a], idelalisib, nefazodone
Protease inhibitors: indinavir, lopinavir [a], nelfinavir, ritonavir, saquinavir, tipranavir
Moderate CYP3A Inhibitors (Use with Caution)
Antibiotics: ciprofloxacin, erythromycin
Antifungals: fluconazole
Protease inhibitors: amprenavir, atazanavir, darunavir/ritonavir combination, fosamprenavir
Calcium channel blockers: diltiazem, verapamil
Tyrosine kinase inhibitors (anticancer): imatinib, crizotinib
Food products: grapefruit (*Citrus paradisi*) or grapefruit juice [a], Seville oranges, starfruit (*Averrhoa carambola*)
Herbal medications: *Schisandra sphenanthera*
Others: amiodarone, aprepitant, casopitant, cimetidine, cyclosporine, dronedarone, tofisopam
Strong CYP3A Inducers (Avoid)
Antibiotics: rifampin (rifampicin)
Anticonvulsants: carbamazepine, phenytoin
Herbal medications: St. John's wort (*Hypericum perforatum*)
Other: avasimibe, enzalutamide, mitotane
Moderate CYP3A Inducers (Use with Caution)
Antibiotics: rifabutin
Anticonvulsants: phenobarbital
Antivirals: efavirenz, etravirine
Other: bosentan, modafinil

Source: FDA Drug Development and Drug Interactions: Table of Substrates, Drug Development and Drug Interactions and Inducers: https://www.fda.gov/drugs/drug-interactions-labeling/drug-development-and-drug-interactions-table-substrates-inhibitors-and-inducers.

[a] elvitegravir and lopinavir are only strong CYP3A inhibitors if used in combination.

Note: For a more complete list, refer to the Flockhart Table: https://drug-interactions.medicine.iu.edu/MainTable.aspx. Additionally, this resource by the FDA provides further examples of drugs and their interaction potential: FDA's Examples of Drugs that Interact with CYP Enzymes and Transporter Systems: https://www.fda.gov/drugs/drug-interactions-labeling/healthcare-professionals-fdas-examples-drugs-interact-cyp-enzymes-and-transporter-systems.

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APPENDIX 5. EASTERN COOPERATIVE ONCOLOGY GROUP PERFORMANCE STATUS

Grade	Description
0	Fully active, able to carry on all predisease performance without restriction.
1	Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature, eg, light housework, office work.
2	Ambulatory and capable of all self-care but unable to carry out any work activities. Up and about more than 50% of waking hours.
3	Capable of only limited self-care, confined to bed or chair more than 50% of waking hours.
4	Completely disabled. Cannot carry on any self-care. Totally confined to bed or chair.
5	Dead.

As published by Oken et al, 1982.

APPENDIX 6. NEW YORK HEART ASSOCIATION FUNCTIONAL CLASSIFICATION

Class	Symptoms
I	No limitation of physical activity. Ordinary physical activity does not cause undue fatigue, palpitation, dyspnea (shortness of breath).
II	Slight limitation of physical activity. Comfortable at rest, but ordinary physical activity results in fatigue, palpitation, dyspnea (shortness of breath).
III	Marked limitation of physical activity. Comfortable at rest, but less than ordinary activity causes fatigue, palpitation, or dyspnea.
IV	Unable to carry on any physical activity without discomfort. Symptoms of heart failure at rest. If any physical activity is undertaken, discomfort increases.

Adapted from Dolgin et al 1994.
Original source: Criteria Committee, NYHA, Inc. Diseases of the Heart and Blood Vessels. Nomenclature and Criteria for diagnosis, 6th edition Boston, Little, Brown and Co. 1964, p 114.

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APPENDIX 7. RESPONSE DEFINITION AFTER TREATMENT FOR PATIENTS WITH CHRONIC LYMPHOCYTIC LEUKEMIA

From Modified iwCLL Guidelines (Hallek et al 2018 and Cheson et al 2012)

Parameter	CR [a]	PR	PR-L [b]	Progressive disease	Stable disease
Group A					
Lymph Nodes [c, d]	None ≥ 1.5 cm	Decrease of ≥ 50% (from baseline)	Decrease of ≥ 50% (from baseline)	Increase of ≥ 50% from baseline or from response	Change of -49% to +49%
Liver and/or spleen size [e]	Spleen size < 13 cm; liver size normal	Decrease of ≥ 50% (from baseline)	Decrease of ≥ 50% (from baseline)	Increase of ≥ 50% from baseline or from response	Change of -49% to +49%
Constitutional symptoms	None	Any	Any	Any	Any
Circulating lymphocyte count	ALC < 4.0 x 10^9/L	Decrease of ≥ 50% from baseline	Decrease of < 50% or increase from baseline	Increase of ≥ 50% [f]	Change of -49% to +49%
Group B					
Platelet count	≥ 100 x 10^9/L	≥ 100 x 10^9/L or increase of ≥ 50% over baseline	≥ 100 x 10^9/L or increase of ≥ 50% over baseline	Decrease of ≥ 50% from baseline secondary to CLL [h]	Change of -49% to +49%
Hemoglobin	≥ 11.0 g/dL (untransfused and without erythropoietin)	≥ 11 g/dL or increase of ≥ 50% over baseline	≥ 11 g/dL or increase of ≥ 50% over baseline	Decrease of ≥ 2 g/dL from baseline secondary to CLL [h]	Increase of < 11.0 g/dL or of < 50% over baseline, or decrease of < 2 g/dL
Bone marrow biopsy	No CLL cells, no B-lymphoid nodules	Presence of CLL cells, or of B-lymphoid nodules [g], or not done	Presence of CLL cells, or of B-lymphoid nodules, or not done	Increase of CLL cells by ≥ 50% on successive biopsies	No change in marrow infiltrate

Source: Hallek et al 2018 and Cheson et al 2012.
Abbreviations: ALC, absolute lymphocyte count; BTK, Bruton tyrosine kinase; CLL, chronic lymphocytic leukemia; CR, complete remission (response); CRi, complete response with incomplete bone marrow recovery; CT,

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computed tomography; iwCLL, International Workshop on Chronic Lymphocytic Leukemia; PR, partial remission (response); PR-L, partial response with lymphocytosis; PTFU, posttreatment follow-up; PTFU 1, first Posttreatment Follow-up Visit; SD, stable disease.

Note: The response category of **indeterminate response due to study drug hold (IND)** should be used in the following cases during the treatment period:

Patient meets criteria for progressive disease due to temporary withholding of study drug (eg, for drug-related toxicity, surgery, or intercurrent illness) can cause a transient worsening of disease and/or of constitutional symptoms. In such circumstances, and if medically appropriate, patients may resume therapy and relevant clinical, laboratory, and/or radiologic assessments should be performed to document whether tumor control can be maintained or whether actual disease progression has occurred.

Isolated increase in lymph nodes and/or splenomegaly (defined as craniocaudal spleen length ≥ 13 cm) during periods of study drug hold will be considered IND. For each code of IND CT scans/MRI must be performed ≥ 6 weeks and no more than 12 weeks after restarting study drug administration: if disease progression is confirmed the previously recorded IND should be replaced with progressive disease.

Parameters of Group A assess the lymphoid tumor load and constitutional symptoms; parameters of Group B assess the hematopoietic system.

Important: To assess group A parameters Lymph nodes, Liver and Spleen for CR/CRi/nPR/PR/PR-L/stable disease compare each timepoint against the baseline. To assess group A parameters for progressive disease, compare against the best response achieved (nadir).

CR: all of the criteria have to be met, and in addition, absolute neutrophil count must be ≥ 1.5 x 10^9/L; PR: ≥ 2 of the parameters of Group A and 1 parameter of Group B need to improve if previously abnormal; if only 1 parameter of both Groups A and B is abnormal before therapy, only 1 of either Group A or Group B needs to improve; progressive disease: ≥ 1 of the criteria of Group A or Group B has to be met; stable disease: all of the criteria have to be met, and constitutional symptoms alone do not define progressive disease.

[a] For CR: bone marrow cellularity is not part of the definition of CR for this study; the bone marrow must have no CLL cells and no B-lymphoid nodules. Some patients fulfill all the criteria for a CR except they have a persistent anemia, thrombocytopenia, or neutropenia apparently unrelated to CLL but related to drug toxicity. These patients should be considered as a different category of remission: CR with incomplete bone marrow recovery (CRi).

[b] During treatment period only, PR-L (Partial response with lymphocytosis): PR criteria are otherwise met, while blood lymphocytes are only decreased < 50% from BL or are or increased from BL. PR-L would override disease progression by ALC alone during treatment period.

[c] For determination of response categories except progressive disease, the percent change is calculated from the sum of the products (SPD) of target lesions as evaluated by CT scans/MRI, or by physical examination against baseline. For CR, all CLL related lesions must resolve. For progressive disease, LDi is compared to nadir. Additionally for progressive disease: any appearance of new lymph node of ≥ 15 mm is progressive disease, note that a transient single LN increase during treatment should not be counted as progressive disease; a new extranodal lesion of any size felt to be unequivocal for CLL or unequivocal reappearance of an extranodal lesion in an area where disease had previously resolved is progressive disease; any previous lymph node lesion increase by ≥ 50% from nadir (from the baseline or response, whichever is lower) in LDi and ≥ 15 mm. Progressive disease based on physical examination must be confirmed by CT scans/MRI within 28 days of the physical examination date. Richter transformation must be confirmed by lymph node or other biopsy and recorded in EDC. A PET-CT showing hyper metabolic areas is not sufficient for confirmation of RT. A biopsy confirmed Richter transformation fulfills criteria for progressive disease.

[d] Target lesions (up to 6 lesions) at baseline are to be recorded twice: once based on physical examination only and separately based on imaging only. For clinical response evaluation include only target lesions identified by physical examination and evaluate based on the longest diameter, for overall response SPD evaluation include target lesions identified by imaging.

[e] Spleen size is considered normal if craniocaudal spleen length < 13 cm. For example, a patient with a total craniocaudal spleen length at baseline of 18 cm has a 5 cm abnormal spleen length: 18 cm – 13 cm = 5 cm abnormal portion. If the total length of the spleen at C10D1 visit is 15 cm, then the % change from baseline is 3/5 = 60% decrease, which meets the splenomegaly “A” parameter definition for partial response. Disease progression requires a ≥ 50% increase in the abnormal portion from nadir (if an abnormal portion exists at nadir) and a minimum absolute increase in the abnormal portion by ≥ 2 cm. NOTE: Other clinical factors that could result in temporary splenomegaly (ie, infection) should be recorded. Progressive disease based on physical examination must be confirmed by CT scans/MRI within 28 days of the physical examination date. NOTE: If the

spleen is barely palpable but still considered enlarged on physical examination, enter "0.5 cm" as the measurement for "centimeters below left costal margin" on the corresponding eCRF page.

[f] An increase in the number of absolute blood lymphocyte count by ≥ 50% with ≥ 5 x 10^9/L absolute lymphocyte count. During the PTFU period increases should be assessed against the lowest absolute lymphocyte count value (nadir) starting from the PTFU 1 and onwards. During the treatment period and up to PTFU 1, the increase of absolute lymphocyte count alone is not sufficient for the criteria of progressive disease.

[g] Lymphoid nodules that may reflect residual disease can be found. These nodules may be recorded as "nodular partial remission (nPR)"; however, immunohistochemistry may be performed to define whether the nodules comprise primarily T cells, B cells other than CLL cells, or CLL cells. If nodules are not composed of CLL cells, a CR can be documented provided all other criteria are met.

[h] The progression of any cytopenia defines disease progression (unrelated to autoimmune cytopenia), which occurs ≥ 3 months after treatment, if the marrow biopsy demonstrates an infiltrate of clonal CLL cells. During therapy, cytopenias cannot be used to define disease progression (see Section 8.2.4).

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APPENDIX 8. SELECTION AND EVALUATION OF TARGET AND NON-TARGET LESIONS BASED ON IMAGING

All chronic lymphocytic leukemia (CLL) lesions identified by imaging at baseline and meeting the following requirements will be recorded in the electronic case report form (eCRF): lymph node ≥ 1.5 cm in ≥ 1 dimension, or non-nodal lesion measuring ≥ 1 cm x 1 cm. Up to 6 of the lesions with bi-dimensional measurements will be identified as 'target' lesions, and ≥ 1 lymph node target lesion must be selected. The remaining lesions, whether measurable or non-measurable, will be recorded as 'non-target' lesions. For target lesions, measurements will be recorded in the eCRF at each imaging timepoint in 2 perpendicular dimensions: LDi and SDi.

LDi is the longest diameter, and SDi is the longest diameter perpendicular to LDi. Multiple non-target lymph nodes co-located in the same anatomical region may be classified under a single non-target annotation. Examples of non-target lymph nodes include:

- Any measurable nodal disease beyond the maximum number of 6 target lymph nodes
- Extranodal disease beyond the maximum number of 6 target lymph nodes
- Assessable disease that is non-measurable
- All bone lesions
- Cutaneous lesions
- Gastrointestinal disease
- Kidneys
- Irradiated lymph nodes
- Groups of lymph nodes that are small and numerous
- Pleural/pericardial effusions and/or ascites

Spleen and liver are not to be selected as target or non-target lesions, instead, they are to be evaluated and recorded separately in the electronic data capture (EDC) system (details refer to Imaging Manual). However, focal lesions within spleen or liver that are considered to be related to CLL can be selected as target or non-target lesions.

At each imaging timepoint, target lymph nodes (nodal and non-nodal) recorded at baseline should have their actual measurements recorded, even when very small (eg, 2 mm). However, sometimes, lymph nodes become so faint on a scan that the radiologist may not feel comfortable assigning an exact measurement and may report them as being "too small to measure (TSTM)." When this occurs, it is important that a value be recorded on the eCRF. If it is the opinion of the radiologist that the lymph node has likely disappeared, the measurement should be recorded as 0 mm. If the lymph node is believed to be present and is faintly seen but TSTM, a default value of 5 mm should be assigned (NOTE: it is less likely that this rule will be used for lymph nodes because they usually have a definable size when normal and are frequently surrounded by fat such as in the retroperitoneum; however, if a lymph node is believed to be present and is faintly seen but is TSTM, a default value of 5 mm should be assigned in this circumstance as well). For lymph nodes where the longest diameter is measurable but the longest perpendicular

measurement is not, enter the longest diameter into the EDC system; the longest perpendicular measurement should be recorded as “TSTM” with a default value of 5 mm.

At each timepoint, the presence of new lymph nodes should be assessed as per the criteria below:

- A new lymph node ≥ 1.5 cm in any axis
- A new extranodal site > 1.0 cm in any axis; OR if ≤ 1.0 cm in any axis, its presence must be unequivocal and must be attributable to CLL.
- Assessable disease of any size unequivocally attributable to CLL

Other clinical factors that could result in the temporary appearance of a new lymph node (ie, infection) should be ruled out before reporting disease progression.

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APPENDIX 9. DIAGNOSIS OF CHRONIC LYMPHOCYTIC LEUKEMIA

A short summary of the blood smear, the immunophenotype, and genetic features of the circulating lymphoid cells to verify the patient has CLL and not another lymphoproliferative disease is listed below. See details in Hallek et al 2018.

Laboratory tests	**Description**	
Blood [a]	The diagnosis of CLL requires the presence of B lymphocytes ≥ 5 x 10^9 cells/L in the peripheral blood, sustained for ≥ 3 months	
	The clonality of these B lymphocytes needs to be confirmed by demonstrating immunoglobulin light-chain restriction using flow cytometry	
Immunophenotype [a]	CLL cells coexpress the surface antigen CD5 together with the B-cell antigens CD19, CD20, and CD23	
	A panel of CD19, CD5, CD20, CD23, κ, and λ is usually sufficient to establish the diagnosis; in borderline cases, markers such as CD43, CD79b, CD81, CD200, CD10, or ROR1 may help to refine the diagnosis	
Other tests [b]	Molecular genetics	Molecular cytogenetics (FISH) [c] for del(13q), del11q), del(17p), and add(12) in peripheral blood lymphocytes
		Conventional karyotyping [c] in peripheral blood lymphocytes (with specific stimulation)
		The assessment of both del(17p) and *TP53* mutation has prognostic and predictive value and should guide therapeutic decision in routine practice
	Mutational status of IGHV and variable heavy stereotypes	
	Immunophenotypic markers	CD49d has been associated with an unfavorable prognosis in CLL and was shown to be the strongest flow cytometry-based predictor of overall survival
		ZAP-70 and CD38
	Serum markers	CD23, thymidine kinase, and β_2-microglobulin are associated with overall survival
	Marrow examination	A marrow aspirate and biopsy generally are not required for the diagnosis of CLL; a marrow biopsy is mandatory to confirm a complete remission

Abbreviations: CLL, chronic lymphocytic leukemia; FISH, fluorescence in situ hybridization; *IGHV*, immunoglobulin heavy-chain variable region.

[a] Tests to establish the diagnosis.

[b] Tests are not essential to diagnose CLL, but may help to predict the prognosis or assess the tumor burden.

[c] FISH and conventional karyotyping help distinguish CLL from other lymphoproliferative diseases.

APPENDIX 10. CHRONIC LYMPHOCYTIC LEUKEMIA CLINICAL STAGING

Binet System[a]

Stage	Description
A	Hemoglobin ≥ 10 g/dL and platelets ≥ 100,000/mm^3 and < 3 enlarged areas [c]
B	Hemoglobin ≥ 10 g/dL and platelets ≥ 100,000/mm^3 and ≥ 3 enlarged areas [c]
C [b]	Hemoglobin < 10 g/dL and/or platelets < 100,000/mm^3 and any number of enlarged areas [c]

Abbreviations: NCCN, National Comprehensive Cancer Network.
Source: NCCN Clinical Practice Guidelines in Oncology Chronic Lymphocytic Leukemia Version 3.2018.
[a] From Binet JL, Auquier A, Dighiero G, et al. A new prognostic classification of chronic lymphocytic leukemia derived from a multivariate survival analysis. Cancer 1981;48:196-206.
[b] Immune-mediated cytopenias are not the basis for these stage definitions.
[c] Areas include liver, spleen, axillary region, inguinal region, and cervical region

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APPENDIX 11. RECOMMENDATIONS FOR INITIAL MANAGEMENT OF ELECTROLYTE IMBALANCES AND PREVENTION OF TUMOR LYSIS SYNDROME

FIRST DOSE OF SONROTOCLAX OR DOSE INCREASE

- Within the first 24 hours after either the first dose or dose increase, if any laboratory criteria below are met, the patient should be hospitalized for monitoring and the investigator notified. No additional doses should be administered until resolution. A rapidly rising serum potassium level is a medical emergency.
- Nephrology (or acute dialysis service) must be consulted/contacted on admission (per institutional standards to ensure emergency dialysis is available).
- Intravenous fluids (eg, D5 1/2 normal saline) should be initiated at a rate of ≥ 1 mL/kg/h rounded to the nearest 10 mL (target 150 to 200 mL/h; not < 50 mL/h). Modification of fluid rate should also be considered for individuals with specific medical needs.
- Monitor for symptoms or signs of tumor lysis syndrome (TLS) (eg, fever, chills, tachycardia, nausea, vomiting, diarrhea, diaphoresis, hypotension, muscle aches, weakness, paresthesias, mental status changes, confusion, and seizures). If any clinical features are observed, recheck potassium, phosphorus, uric acid, calcium, and creatinine within 1 hour STAT.
- Vital signs should be taken at time of all blood draws or any intervention.

  The management recommendations below focus on the minimum initial responses required. If a diagnosis of TLS is established, ongoing intensive monitoring and multi-disciplinary management will be per institutional protocols.

Recommendations for Initial Management of Electrolyte Imbalances and Prevention of Tumor Lysis Syndrome

Abnormality	Management Recommendations
Hyperkalemia (including rapidly rising potassium)	
Potassium ≥ 0.5 mmol/L increase from prior value (even if potassium is WNL)	• Recheck potassium, phosphorus, uric acid, calcium, and creatinine in 1 hour STAT. If further ≥ 0.2 mmol/L increase in potassium, but still < ULN, manage per potassium ≥ ULN. Otherwise recheck in 1 hour. • Resume per protocol testing if change in potassium is < 0.2 mmol/L, potassium < ULN, and there is no other evidence of tumor lysis. • At the discretion of the investigator, may recheck before hospitalization.

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Abnormality	Management Recommendations
	• If stable or decreased and still WNL, hospitalization is at the discretion of the investigator. Potassium, phosphorus, uric acid, calcium, and creatinine must be rechecked within 24 hours.
Potassium > upper limit of normal	• Perform STAT ECG and commence telemetry. • Nephrology notification with consideration of initiating dialysis. • Administer Kayexalate 60 g (or Resonium A 60 g). • Administer furosemide 20 mg intravenously x 1. • Administer calcium gluconate 100 to 200 mg/kg intravenously slowly if there is ECG/telemetry evidence of life-threatening arrhythmias. • Recheck potassium, phosphorus, uric acid, calcium, and creatinine in 1 hour STAT. • If potassium < ULN 1 hour later, repeat potassium, phosphorus, uric acid, calcium, and creatinine 1, 2, and 4 hours later if there is no other evidence of tumor lysis.
Potassium ≥ 6.0 mmol/L (6.0 mEq/L) and/or symptomatic (eg, muscle cramps, weakness, paresthesias, nausea, vomiting, or diarrhea)	• Perform STAT ECG and commence telemetry. • Nephrology assessment with consideration of initiating dialysis. • Administer Kayexalate 60 g (or Resonium A 60 g). • Administer furosemide 20 mg intravenously x 1. • Administer insulin 0.1 U/kg intravenously + D25 2 mL/kg intravenous. • Administer sodium bicarbonate 1 to 2 mEq/kg intravenously push. – If sodium bicarbonate is used, rasburicase should not be used as this may exacerbate calcium phosphate precipitation. • Administer calcium gluconate 100 to 200 mg/kg intravenously slowly if there is ECG/telemetry evidence of life-threatening arrhythmias. Do not administer in same intravenous line as sodium bicarbonate. • Recheck potassium, phosphorus, uric acid, calcium, and creatinine every hour STAT.

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Abnormality	Management Recommendations
Hyperuricemia	
Uric acid ≥ 8.0 mg/dL (476 µmol/L)	• Consider rasburicase (dose per institutional guidelines). – If rasburicase is used, sodium bicarbonate should not be used as this may exacerbate calcium phosphate precipitation. • Recheck potassium, phosphorus, uric acid, calcium, and creatinine in 1 hour STAT
Uric acid ≥ 10 mg/dL (595 µmol/L) OR Uric acid ≥ 8.0 mg/dL (476 µmol/L) with 25% increase and creatinine increase ≥ 0.3 mg/dL (≥ 0.027 mmol/L) from predose level	• Administer rasburicase (dose per institutional guidelines). – If rasburicase is used, sodium bicarbonate should not be used as this may exacerbate calcium phosphate precipitation. • Consult nephrology. • Recheck potassium, phosphorus, uric acid, calcium, and creatinine in 1 hour STAT. • If uric acid < 8.0 mg/dL 1 hour later, repeat potassium, phosphorus, uric acid, calcium, and creatinine 2 and 4 hours later if there is no other evidence of tumor lysis.
Hypocalcemia	
Corrected calcium ≤ 7.0 mg/dL (1.75 mmol/L) OR Patient symptomatic (eg, muscle cramps, hypotension, tetany, or cardiac arrhythmias) in the presence of hypocalcemia	• Administer calcium gluconate 50 to 100 mg/kg intravenously slowly with ECG monitoring. • Telemetry. • Recheck potassium, phosphorus, uric acid, calcium, and creatinine in 1 hour STAT. • If calcium normalizes 1 hour later, repeat potassium, phosphorus, uric acid, calcium, and creatinine 2 and 4 hours later if no other evidence of tumor lysis.
Hyperphosphatemia	
Phosphorus ≥ 5.0 mg/dL (1.615 mmol/L) with ≥ 0.5 mg/dL (0.16 mmol/L) increase	• Administer a phosphate binder (eg, aluminum hydroxide, calcium carbonate, sevelamer hydroxide, or lanthanum carbonate). • Nephrology notification (dialysis required for phosphorus > 10 mg/dL). • Recheck potassium, phosphorus, uric acid, calcium, and creatinine in 1 hour STAT.

Abnormality	Management Recommendations
	• If phosphorus < 5.0 mg/dL 1 hour later, repeat potassium, phosphorus, uric acid, calcium, and creatinine 2 and 4 hours later, if no other evidence of tumor lysis.
Creatinine	
Increase ≥ 25% from baseline	• Start or increase rate of intravenous fluids. • Recheck potassium, phosphorus, uric acid, calcium, and creatinine in 1 to 2 hours STAT.

Abbreviations: ECG, electrocardiogram; STAT, <definition>; ULN, upper limit of normal; WNL, within normal limits.

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APPENDIX 12. DEFINITION AND GRADING OF TUMOR LYSIS SYNDROME

Tumor lysis syndrome (TLS) is an oncologic emergency that is caused by rapid tumor cell lysis leading to a release of large amounts of potassium, phosphate, and nucleic acids into systemic circulation.

TLS can be separated into laboratory TLS and clinical TLS as defined per the Howard Criteria (Howard et al 2018).

Metabolic Abnormality	**Criteria for Classification of Laboratory TLS [a]**	**Criteria for Classification of Clinical TLS [a]**
Hyperuricemia	Uric acid > 8.0 mg/dL (475.8 µmol/L) in adults or above the upper limit of the normal range for age in children	
Hyperphosphatemia	Phosphorus > 4.5 mg/dL (1.5 mmol/L) in adults or > 6.5 mg/dL (2.1 mmol/L) in children	
Hyperkalemia	Potassium > 6.0 mmol/L	Cardiac dysrhythmia or sudden death probably or definitely caused by hyperkalemia
Hypocalcemia	Corrected calcium < 7.0 mg/dL (1.75 mmol/L) or ionized calcium < 4.5 mg/dL (1.12 mmol/L) [b]	Cardiac dysrhythmia, sudden death, seizure, neuromuscular irritability (tetany, paresthesias, muscle twitching, carpopedal spasm, Trousseau's sign, Chvostek's sign, laryngospasm, or bronchospasm), hypotension, or heart failure probably or definitely caused by hypocalcemia
Acute kidney injury [c]	Not applicable	Increase in the serum creatinine level of 0.3 mg/dL (26.5 µmol/L) (or a single value > 1.5 times the upper limit of the age appropriate normal range if no baseline creatinine measurement is available) or the presence of oliguria, defined as an average urine output of < 0.5 mL/kg/hr for 6 hours

Abbreviations: TLS, tumor lysis syndrome.
Source: Howard et al 2018.

[a] In laboratory TLS, 2 or more metabolic abnormalities must be present during the 24-hour period within 3 days before the start of treatment or up to 7 days afterward. Clinical TLS requires the presence of laboratory TLS plus an increased creatinine level, seizures, cardiac dysrhythmia, or death.

[b] The corrected calcium level in milligrams per deciliter = measured calcium level in milligrams per deciliter + 0.8 x (4 – albumin in grams per deciliter).

[c] Acute kidney injury is defined as an increase in the creatinine level of ≥ 0.3 mg per deciliter (26.5 μmol per liter) or a period of oliguria lasting 6 hours or more. By definition, if acute kidney injury is present, the patient has clinical TLS.

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APPENDIX 13. THE MODIFIED CUMULATIVE ILLNESS RATING SCALE (CIRS)

Body system	Score				
1. Cardiac (heart only)	0	1	2	3	4
2. Hypertension (rating is based on severity; organ damage is rated separately)	0	1	2	3	4
3. Vascular (blood, blood vessels and cells, bone marrow, spleen, lymphatics)	0	1	2	3	4
4. Respiratory (lungs, bronchi, trachea below the larynx)	0	1	2	3	4
5. EENT (eye, ear, nose, throat, larynx)	0	1	2	3	4
6. Upper GI (esophagus, stomach, and duodenum; pancreas; do not include diabetes)	0	1	2	3	4
7. Lower GI (intestines, hernias)	0	1	2	3	4
8. Hepatic (liver and biliary tree)	0	1	2	3	4
9. Renal (kidneys only)	0	1	2	3	4
10. Other GU (ureters, bladder, urethra, prostate, genitals)	0	1	2	3	4
11. Muscolo-skeletal-integumentary (muscle, bone, skin)	0	1	2	3	4
12. Neurological (brain, spinal cord, nerves, do not include dementia)	0	1	2	3	4
13. Endocrine-Metabolic (includes diabetes, thyroid; breast; systemic infections; toxicity)	0	1	2	3	4
14. Psychiatric/Behavioral (includes dementia, depression, anxiety, agitation/delirium, psychosis)	0	1	2	3	4

PHILOSOPHY AND DEVELOPMENT OF THE SCALE

Compiling and quantifying medical problems in the elderly population would allow meaningful comparison of medical burden and treatment outcomes in elderly patients with variable and complex medical problems.

The Cumulative Illness Rating Scale (CIRS) was initially developed by Linn et al. and published in JAGS 1968; it appeared immediately a user friendly but comprehensive review of medical problems by organ systems, based on a 0 thru 4 rating, yielding a cumulative score.

This scale was successively revised by Miller et al. to reflect common problems of the elderly with an emphasis on morbidity using specific examples and was renamed CIRS for Geriatrics (CIRS-G) (Miller et al. 1992); moreover, Miller and Towers also provided a manual of guidelines for scoring their version (Miller et al. 1991).

Then, in 1995, Parmelee et al. validated a modified CIRS version, based on a 1 thru 5 rating and with some differences in categories, in a geriatric residential population (Parmelee et al. 1995).

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Finally, Mistry et al. used this latter modified CIRS version with a 0 through 4 rating to measure medical burden in psychogeriatric participants of the UPBEAT program, showing that inclusion of acute medical conditions did not undermine the usefulness of the CIRS (Mistry et al. 2004).

Based on the version of Miller and Towers, current guidelines were adapted to the modified CIRS version and updated.

EDUCATION OF RATER

Nurses, physician assistants, nurse practitioners, or physicians are required to have a necessary background for completing this scale. Due to the judgment required, some physician consultation may be necessary to clarify complex medical problems or their severity.

THE MINIMUM DATABASE REQUIRED

It is expected that every patient have a complete history and physical examination with a detailed problem list, height and weight (to calculate body mass index [BMI]), and baseline laboratory values including a complete blood count and differential, chemistry profile to include electrolytes, liver and kidney function, serum B12, thyroid function, cholesterol level, hemoglobin A1c (for diabetic patients), and an EKG. For rating psychiatric conditions, the rater is expected to be familiar with the Mini Mental Status Examination (Folstein et al 1975), and the Diagnostic and Statistical Manual IV (DSM-IV) APA 1994. A check list of needing information is available in the Appendix.

RATING SUGGESTIONS (GENERAL PRINCIPLES)

Every single disease must be classified in the appropriate system. If there are several problems in the same system, only the most severe is rated. For example, for a patient suffering from a well-controlled angina (Rated 2) and terminal heart failure (Rated 4), only the higher-rated condition would be scored in the Cardiac system (eg, rating is 4).

The spread of a cancer may lead to rate the condition in more than 1 category. For example, a lung cancer with bone metastases treated with nonsteroidal anti-inflammatory drugs (NSAID) is Rated 4 in Respiratory and Rated 2 in Musculoskeletal.

General rules for severity rating:

0. No problem affecting that system or past problem without clinical relevance
1. Current mild problem or past significant problem
2. Moderate disability or morbidity and/or requires first-line therapy
3. Severe problem and/or constant and significant disability and/or hard to control chronic problems (complex therapeutic regimen)
4. Extremely severe problem and/or immediate treatment required and/or organ failure and/or severe functional impairment

LEVEL 0

No problem or healed minor injuries; past childhood illnesses (chickenpox); minor surgery (carpal tunnel completely healed, cesarean); uncomplicated healed fractures; other past problems healed without sequel, residual or complication (pneumonia).

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LEVEL 1

Any current medical problem that causes mild discomfort or disability, or has occasional exacerbations, having only minor impact on morbidity (asthma controlled with PRN bronchodilators, occasional heartburn relieved with PRN antiacids). Medical problems that are not currently active but were significant problems in the past (passage of a kidney stone) or required major surgery (hysterectomy, cholecystectomy, appendectomy).

LEVEL 2

Medical conditions that require daily treatment or first-line therapy (asthma controlled with inhaled steroids, gastro-esophageal reflux treated with daily medication, osteoarthritis requiring daily NSAID, etc) and/or have moderate disability or morbidity.

LEVEL 3

Chronic conditions that are not controlled with first-line therapy (asthma needing continuous corticosteroid therapy, symptomatic angina despite medical regimes, heart failure with symptoms, or uncontrolled hypertension despite complex therapeutic regimen) and/or constant significant disability, but not severe disability.

LEVEL 4

Any acute condition that requires immediate treatment or hospitalization (unstable angina, acute myocardial infarction, stroke, but also bladder outlet obstruction) and/or extremely severe problems; organ failure (end-stage renal disease needing dialysis, oxygen-dependent chronic obstructive pulmonary disease, terminal heart failure); severe sensory impairment (almost complete blindness or deafness, being wheelchair bound) and/or severely affected quality of life, severe impairment in function; delirium by medical (organic) conditions.

RATING MALIGNANCIES

Consistent scoring of severity ratings for various malignancies is a difficult problem. Each malignancy has its own rating system and prognostic indicators, the complexity of which would quickly exceed the aim of the intended simplicity and ease of use of CIRS.

The following general guidelines are intended to provide a reasonably accurate delineation of medical burden for cancer without excessive complexity.

- Level 1: Cancer diagnosed in the remote past without evidence of recurrence or sequel in the past 10 years or skin cancer excised in the past without major sequel (other than melanoma)
- Level 2: No evidence of recurrence or sequel in the past 5 years
- Level 3: Required chemotherapy, radiation, hormonal therapy, or surgical procedure for cancer in the past 5 years
- Level 4: Recurrent malignancy or metastasis (other than to lymph glands) or palliative treatment stage

These ratings are to be made in the appropriate organ category for a given malignancy.

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ORGAN SPECIFIC CATEGORIES

The following organ specific categories will attempt to provide guidelines for consistent rating of comparable severity. Common conditions will be stressed with the focus on the "judgment strategy" that can be applied to other problems not listed.

If there are several problems in the same system, only the most severe is rated.

HEART

In this category, only heart and coronary disease have to be considered (not vascular): coronary arteries disease, heart failure, valvular heart diseases, heart disease secondary to hypertension, endocarditis, myocarditis, pericarditis, arrhythmias (extrasystoles, bundle-branch blocks, atrial fibrillation, PMK placement), and heart malignancies. Functional impact must be considered too (eg, New York Heart Association Classification [NYHA] II heart failure has different value between dependent and independent persons).

0. No problems
1. Remote MI (> 5 years ago); occasional [exertion] angina; asymptomatic valvular disease
2. CHF compensated with medications (NYHA I-II); daily anti-angina medications; left ventricular hypertrophy; atrial fibrillation, bundle-branch block, daily anti-arrhythmic drugs (even for prophylaxis); PMK placement for asymptomatic bradycardia (relieved by Holter EKG monitoring); valvular disease requiring medical treatment
3. Previous MI (< 5 years ago); abnormal stress test; status post (previous) percutaneous coronary angioplasty, coronary artery bypass graft surgery or other cardiac surgery (valve replacement); moderate CHF (NYHA II−III) or complex medical treatment; bifascicular block; PMK placement for cardiogenic syncope; pericardial effusion or pericarditis
4. Acute coronary syndrome, unstable angina or acute MI; intractable CHF (NYHA III−IV acute or chronic); marked restriction to the normal activity of daily living secondary to cardiac status

HYPERTENSION

Consider only hypertension severity; organ damage (complications) should be considered into the respective categories.

0. Normotension
1. Borderline hypertension; hypertension compensated with salt restriction and weight loss, drug free (when drug therapy is indicated, but the patient does not take medications, the score is ≥ 2)
2. Daily antihypertensive medications: hypertension controlled by 1 pill therapy (even fixed doses combinations)
3. Hypertension requiring 2 or more pills for control
4. Malignant hypertension, or hypertension non-controlled by complex therapeutic regimen

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VASCULAR-HEMATOPOIETIC

Artery disease: carotid atherosclerosis, peripheral arteries disease (PAD), aneurysms (every site);

Venous disease: venous insufficiency, varices, deep venous thrombosis (DVT), pulmonary embolism, primary pulmonary hypertension;

Hematopoietic disease: anemia, leucopenia, thrombocytopenia, hematological malignancy;

Lymphopoietic disease: chronic lymphatic edema, lymphoma, spleen and thymus disease;

Immunologic disease: systemic lupus erythematosus, systemic sclerosis (scleroderma), sarcoidosis, hypersensitivity

0. No problem
1. Venous insufficiency, varices, lymphedema; carotid stenosis < 70%; hemoglobin 10 to 12 g/dl (in females), 12 to 14 g/dl (in males); anemia of chronic "inflammatory" disease
2. Previous DVT; 1 symptom of atherosclerosis disease (claudication, bruit, amaurosis fugax, absent pedal pulses) or daily medications (eg, anti-platelets drugs); PAD IIa to IIb by Fontaine; carotid stenosis > 70%; aortic aneurysm < 4 cm; hemoglobin 8 to 10 g/dl (in females), 10 to 12 g/dl (in males); anemia secondary to iron, B12 vitamin or folate deficiency, or to chronic renal failure; total white blood cell (WBC) 2000 to 4000/mmc; mild thrombocytopenia (50000 to 150000/mmc)
3. DVT or recent DVT (< 6 months ago); 2 or more symptoms of atherosclerosis (see above); PAD Fontaine III or recent/previous angioplasty (with or without stenting); hemoglobin < 8g/dl (in females), < 10 g/dl (in males); dyserythropoietic anemia; WBC < 2000/mmc; severe thrombocytopenia (< 50000/mmc)
4. Pulmonary embolism (acute or recent/previous); atherosclerosis requiring surgical intervention (eg, aortic aneurysm > 4 cm, symptomatic carotid stenosis > 70%, PAD Fontaine IV or amputation for vascular causes); recent/previous vascular surgery; any hematological or vascular malignancy (including multiple myeloma)

In case of immunological disease, score should be assigned by considering blood abnormalities, stadium of organ damage and/or functional disability (2: symptoms controlled by daily medications; 3: symptoms not well controlled; 4: symptoms impossible to be controlled or short time poor prognosis).

RESPIRATORY

In this category, we consider COPD, asthma, emphysema, restrictive pulmonary interstitial lung diseases, malignancies of lung and pleura, pneumonia, and smoking status too.

0. No problem
1. Recurrent episodes of acute bronchitis; currently treated asthma with prn inhalers when required; cigarette smoker > 10 but < 20 pack years
2. Instrumental diagnosis of COPD or pulmonary interstitial disease (X-ray, TC, spirometry); daily prn inhalers (≤ 2 pharmacological classes); 2 or more episodes of pneumonia in the last 5 years; cigarette smoker < 20 but < 40 pack years

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3. Exertion dyspnea secondary to limited respiratory capacity, not well controlled by daily medications; required oral steroids for lung disease; daily prn inhalers (3 pharmacological classes); acute pneumonia treated as an outpatient
4. Chronic supplementation of oxygen; respiratory failure requiring assisted ventilation, or previous (≥ 1 episode); any lung or pleural neoplasm; acute pneumonia requiring hospitalization

Smoking is an important respiratory and cardiovascular risk, so it is considered as a disease, and it is rated according to *lifetime pack years*:

Number of cigarette packs smoked per day x Number of years smoked in their lifetime, eg, 1 pack year = 20 cigarettes/die (1 pack) x 1 year.

Ex-smokers should be rated too, but those who have been smoke free for the most recent 20 years would merit a lower rating than currently smoking.

Examples:

a. Patient smoking 20 cig/die (1 pack) for 25 years = 25 pack years − CIRS score: 2
b. Patient smoking 40 cig/die (2 packs) for 25 years = 50 pack years − CIRS score: 3
c. Ex-smoker of 20 cig/die (1 pack) for 25 years, patient stopped 5 years ago −CIRS score: 2
d. Ex-smoker of 20 cig/die (1 pack) for 25 years, patient stopped 20 years ago −CIRS score: 1

Classification of COPD could be more specific when instrumental data (objective evidence) are available: blood gases, forced expiratory volume in 1 second (FEV1), etc.

EYES, EARS, NOSE AND THROAT, AND LARYNX

To simplify the potential complexity of this category it was decided to score according to the severity of the disability created by sensory diseases (degree of limited autonomy and communication) and to avoid rating each type of pathology. Sensory impairments should be rated after instrumental correction (corrective lenses, hearing aid, etc).

Eyes: glaucoma, cataracts, macular degeneration (diabetic/hypertensive retinopathy), any other pathology.

Ears: otitis, dizziness, any cause of hearing impairment.

Nose and throat: rhinitis, pharyngitis, nasal polyps, sinusitis, malignancies.

Larynx: dysphonia, acute and chronic laryngitis, malignancies.

0. No problems
1. Corrected vision with glasses; mild hearing loss; chronic sinusitis
2. Difficulty in reading newspaper or drive although glasses; required hearing aid; chronic sinonasal complaints requiring medication; vertigo/dizziness requiring daily medications
3. Severe low vision, partially blind (required an escort to venture out, unable to read newspaper); severe ear impairment (conversational heading still impaired with hearing aid); laryngeal dysphonia (not neurological dysarthria)

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4. Functional blindness/deafness: unable to read, recognize a familiar face, unable to conversational heading, even if "organically" he is not completely blind or deaf; laryngectomy (every cause, especially malignancies); required surgical intervention for vertigo; aphonia secondary to laryngeal impairment.

UPPER GASTROINTESTINAL SYSTEM

This category is comprehensive of the intestinal tract from esophagus to duodenum, and pancreatic trees: dysphagia, GERD, hiatal hernia, esophageal diverticula, any type of gastritis (also consider *Helicobacter pylori* eradication or not), gastric/duodenal ulcer, acute or chronic pancreatitis, malignancies (comprehensive of gastric lymphoma).

Pay attention that Type 1 diabetes is rated under "metabolic."

0. No problem
1. Hiatal hernia, GERD or gastritis requiring prn medications; previous ulcer (> 5 years ago); previous H. pylori eradication therapy (> 5 years ago)
2. Daily proton-pump inhibitor/anti-acid medications; documented gastric or duodenal ulcer or H. pylori eradication therapy within 5 years
3. Active gastric or duodenal ulcer; positive fecal occult blood test; any swallowing disorder or dysphagia; chronic pancreatitis requiring supplemental pancreatic enzymes for digestion; previous episode of acute pancreatitis
4. Any type of malignancies (see "Rating Malignancies" above); previous gastric surgery because of cancer; history of perforated ulcer (gastric surgery not because of cancer, ulcorrhaphy); melena/heavy bleeding from upper gastrointestinal (GI) source; acute pancreatitis

LOWER GASTROINTESTINAL SYSTEM

This category is comprehensive of the rest of the GI system, from small bowel to anus: Whipple's disease, diverticulosis, irritable bowel, malignancies. Constipation is rated, too, by type and frequency of laxatives required, or by history of impaction.

0. No problems, previous appendectomy, previous hernia repair (without complications)
1. Constipation managed with prn medications; active hemorrhoids; intestinal hernia requiring surgery; previous hernia repair with complications (intestinal adherences, laparocele, etc); irritable bowel syndrome (few symptoms)
2. Constipation requiring daily bulk laxatives (psyllium, policarbophil, sterculia, guar gum, etc), or stool softeners; diverticulosis (previous diverticulitis); inflammatory bowel disease in remission with medications (> 5 years ago)
3. Bowel impaction/diverticulitis within the last year; daily use of stimulant (irritant) or osmotic laxatives (bysacodil, senna, glycerol, sodium docusate; lactulose, polyethylene glycol) or enemas; chronic bowel inflammation in remission with medications (< 5 years ago)
4. Diverticulitis flare up; active inflammatory disease; current impaction; hematochezia/active bleeding from lower GI source; bowel carcinoma

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LIVER AND BILIARY TREES

This category is comprehensive of liver, gallbladder, biliary trees, and portal system: acute and chronic hepatitis (viral, alcoholic, toxic, autoimmune, idiopathic), cirrhosis, portal hypertension, hemochromatosis, primary biliary cirrhosis, cholelithiasis, cholangitis, and primary malignancies. As the hepato-biliary system is difficult to assess through the physical examination, therefore, laboratory results must be used.

0. No problem
1. History of hepatitis (actually normal values of transaminases); cholecystectomy
2. Cholelithiasis; chronic hepatitis or previous hepatitis (< 5 years ago) or any other liver disease (hemochromatosis, primary biliary cirrhosis) with mildly elevated transaminases (within 3-times normal values); heavy alcohol use within 5 years (to rate in "psychiatric," too)
3. Chronic hepatitis or any other liver disease with marked elevation of transaminases (> 3-times normal values); elevated bilirubin
4. Acute cholecystitis; any biliary obstruction; active hepatitis/liver cirrhosis; any liver or biliary tree carcinoma

RENAL

This category is exclusive of kidney: kidney stones, acute/chronic renal failure, glomerulonephritis; nephrosic/nephritic syndrome; active/chronic pyelonephritis, diabetic or hypertensive nephropathy (albuminuria/proteinuria), and renal carcinoma. Bence-Jones proteinuria in multiple myeloma should not be considered.

0. No problem
1. Asymptomatic kidney stone; kidney stone passage within the last 10 years; pyelonephritis within 5 years; kidney cysts without hematuria
2. Serum creatinine > 1.5 but < 3 mg/dl without diuretic or antihypertensive medication (particularly ACE-inhibitors or SRAA blockers); kidney calculi requiring daily medications
3. Serum creatinine > 3 mg/dl or < 1.5 mg/dl in conjunction with diuretics, antihypertensive, or bicarbonate therapy; active pyelonephritis; nephrosic syndrome; colic symptoms treated as an outpatient
4. Required dialysis; renal carcinoma; colic symptoms requiring hospitalization

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GENITOURINARY

Ureters, bladder, urethra. Genitals, prostate, testicles, penis, seminal vesicles.

Uterus, ovaries. *Mammary gland is rated under "metabolic."*

This category is comprehensive of all GU tract impairments: ureteral or bladder stones, benign prostate hypertrophy (BPH), urinary tract infections (UTIs), prolapses, etc. Urinary incontinence and indwelling catheter should also be considered.

0. No problem
1. Stress incontinence; BPH without urinary symptoms; hysterectomy or ovariectomy (uterine fibroma, benign neoplasm)
2. Pathological pap smear (or 2 consecutives abnormal); frequent UTIs (3 or more in the past year) in female or current UTIs; urinary incontinence (not stress) in females; BPH with urinary symptoms (frequency, urgency, hesitancy); status post TURP; any urinary diversion procedure; indwelling catheter; bladder calculi
3. Prostatic cancer in situ (eg, incidentally found during TURP); vaginal bleeding; cervical carcinoma in situ; hematuria (any cause); urinary incontinence (not stress) in males; bladder polyps
4. Acute urinary retention; current urosepsis; any GU malignancies except as above

MUSCULOSKELETAL/INTEGUMENT

This is a very wide category, including osteoarthritis, osteoporosis, any bone fracture; primary neoplasm (bone, muscle, connective tissue, skin), distinguishing melanoma from other localized skin cancers; rheumatoid arthritis and polymyalgia rheumatica; muscular injuries (rotator cuff, long head of the biceps); pressure sores; any dermatological disease.

The scores of this category are strictly correlated to the disability they cause; for the evaluation of the level of disability, refer to BADL and IADL.

NOTICE: score the severity of each illness according to the level of disability caused by the same illness in this category, without considering the disability caused by other diseases. For example, a patient affected both by osteoarthritis and hemiplegia from a previous stroke has a high level of disability, but you have to score 2 for disability by osteoarthritis (in this category) and 4 for disability by stroke (in the neurological category); for a patient with both a deforming rheumatoid arthritis and a previous stroke without remaining outcomes you have to score 4 for disability from arthritis (in this category) and 2 for disability from stroke (in the neurological category).

0. No problem
1. Requires PRN medications for osteoarthritis (NSAID) or has mildly limited IADL from joint pathology; excised skin cancers (except melanoma); skin infections requiring antibiotics within a year
2. Daily anti-osteoarthritis medications (NSAID) or use of assistive devices or little limitation in ADL (previous arthroprosthesis or treated fracture with a low level of remaining disability); osteoporosis without vertebral fractures; daily medications for

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chronic skin diseases (even local, as psoriasis or pressure sores); non-metastatic melanoma; daily medications for rheumatoid arthritis (except steroids) with a low level of disability

3. Osteoarthritis with a moderate level of disability in ADL; requires chronic treatment with steroids for arthritic conditions or joints' deformities or severely impaired; osteoporosis with vertebral compression fractures
4. Wheelchair bound for osteomuscular disease; severe joint deformities or severely impaired usage; osteomyelitis; any bone or muscle or connective tissue neoplasm (see "Rating Malignancies"); metastatic melanoma

Fractures and/or arthroprosthesis (both recent and old) have to be scored according to the level of disability they cause (considering outcomes too) to avoid confusion about possible classifications of different fractures or joints. The same is true for muscular diseases.

CENTRAL AND PERIPHERAL NERVOUS SYSTEM

This category includes the "somatic" pathologies of the central and peripheral nervous system: any kind of stroke, neurodegenerative diseases (Parkinson's disease and parkinsonism, multiple sclerosis, amyotrophic lateral sclerosis, etc), myelopathies, traumas with neurological outcomes, primary or secondary epilepsy, neuropathies (diabetic, alcoholic, any other etiology), primary tumors, chronic headaches (migraine), insomnia, etc. It must carefully estimate the severity and prognosis of the illness but also the functional impairment that the illness causes.

0. No problem (or fewer convulsions in childhood)
1. Frequent headaches requiring PRN medications without impairment in Advanced ADL; previous TIA (one event); previous epilepsy, actually not treated, without crisis since more than 10 years ago
2. Chronic headache requiring daily medications (even for prophylaxis) or with regularly functional impairment in Advanced ADL (bed rest, job withdrawal, etc); actual TIA or more than 1 previous TIA; previous stroke without significant residual; mild severity neurodegenerative diseases (see above), treated and well controlled; epilepsy controlled with drugs
3. Previous stroke with mild residual dysfunction (hemiparesis, dysarthria); any neurosurgical procedure; moderate severity neurodegenerative diseases (see above), not well controlled by medications; epilepsy in treatment but with periodic crisis
4. Acute stroke or previous stroke with severe residual dysfunction (hemiplegia, aphasia, severe vascular dementia) or more than 1 previous stroke (multi-infarct encephalopathy); severe neurodegenerative diseases (see above) causing disability in ADL; neurological coma

Alzheimer's disease and dementia should not be rated into this category (Psychiatric and behavioral diseases): Alzheimer's disease should be listed only under psychiatric disorders; if dementia stems from vascular and/or mixed dementia and/or other neurological condition (eg, Parkinson's Disease), both "neurologic" and "psychiatric" categories should be endorsed at the appropriate level for severity, considering in this category the stroke and the multi-infarct

encephalopathy responsible for the cognitive impairment (score 3 for stroke with remaining outcomes, score 4 for multi-infarct encephalopathy).

ENDOCRINE-METABOLIC SYSTEM AND BREAST (SYSTEMIC INFECTIONS AND POISONINGS)

Type 1 and Type 2 diabetes (organ damage should be considered into the respective categories, like for hypertension), obesity and dyslipidemia (hypercholesterolemia) represent the core of this category; it also includes hypo- and hyperthyroidism, hypo- and hyperparathyroidism, adrenal pathologies (Cushing' or Addison' disease), hypogonadism, hypopituitarism, etc. Malignancies of these glands, both benignant (like thyroid nodules) and malignant (like thyroid or adrenal cancer, vipoma) are included too.

Even if it is an exocrine gland, breast was included in this category because the authors did not find a more appropriate one, so it includes the breast cancer too.

Moreover, it includes electrolyte disorders, sepsis, systemic infections (like tuberculosis, syphilis, AIDS) scored according to their severity and the functional impairment they cause (see general indications), and poisonings (chronic by metals or acute by pesticides or carbon monoxide).

0. No problem
1. Diabetes and/or dyslipidemia compensated with diet; mild obesity (BMI 30-35 kg/m^2); hypothyroidism in replacement therapy (L-thyroxin); hyperthyroidism caused by Plummer' adenoma surgically treated
2. Diabetes compensated with oral hypoglycemic drugs or insulin (hemoglobin A1c < 7%); dyslipidemia well controlled by daily medications (c-LDL lower than the recommended target according to the individual global cardiovascular risk); moderate obesity (BMI 35−45 kg/m^2); hyperthyroidism in pharmacologic treatment; asymptomatic or surgically treated hyperparathyroidism; fibrocystic breast disease
3. Diabetes not well compensated by therapy (hemoglobin A1c 7−8.5%, presence of complications); dyslipidemia not well controlled (c-LDL higher than the recommended target according to the individual global cardiovascular risk; for instance, c-LDL > 100 mg/dl in patients with previous myocardial infarction or stroke); severe obesity (BMI > 45 kg/m^2); symptomatic hyperparathyroidism (eg, hypercalcemia); replacement therapy for adrenal failure; any electrolytes disorder requiring hospitalization
4. Uncontrolled diabetes (hemoglobin A1c > 8.5%) or 1 diabetic ketoacidosis or nonketotic hyperosmolar coma during the past year; genetic uncontrolled dyslipidemia; acute adrenal failure during hormonal replacement therapy; any neoplasm of thyroid, breast, adrenal gland (see "Rating Malignancies")

NOTICE: when the patient is not treated with drug therapy for diabetes or dyslipidemia but he should be for the optimal control of the pathology (for instance, hemoglobin A1c > 7%, total cholesterol > 250 mg/dl), score the pathology according to the laboratory values, which really define its severity.

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PSYCHIATRIC AND BEHAVIORAL DISEASES

This category includes both dementia and related behavioral disorders (psychosis, anxiety, depression, agitation) and all the pre-existing and/or not related to dementia psychiatric disorders. Because this is the only item analyzing patient's mental status (all the others refer to physical status), it is very important to evaluate it considering carefully further information derived from the Comprehensive Geriatric Assessment (MMSE; Geriatric Depression Scale, Neuro-Psychiatric Inventory if available).

0. No psychiatric problem or history thereof
1. Minor psychiatric condition or history thereof: previous (occasional) psychiatric treatment without hospitalization; major depressive event and/or use of antidepressants more than 10 years ago without hospitalization; occasional use of minor tranquilizers (eg, BDZ; even if as hypnotherapy for insomnia); mild cognitive impairment (MMSE 25-28)
2. A history of major depression (according to DSM-IV criteria) within the last 10 years (treated or untreated); mild dementia (MMSE 20-25); previous admission to Psychiatric Department for any reason; history of substance abuse (more than 10 years ago, including alcoholism).
3. Current major depression (according to DSM-IV criteria) or more than 2 previous major depression episodes in the past 10 years; moderate dementia (MMSE 15−20); current and usual usage of daily anti-anxiety medications (even as hypnotherapy for insomnia); current or within the past 10 years substance abuse or dependence (according to DSM-IV criteria); requires daily antipsychotic medication; previous attempt at suicide
4. Current mental illness requiring psychiatric hospitalization, institutionalization, or intensive outpatient management (psychiatric emergency, as attempt at suicide or severe depression with suicide purpose, acute psychosis or acute decompensation of chronic psychosis, severe substance abuse; severe agitation from dementia); severe dementia (MMSE < 15); delirium (acute confusion or altered mental status for medical (organic) reasons: in this case you have to codify also the medical cause in its own category with the appropriate level of severity)

It could be requested psychiatric consult for this category; dementia and depression, the most frequent diseases in the elderly, can be scored in detail using the MMSE and GDS. The severity of any mental disorder (dementia, depression, anxiety, psychosis, substance abuse, and all the others) has to be scored according to the level of functional impairment or disability they cause.

CHECK LIST

Medical history

1. Timing of events and/or interventions (how long ago underwent surgery for, how long ago had myocardial infarction or stroke, etc) and evaluation of functional impairment
2. Drugs list (fundamental), including laxatives and tranquilizers (even hypnoinducent)
3. Symptoms of atherosclerotic disease (TIA, angina, claudication, amaurosis)
4. Etiological diagnosis (reasonably reliable) of anemia

5. Degree of vascular stenosis or aneurism dimension (by Doppler and/or ultrasound and/or TC data, when available)
6. Information about smoking status (how many cigarettes per day for how many years, when stopped)
7. Glasses use? With this aid, is the patient able to read a newspaper? Does the patient require an escort to venture out?
8. Any hearing aid? (You should evaluate possibility to communicate with patient.)
9. "Peptic history" of the patient (including previous eradication therapy for H. pylori)
10. Urinary symptoms, incontinence, presence of bladder catheter (even from BADL)

PHYSICAL EXAMINATION

a. Height (m^2) and weight (kg) (measured, not reported, if possible) to calculate BMI
b. Blood pressure, heart rate, cardiac murmurs, peripheral arterial pulses
c. Joint pain or passive stiffness limitation (non−X-ray-based diagnosis of osteoarthritis)
d. Neurological residual (dysarthria/aphasia, hemiparesis/hemiplegia)

BASELINE LABORATORY TESTS

- Blood count: hemoglobin, WBC and platelets
- Creatinine, electrolytes
- Cholesterol levels (total, HDL) and triglycerides
- Aspartate aminotransferase, alanine aminotransferase, fractioned bilirubin
- Thyroid function and serum B12 (when indicated)
- Hemoglobin A1c (for diabetic patients)

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APPENDIX 14. GRADING SCALE FOR HEMATOLOGIC TOXICITY IN CLL STUDIES

Grade[a]	Decrease in platelets[b] or Hgb[c] (nadir) from pretreatment value	Absolute neutrophil count (nadir) x 10^9/L[d]
0	No change to 10%	≥ 2
1	11% - 24%	≥ 1.5 and < 2
2	25% - 49%	≥ 1 and < 1.5
3	50% - 74%	≥ 0.5 and < 1
4	≥ 75%	< 0.5

Source: Hallek et al 2018

Abbreviation: ANC, absolute neutrophil count; CLL, chronic lymphocytic leukemia; G-CSF, granulocyte colony-stimulating factor; Hgb, hemoglobin; WBC, white blood cell.

[a] Grades: 1, mild; 2, moderate; 3, severe; 4, life-threatening; 5, fatal. Death occurring as a result of toxicity at any level of decrease from pretreatment will be reported as Grade 5.

[b] Platelet counts must be below normal levels for Grades 1 to 4. If, at any level of decrease, the platelet count is < 20 × 10^9/L, this will be considered Grade 4 toxicity, unless a severe or life-threatening decrease in the initial platelet count (eg, < 20 × 10^9/L) was present pretreatment, in which case the patient is not evaluable for toxicity referable to platelet counts.

[c] Hgb levels must be below normal levels for Grades 1 to 4. Baseline and subsequent Hgb determinations must be performed before any given transfusions. The use of erythropoietin is irrelevant for the grading of toxicity but should be documented.

[d] If the ANC reaches < 1 × 10^9/L, it should be judged to be Grade 3 toxicity. Other decreases in the WBC count, or in circulating granulocytes, are not to be considered because a decrease in the WBC is a desired therapeutic endpoint. A gradual decrease in granulocytes is not a reliable index in CLL for stepwise grading of toxicity. If the ANC was < 1 × 10^9/L before therapy, the patient is not evaluable for toxicity referable to the ANC. The use of growth factors such as G-CSF is not relevant to the grading of toxicity but should be documented.

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APPENDIX 15. EXAMPLES OF P-GLYCOPROTEIN AND BCRP INHIBITORS

Strong P-gp Inhibitors (To be Avoided)
Antibiotics: clarithromycin, tetracycline, gentamicin, erythromycin
Antifungals: itraconazole, posaconazole
Antivirals: telaprevir and cobicistat [a]
Cardiovascular Drugs: amiodarone, dronedarone, carvedilol, verapamil, captopril, quinidine, propafenone, ranolazine
Protease inhibitors: ritonavir [a], lopinavir and ritonavir, saquinavir and ritonavir, tipranavir and ritonavir
Other: lapatinib, vemurafenib, flibanserin
Strong BCRP Inhibitors (To be Avoided)
Curcumin, cyclosporine A, eltrombopag
Moderate P-gp Inhibitors (To be Used With Caution)
Antibiotics: trimethoprim, telithromycin
Antivirals: saquinavir, velpatasvir
Cardiovascular Drugs: diltiazem, tolvaptan, conivaptan, ticagrelor, atorvastatin
Blood pressure: spironolactone, telmisartan, nifedipine
Proton-pump inhibitor: rabeprazole
Other: indomethacin, nefazodone, quinine, propantheline, mirabegron

Source: University of Washington, drug-drug interaction database and FDA Drug Development and Drug Interactions.
Abbreviation: BCRP, breast cancer resistance protein; P-gp, P-glycoprotein
Note: The list of drugs in this table is not exhaustive. Please refer to the prescribing information and Summary of Product Characteristics to check for P-gp and BCRP inhibition risks or contact the medical monitor.
[a] Ritonavir and cobicistat are usually given in combination with other antiviral drugs in clinical practice.

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APPENDIX 16. KIDNEY FUNCTION CALCULATION

Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI 2021) equation (Inker et al 2021) for glomerular filtration rate (GFR).

$$GFR = 141 \times \min(Scr/\kappa, 1)^{\alpha} \times \max(Scr/\kappa, 1)^{-1.209} \times 0.993^{Age} \times 1.018 \text{ [if female]} \times 1.159 \text{ [if black]}$$

where Scr is serum creatinine in μmol/L (mg/dL), κ is 0.7 for females and 0.9 for males, α is -0.329 for females and -0.411 for males, min indicates the minimum of Scr/κ or 1, and max indicates the maximum of Scr/κ or 1.

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